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ST67W611 Wi-Fi 6 + Bluetooth® LE + Thread

Santa Clara Wireless Team

Agenda

X-Cube-ST67W61 Package

Device Setup and Flashing

BLE Commissioning App

Generating BLE Commissioning app using NUCLEO-U575ZI-Q as host processor (Hands-on)

Porting to a new host mcu using STM32CubeMX

Generating the BLE Commissioning app using NUCLEO-G0B1RE processor (Hands-on)

Documentation

PLEASE DOWNLOAD ALL THE TOOLS LISTED BELOW Prerequisites

- X_CUBE_ST67W61_V1.1.0 (To be installed under C drive)
https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0
- STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack
https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0_CubeMX
- [STM32CubeIDE v1.16.1+](#)
- [STM32CubeProgrammer v2.18.0+](#) To be installed in default directory
(C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer)
- [STM32CubeMX v6.15.0](#)
- [HyperTerminal – TeraTerm v.5.4.0](#)
- [ST BLE Toolbox](#) (To be downloaded on mobile)



X_CUBE_ST67W61_V1.1.0

https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0

← ↻ 🔒 https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0 📄 🗨️ ⭐ ⚙️

☰ STMicroDEMO / ST67_Experts_Training 🔍 Type / to search 🗑️ + 🕒

<> Code 🕒 Issues 🔗 Pull requests 🔄 Actions 📁 Projects 📖 Wiki 🛡️ Security 📈 Insights ⚙️ Settings

[Releases](#) / v1.1.0

X-CUBE-ST67W61_V1.1.0.RC1_Setup.exe

🛡️ STMicroDEMO released this 24 minutes ago 📦 v1.1.0 🔑 4649534 ✅

This is X-CUBE-ST67W61_V1.1.0.RC1_Setup.exe. The package needs to be downloaded and placed under C drive

▼ **Assets** 3

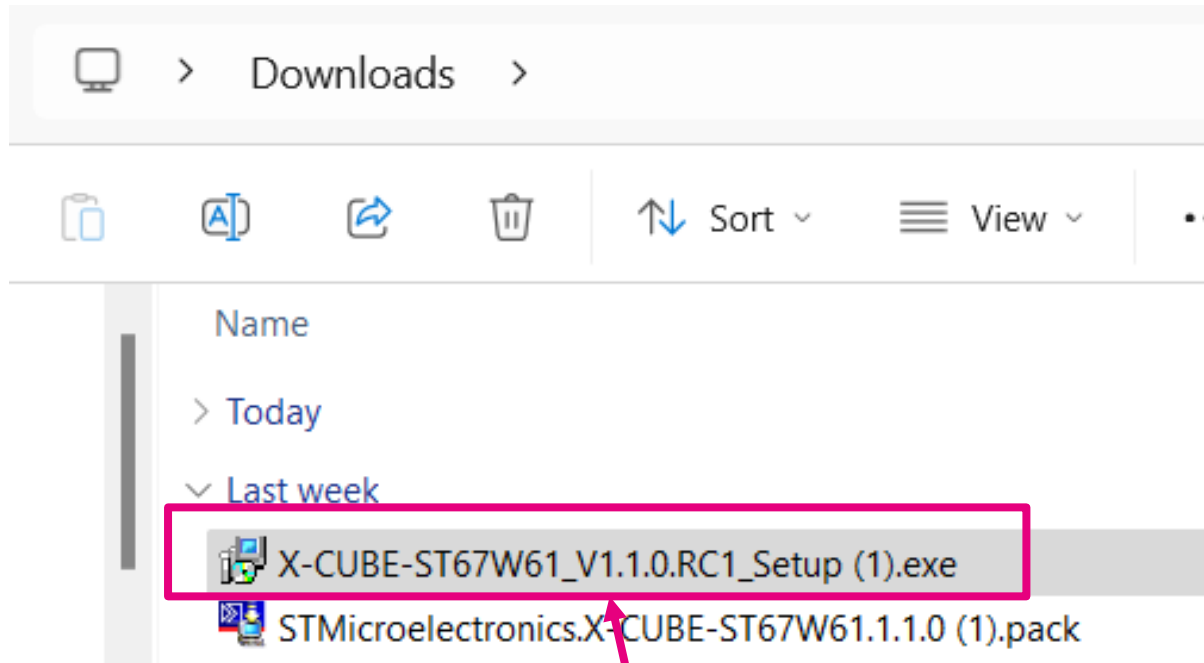
📦 X-CUBE-ST67W61_V1.1.0.RC1_Setup.1.exe	sha256:9428890914a826891034034ac3b7c...	📄	318 MB	19 minutes ago
📄 Source code (zip)				1 hour ago
📄 Source code (tar.gz)				1 hour ago

😊

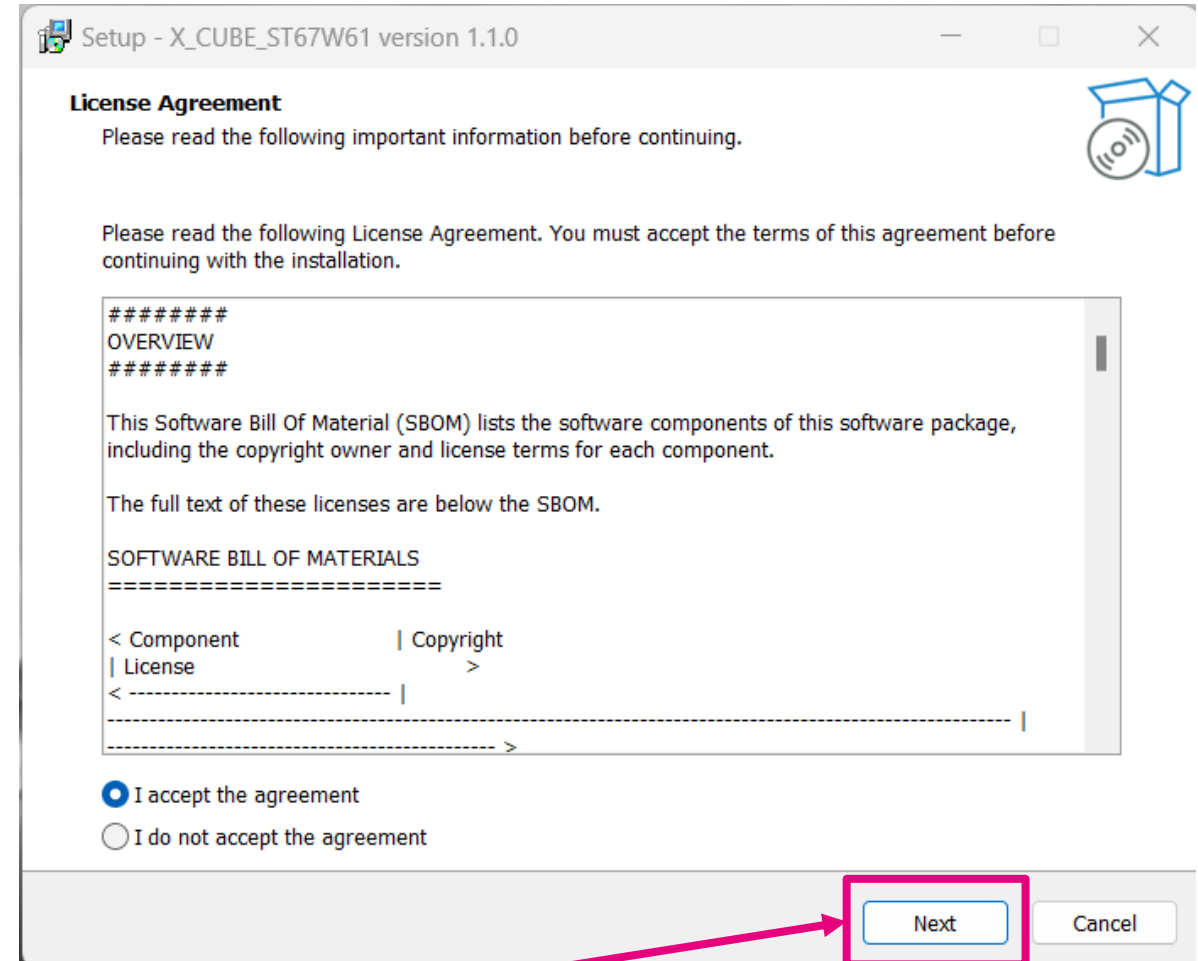
Download the package



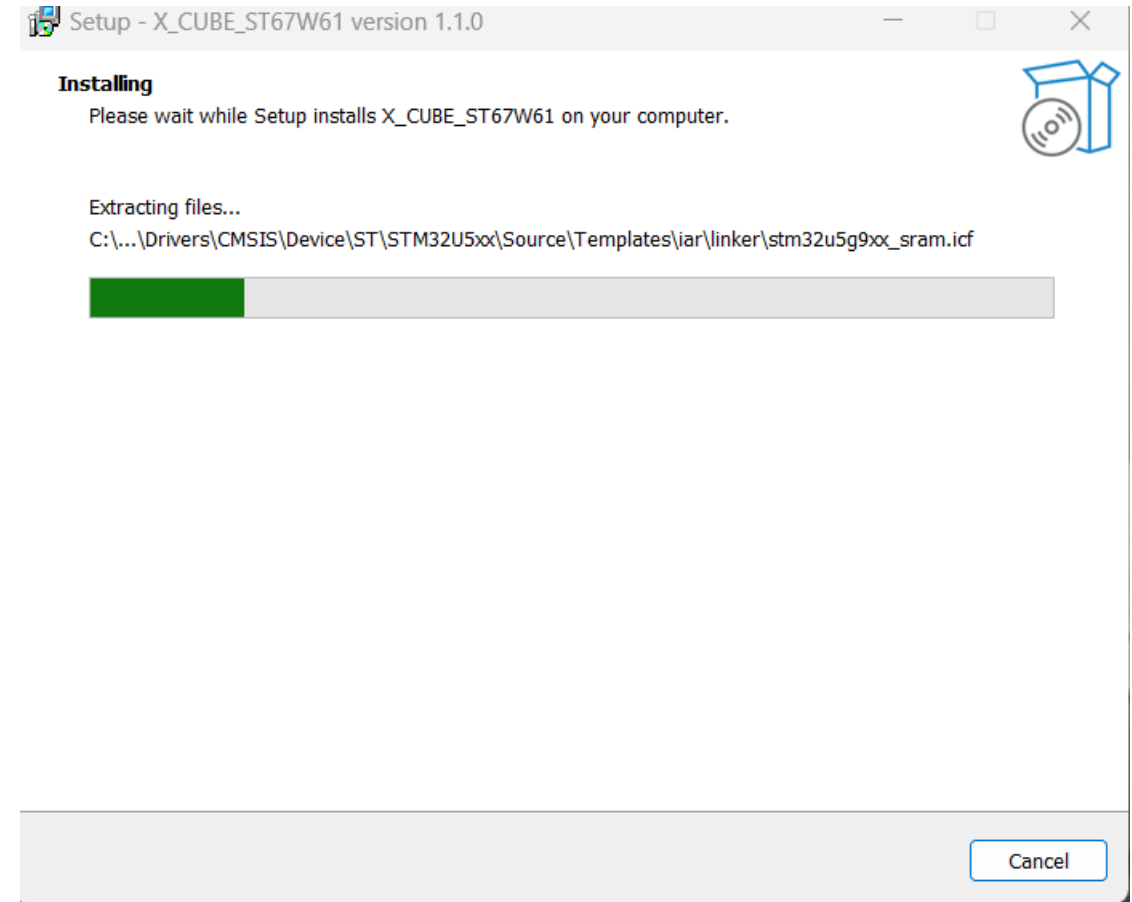
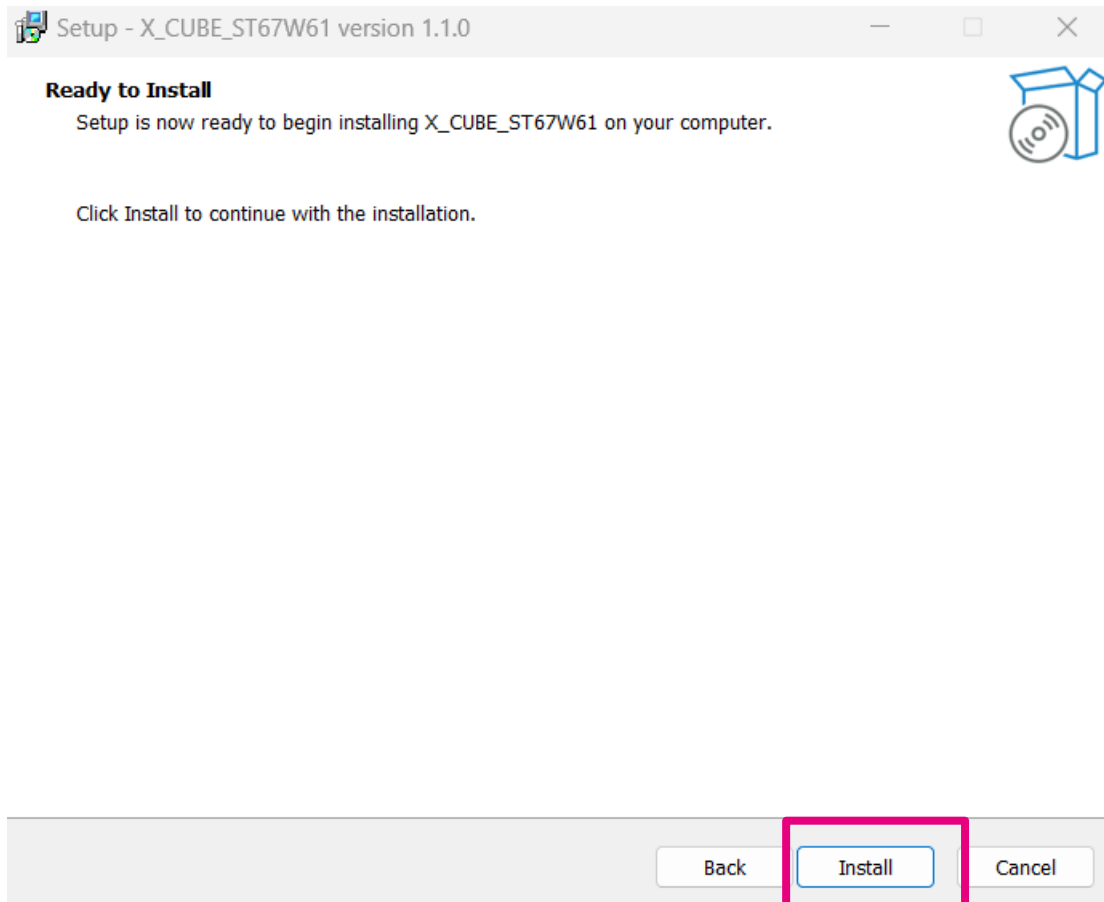
X_CUBE_ST67W61_V1.1.0



Under Download-> Double-click on the package

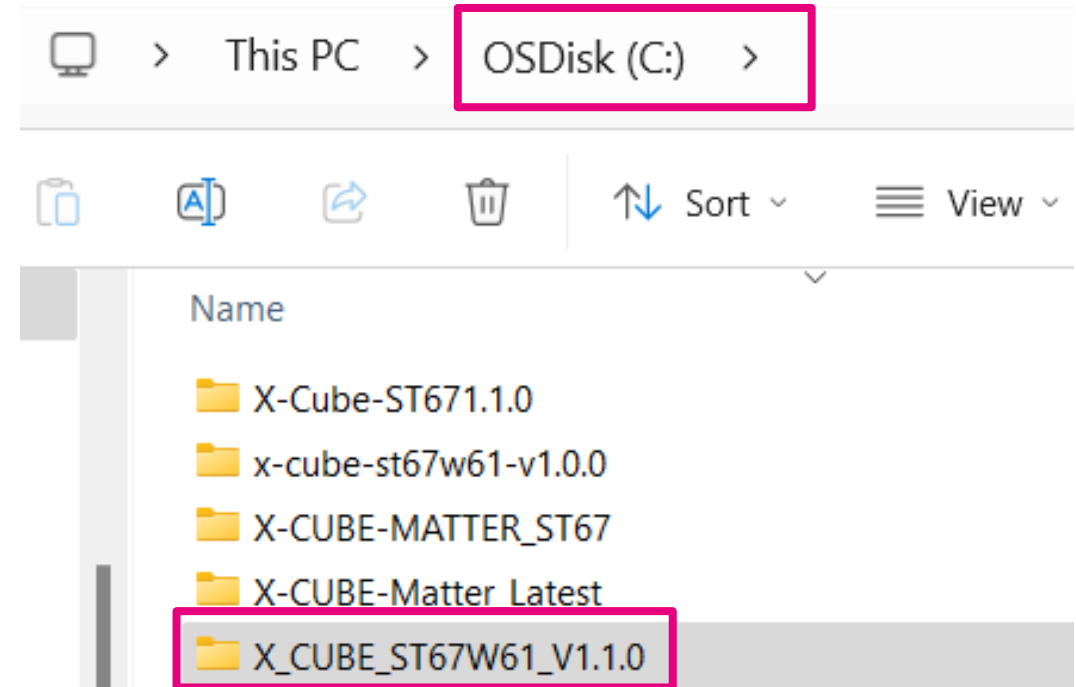
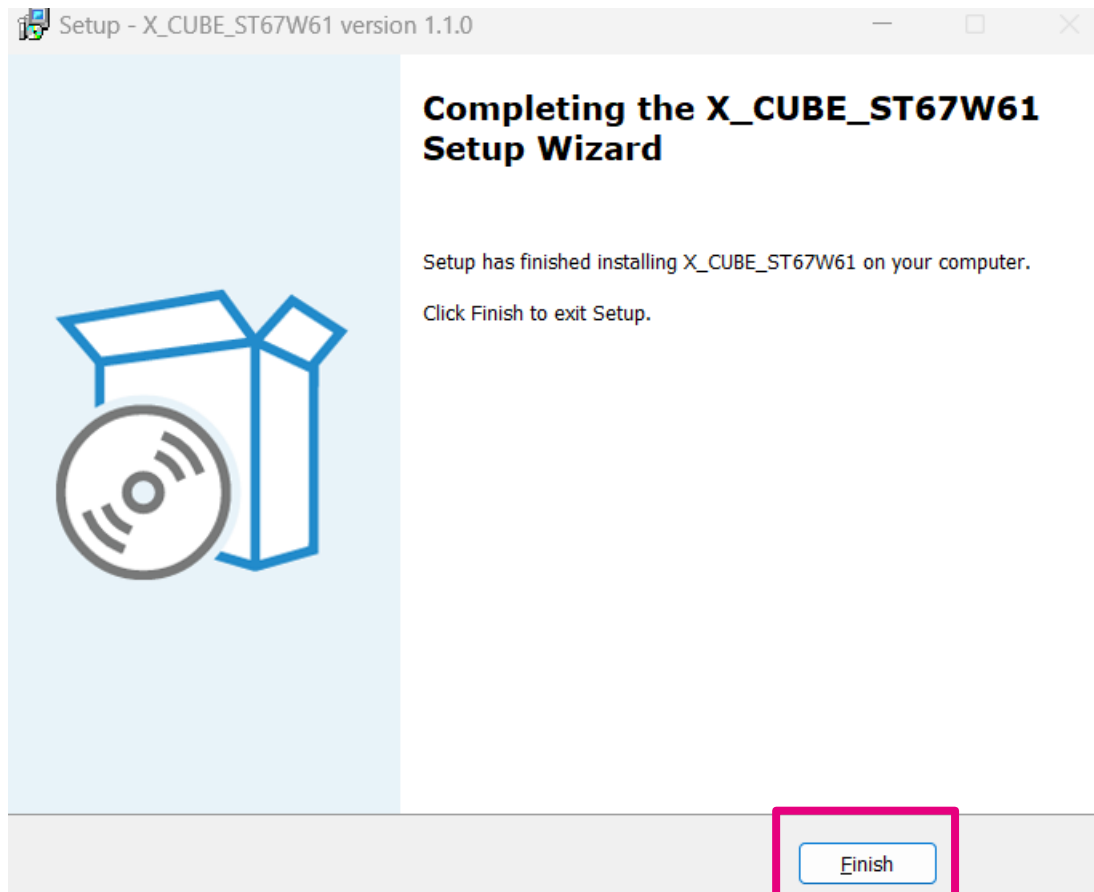


X_CUBE_ST67W61_V1.1.0



Package is installing

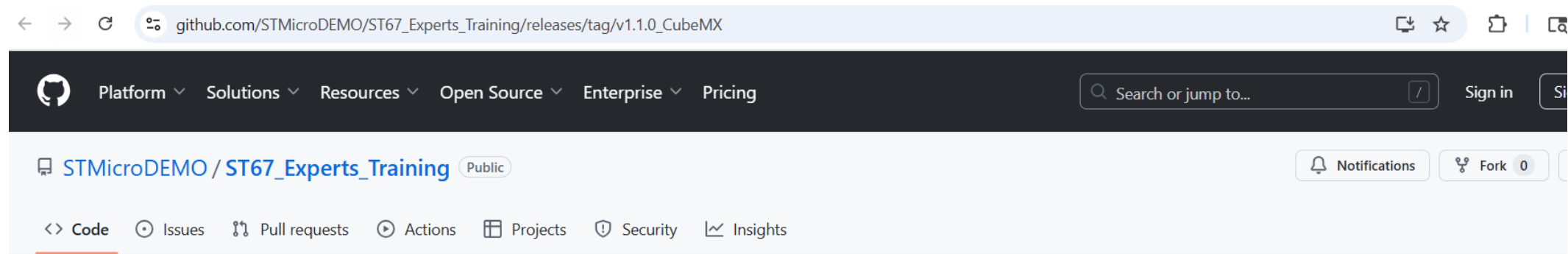
X_CUBE_ST67W61_V1.1.0



Package is installed under C drive

STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack

https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0_CubeMX



Releases / v1.1.0_CubeMX

STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack

Latest

Compare

STMicroDEMO released this 41 minutes ago v1.1.0_CubeMX 4649534

This is STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack. This package needs to be loaded in STM32CubeMX software for the hands-on session

Assets 3

STMicroelectronics.X-CUBE-ST67W61.1.1.0.1.pack

sha256: 211a3841d6c2273128443cc7beb15...



94.3 MB

42 minutes ago

Source code (zip)

1 hour ago

Source code (tar.gz)

1 hour ago

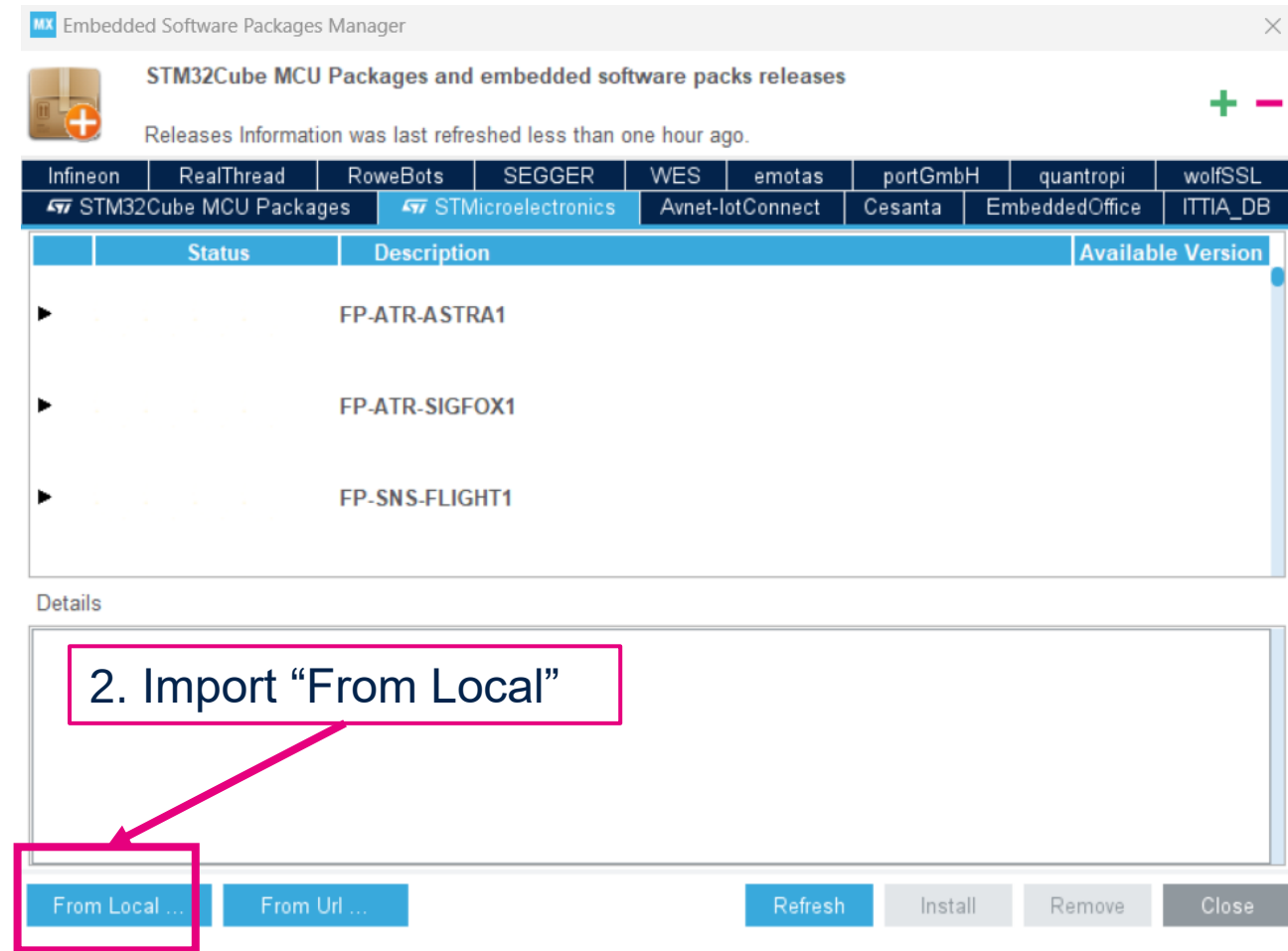
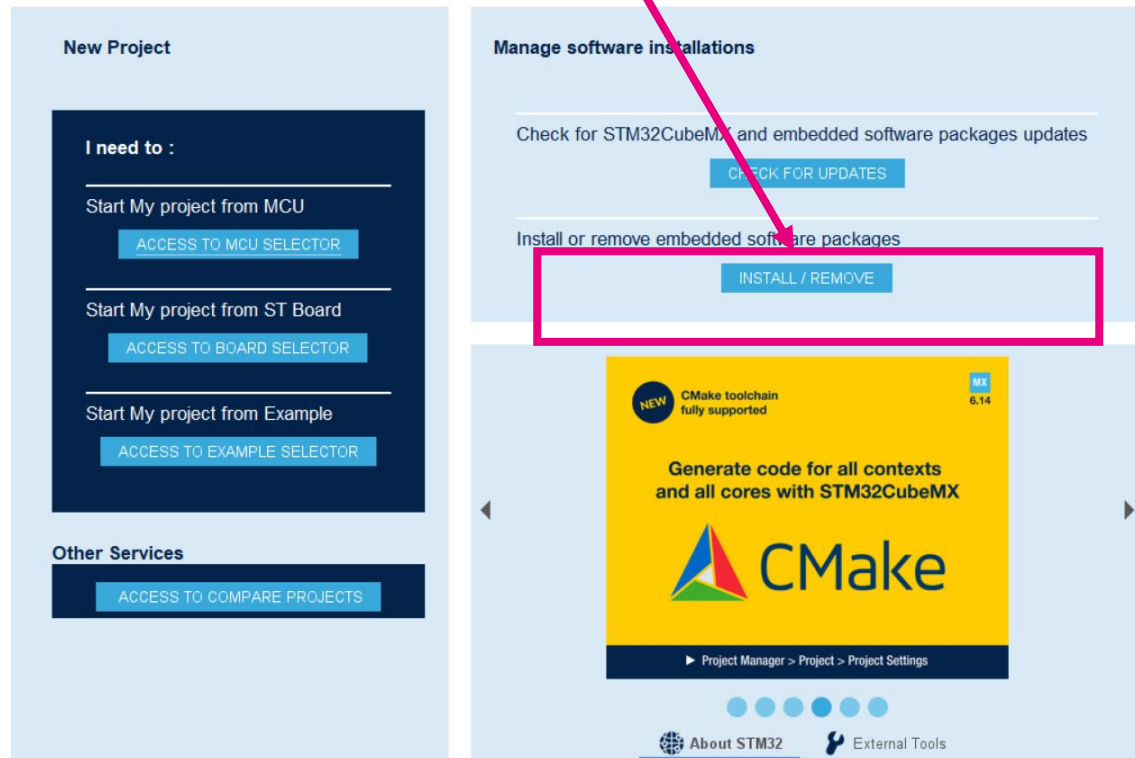
Download the package



STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack

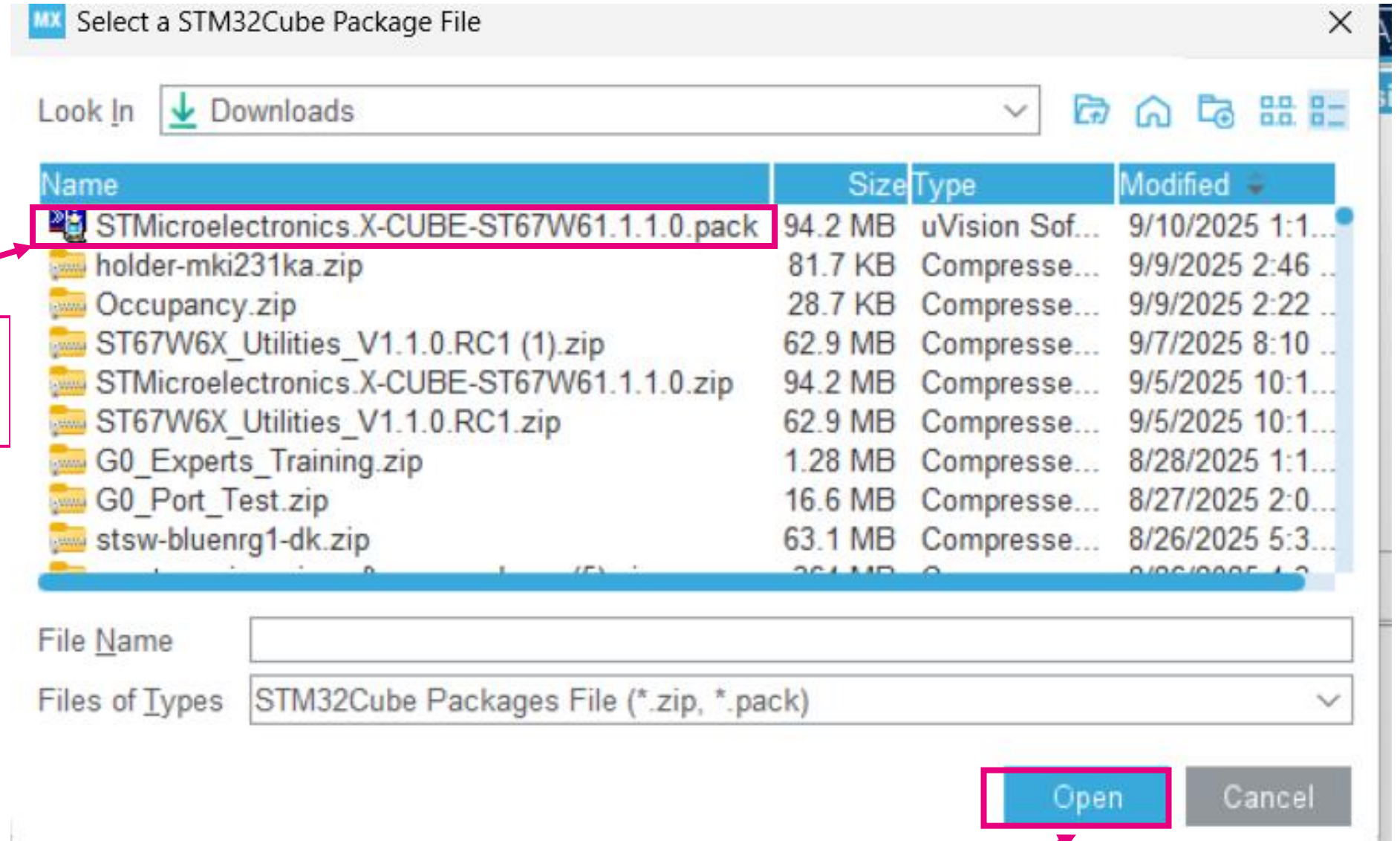
Open STM32CubeMX with admin rights

1. Go to “Install/Remove” software packages



2. Import “From Local”

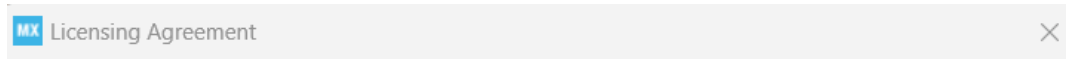
STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack



3. Select
STMicroelectronics.X-CUBE-
ST67W61.1.1.0.pack

4. Click Open

STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack



STMicroelectronics X-CUBE-ST67W61 1.1.0 License Agreement

Please read and accept the following agreement carefully to finish the installation:

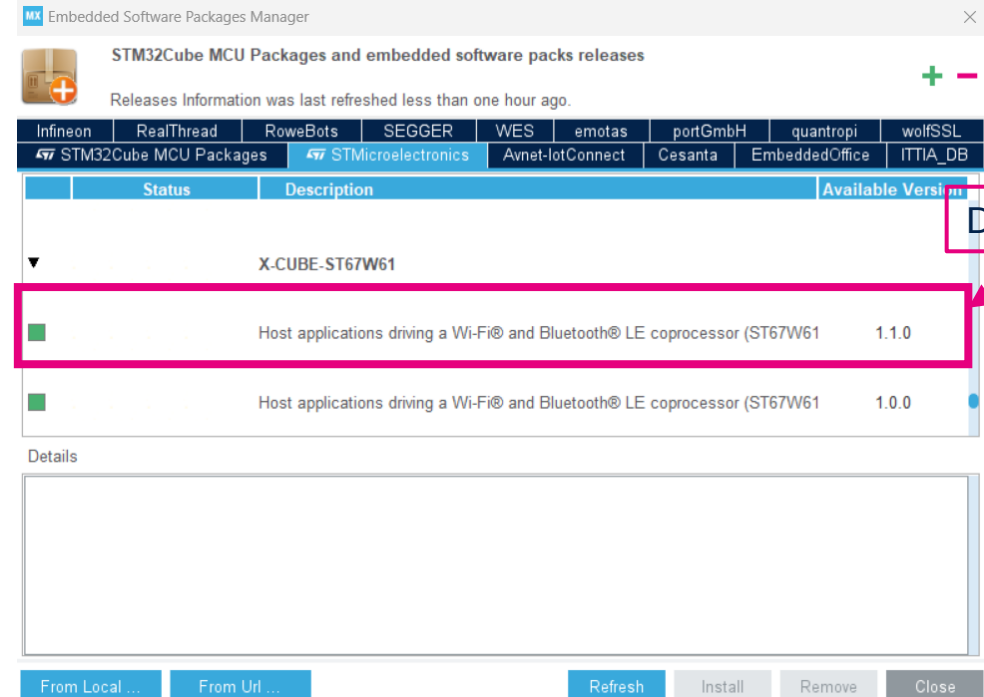
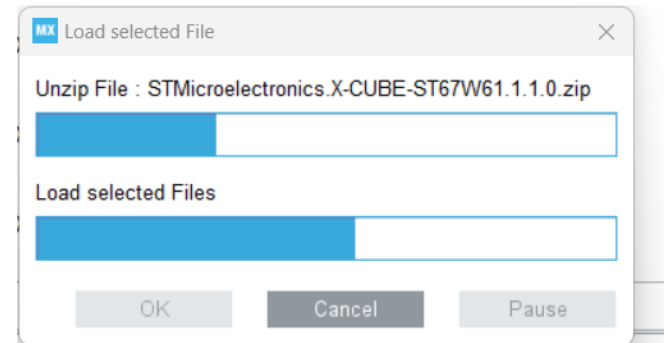
[Click here to open the license agreement](#)

☒ I have read, and I agree to the terms of this license agreement

☐ I do not accept the terms of this license agreement

Finish

Cancel



Downloaded

X-CUBE-FREERTOS

Download latest X-CUBE-FREERTOS

MX Embedded Software Packages Manager

STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed 4 hours ago.

Infinion	RealThread	RoweBots	SEGGER	WES	emotas	portGmbH	quantropi	wolfSSL
STM32Cube MCU Packages	STMicroelectronics	Avnet-lotConnect	Cesanta	EmbeddedOffice	ITTIA_DB			

Status	Description	Available Version
▼	X-CUBE-FREERTOS	
■	FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series	1.3.1
□	FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series (Siz	1.3.0
□	FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series (Siz	1.2.0
□	FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series (Siz	1.1.0

STM32U5

MX Embedded Software Packages Manager

 **STM32Cube MCU Packages and embedded software packs releases** + -

Releases Information was last refreshed 4 hours ago.

Infinion	RealThread	RoweBots	SEGGER	WES	emotas	portGmbH	quantropi	wolfSSL
STM32Cube MCU Packages	STMicroelectronics	Avnet-IotConnect	Cesanta	EmbeddedOffice	ITTIA_DB			
Description	Installed Version	Available Version						
▼ STM32U5								
 STM32Cube MCU Package for STM32U5 Series	1.8.0	1.8.0						
☐ STM32Cube MCU Package for STM32U5 Series (Size : 390 MB)		1.7.0						
☐ STM32Cube MCU Package for STM32U5 Series (Size : 351 MB)		1.6.0						
☐ STM32Cube MCU Package for STM32U5 Series (Size : 306 MB)		1.5.0						

Download STM32U5
(v.1.7.0)

STM32G0

Embedded Software Packages Manager

STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed 20320 days ago.

Infinion	RealThread	RoweBots	SEGGER	WES	emotas	portGmbH	quantropi	wolfSSL
STM32Cube MCU Packages	STMicroelectronics	Avnet-IotConnect	Cesanta	EmbeddedOffice	ITTIA_DB			
Description	Installed Version	Available Version						
▶ STM32F4								
▶ STM32F7								
▼ STM32G0								
■ STM32Cube MCU Package for STM32G0 Series	1.6.2	1.6.2						
□ STM32Cube MCU Package for STM32G0 Series (Size : 204.68 MB)		1.6.1						
□ STM32Cube MCU Package for STM32G0 Series (Size : 203 MB)		1.6.0						

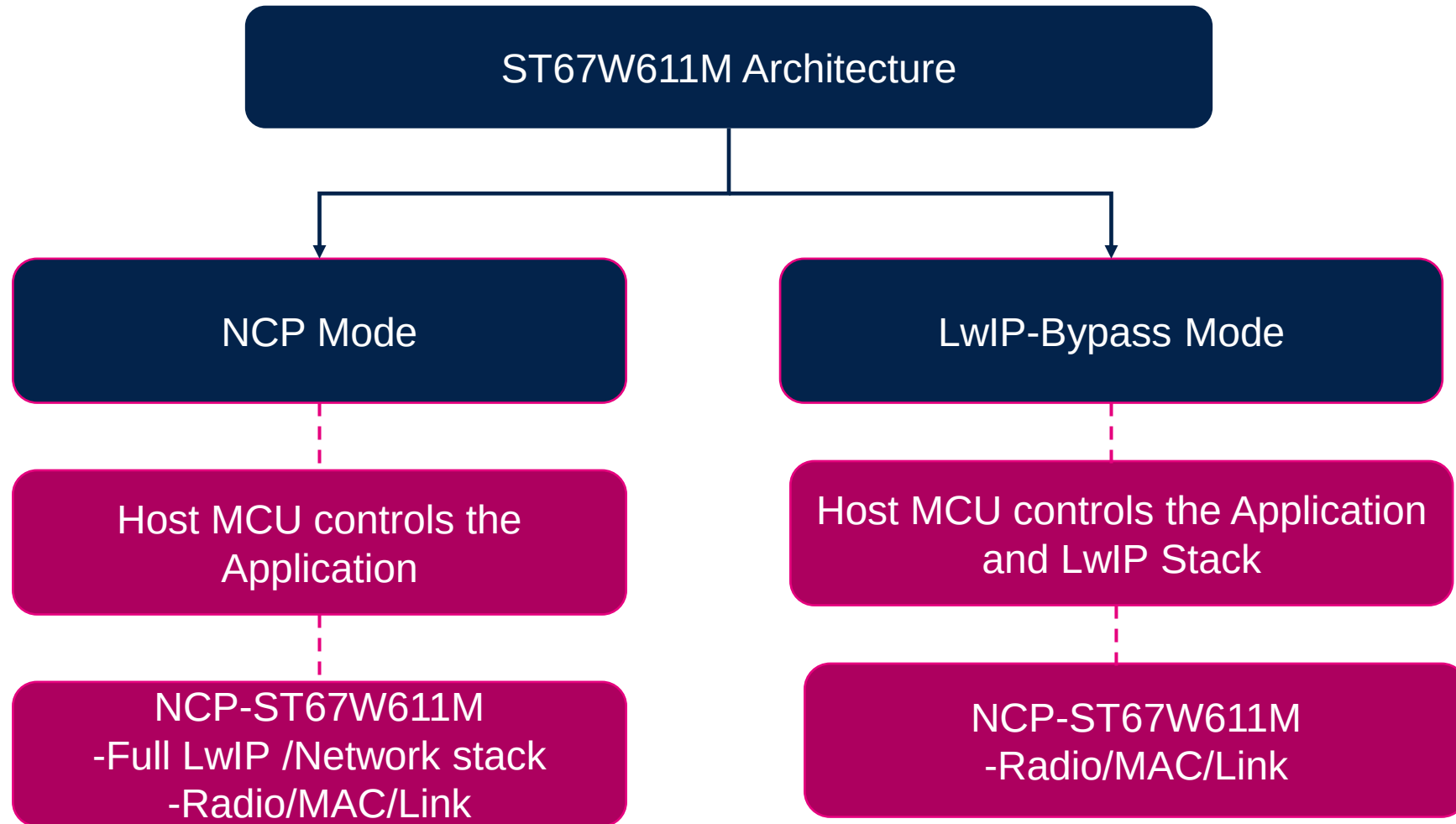
Download latest STM32G0



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X-Cube-ST67W61

ST67W611M Architecture

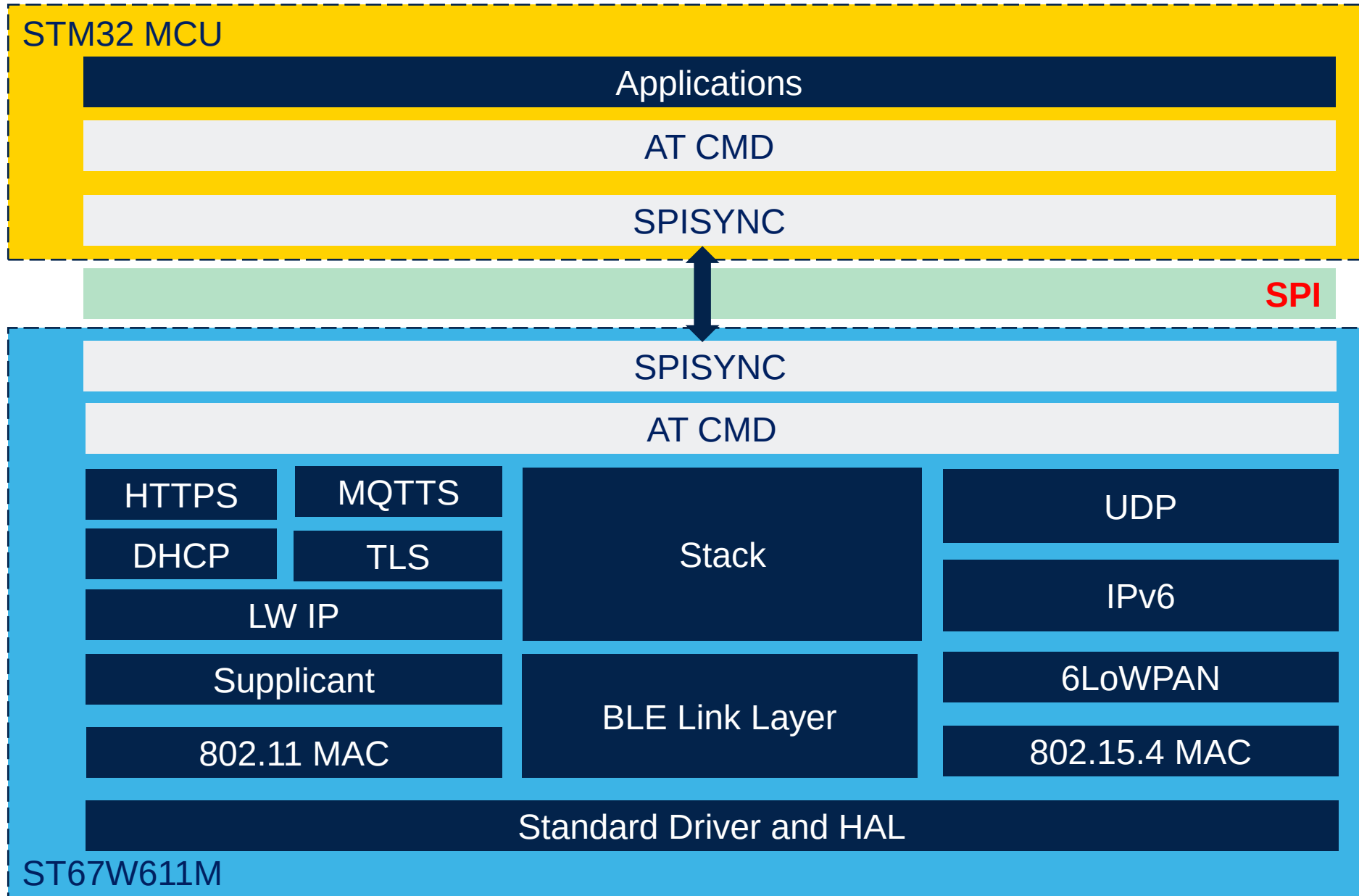


T01:NCP Mode

- **LwIP (TCP/IP stack) and all networking protocols** run on the **NCP (ST67W611M)**, not on the host (STM32 MCU)
- The NCP (ST67W611M) is **responsible for the entire network stack**, including application protocols (HTTPS, MQTT), security (TLS), and wireless operations (radio, MAC, link layer)
- The **host (STM32 MCU) runs only the application logic** and interacts with the NCP using simple AT commands
- Communication between the host and NCP is done via SPI using AT commands and the SPISYNC protocol
- This mode is ideal when the host MCU has limited resources or when the user wants to simplify host software development and integration.



T01:NCP Mode

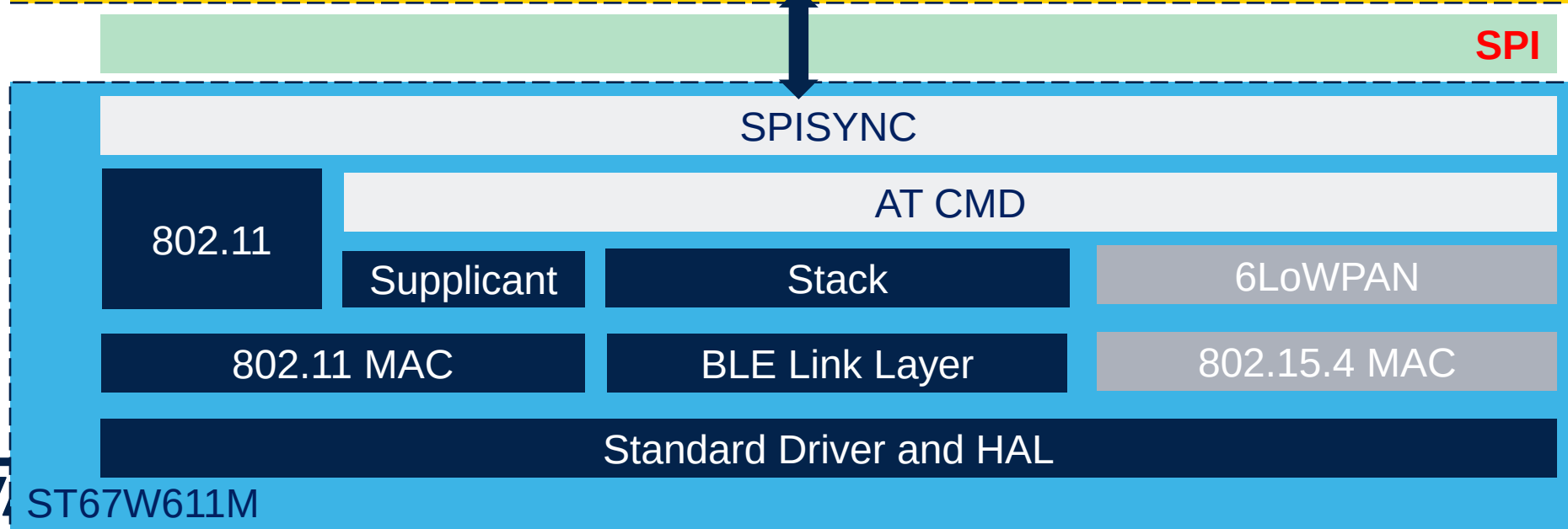
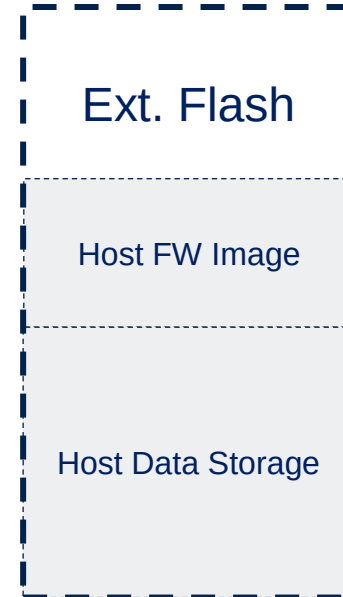
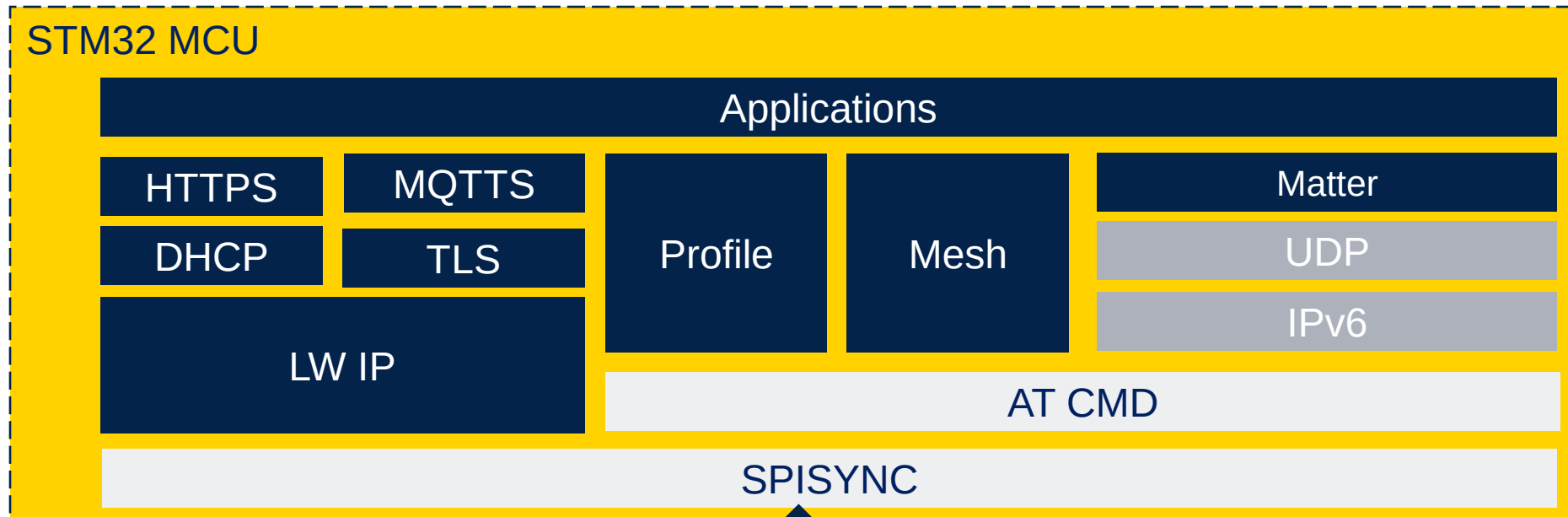


T02: NCP Mode – LwIP bypass

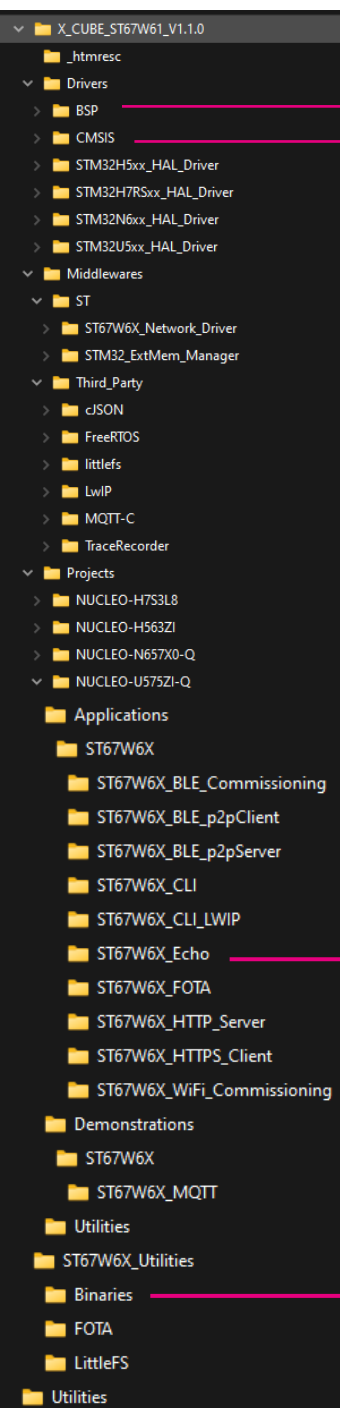
- **LwIP (TCP/IP stack) runs on the host (STM32 MCU), not on the NCP**
- The NCP (ST67W611M) is responsible only for the radio, MAC, and link layer operations.
- **The host, allowing for more flexibility** and has full control over the network stack advanced networking features
- Communication between the host and NCP is done via SPI using AT commands and the **SPISYNC protocol**
- This mode is ideal **when the host MCU is powerful enough to handle the network stack and/ or the user needs advanced networking features or customizations.**



T02:NCP Mode – LwIP bypass

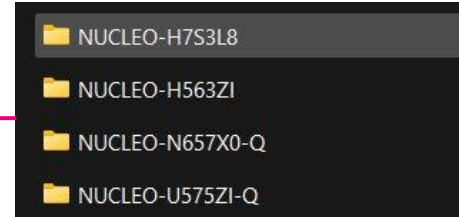


X-CUBE-ST67W61 package overview

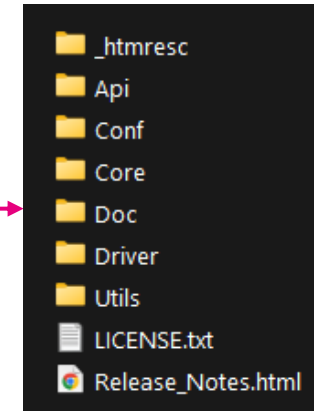


Board support package
CMSIS libraries

Device HAL drivers



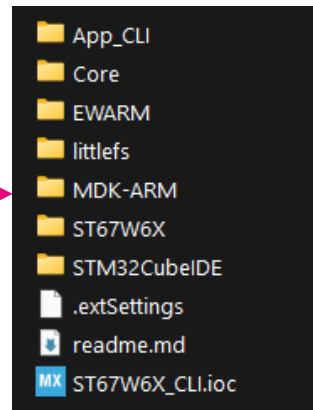
Variety of
Nucleo hosts



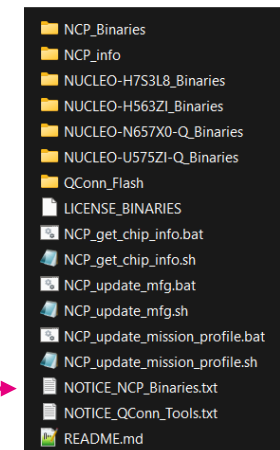
APIs include files
Templates for middleware and driver config files
Integration between generic API and driver

W61 driver
AT commands / responses
SPI driver

UART Output logs
Memory usage statistics
CLI Console



STM32CubeIDE and **EWARM** Project Files



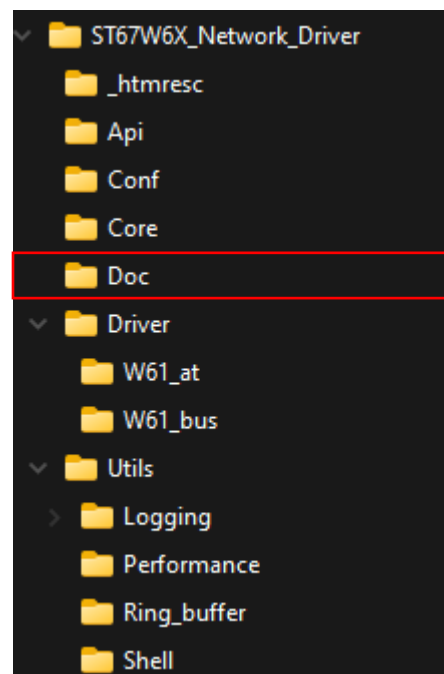
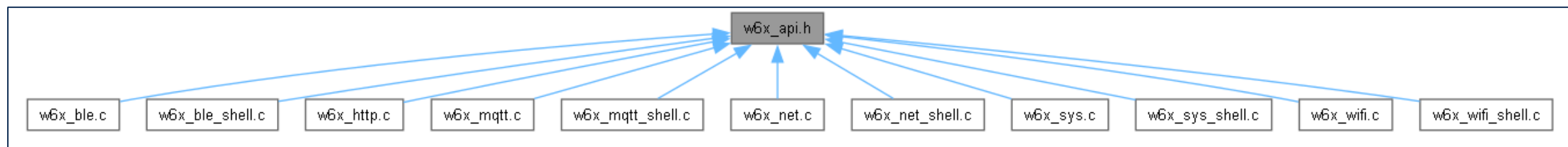
**NCP and
Host binaries**

X-CUBE-ST67W61 middleware

	Scope / description
W61_bus	This is the SPI driver for communication between the host and the ST67W61.
W61_at	It does the formatting of the AT commands to be sent, and it implements a receive tasks to decode the received messages from the W61 and dispatch them to the correspondent module.
W6x service API	The Core directory implements an adaptation layer between the W61 AT driver and the API proposed to the application. The APIs are "Zephyr-like" for the Wi-Fi® and socket API for the network operations.
Utils	Console handling, debug, trace, performance measurements.

All the above submodules make use of FreeRTOS™

API Documentation

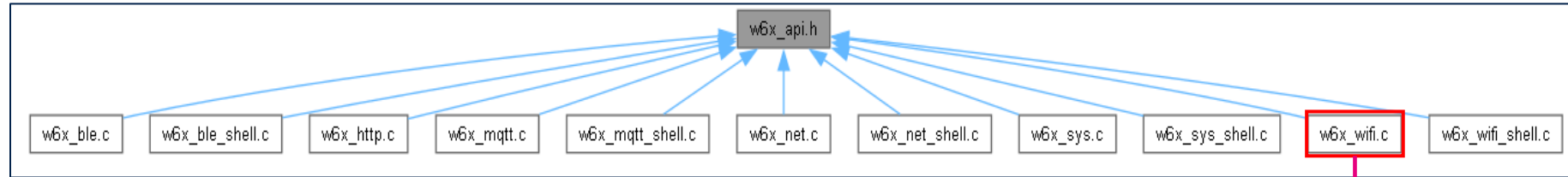


Name
ST67W6X_Network_Driver.chm

Found in \Middlewares\ST\ST67W6X_Network_Driver\Doc

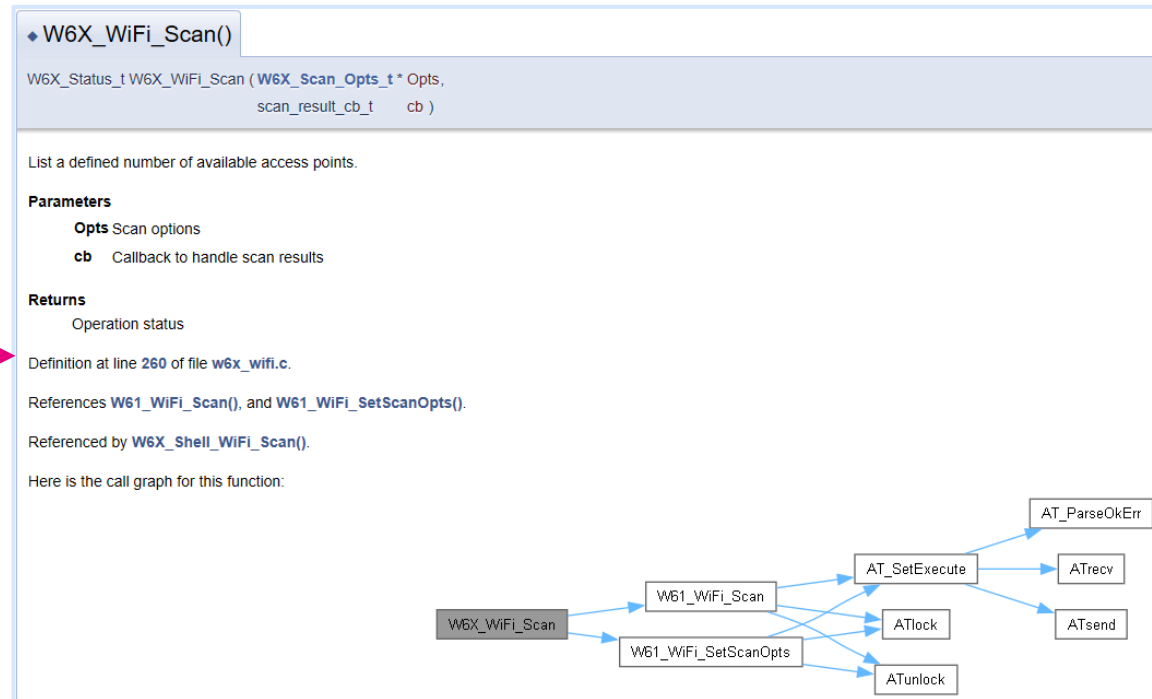
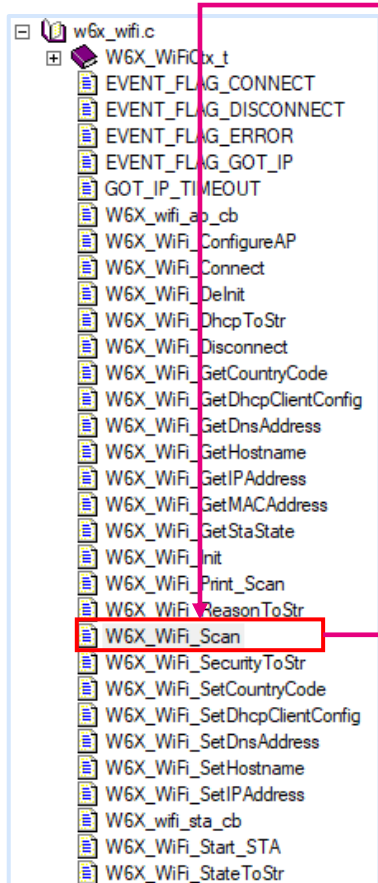
- API Definitions
- Overview of API calls
- Correlated AT Commands

API Documentation cont...



Selecting **w6x_wifi.c** will provide the different API categories

Selecting the API, **W6X_WiFi_Scan()**, will provide information and the call graph



X-CUBE-ST67W61 applications

- Applications
 - ST67W6X
 - ST67W6X_BLE_Commissioning
 - ST67W6X_BLE_p2pClient
 - ST67W6X_BLE_p2pServer
 - ST67W6X_CLI
 - ST67W6X_CLI_LWIP
 - ST67W6X_Echo
 - ST67W6X_FOTA
 - ST67W6X_HTTP_Server
 - ST67W6X_HTTPS_Client
 - ST67W6X_WiFi_Commissioning
- Demonstrations
 - ST67W6X
 - ST67W6X_MQTT

	Scope / description
Bluetooth® LE commissioning	Uses Bluetooth® LE to provide Wi-Fi® SSID and password after scanning available AP
Bluetooth® LE p2p client	Point-to-Point communication demo using Bluetooth® LE (GATT Client) Scan and connect to a p2pServer Write messages & receive notifications from the p2pServer
Bluetooth® LE p2p server	Point-to-Point communication demo using Bluetooth® LE (GATT server) Advertises and wait for a connection from either: - p2pClient app - <i>ST BLE Toolbox</i> smartphone application
Echo	TCP Echo feature over Wi-Fi - Good starting point for generic user development
CLI	ST67 evaluation via CLI and for WTS tests the Wi-Fi solution. - Wi-Fi API: scan, connect, disconnect, etc, NET socket API and iperf, Bluetooth® LE API
CLI_LWIP	Evaluate and to test the X-NUCLEO-67W61M1 Wi-Fi, Bluetooth LE and Network interface with LwIP solutions via command line interface (CLI).
Echo FOTA	FOTA (firmware update over-the-air) feature demo over Wi-Fi - Using HTTP
HTTP server	HTTP server over Network API The server, running on the host board, has been validated with 1 client, using HTTP requests to send and receive data to and from the server.
HTTPS client	HTTPS Client over Network API demo - Basic GET requests to retrieve worldwide weather reports
Wi-Fi® commissioning	Wi-Fi® commissioning project using an HTTP server, hosted by a device configured as a soft access point (SoftAP) and station (STA).
MQTT Demo	Starting point for MQTT user development - Sends IKS sensor data to an MQTT broker

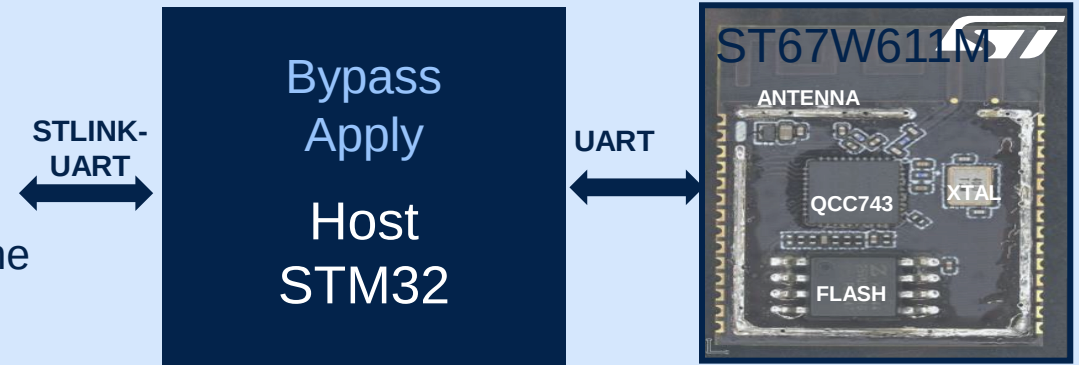


ST67W611M Modes of Operation

Manufacturing and Mission Modes

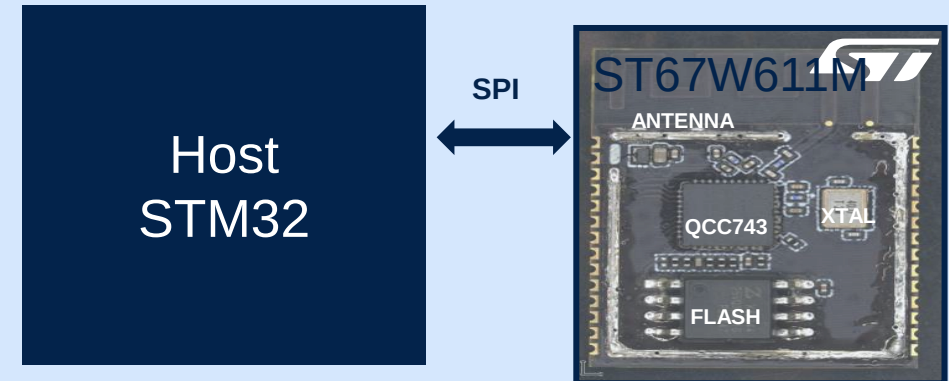
- **Manufacturing mode**

- Host-less Application
- Dedicated to RF performances monitoring and calibration.
- QC provides graphical tools to control the device through the **UART** interface



- **Mission mode (Functional)**

- Implementing the AT command for the different supported protocol (WIFI / BLE / MQTT /..) through the **SPI** interface.
- The ROM boot loader loads and executes the program from SPI Flash to boot the system.

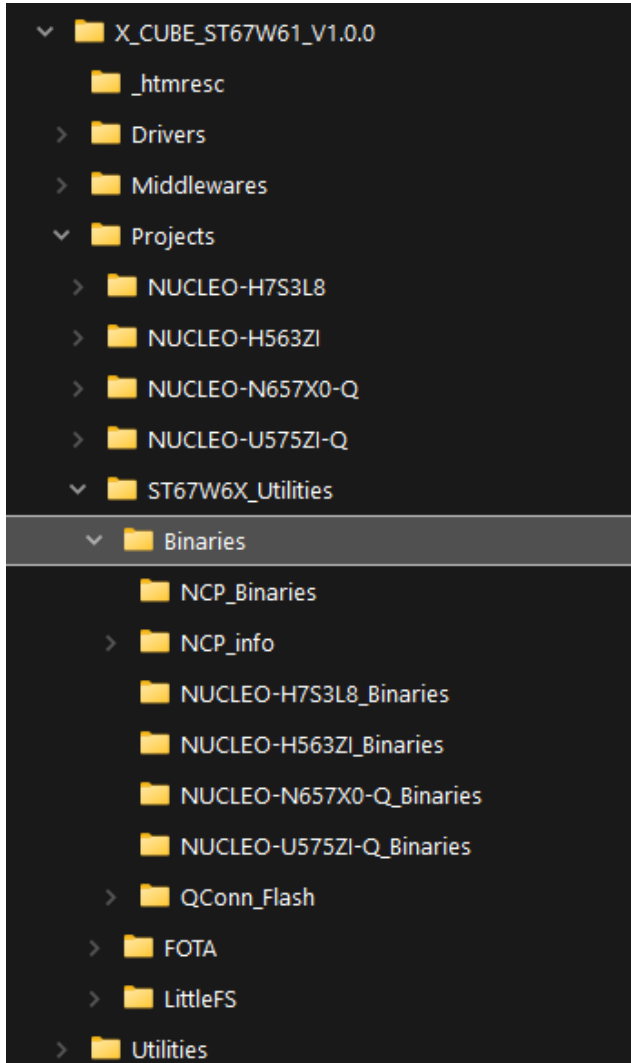


X-CUBE-ST67W1 NCP tools

Scripts for flashing the latest mission or manufacturing firmware:

..\x-cube-st67w61-v1.0.0\Projects\ST67W6X_Uilities\Binaries

Tool	Description
NCP_get_chip_info.bat	Script to retrieve ST67 MAC address and locking information even if no firmware is loaded onto the ST67 device.
NCP_update_mfg.bat	Script to flash the ST67 module with the manufacturing application, which is used for RF evaluation.
NCP_update_mission_profile.bat	Script to flash the ST67 module with the mission profile application, which is used for normal operations and development purposes.



ST67W611M Security Overview

Two modes, each with a header containing a digital signature and versioning info

A. Mission mode:

- **Bootloader binary**
 - Authenticated by the ROM code during initial boot process
 - Versioning info to prevent rollback
- **Mission profile binary**
 - Wi-Fi® & Bluetooth® LE binary used for operational tasks
 - Authenticated by bootloader to ensure integrity & authenticity before execution

B. Manufacturing mode:

- **Bootloader binary**
 - Authenticated by the ROM code during initial boot process
 - Versioning info to prevent rollback
- **MFG firmware binary**
 - Used for test and production configuration
 - Authenticated by bootloader to ensure integrity & authenticity before execution



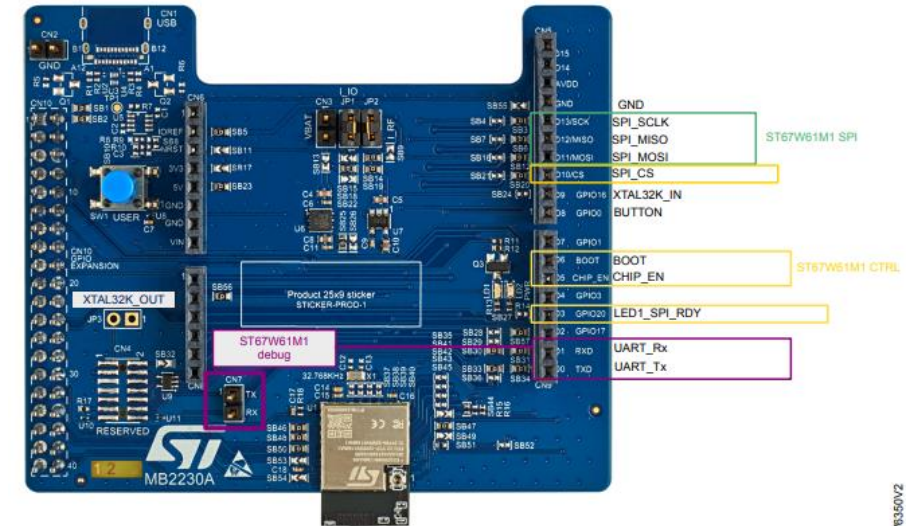


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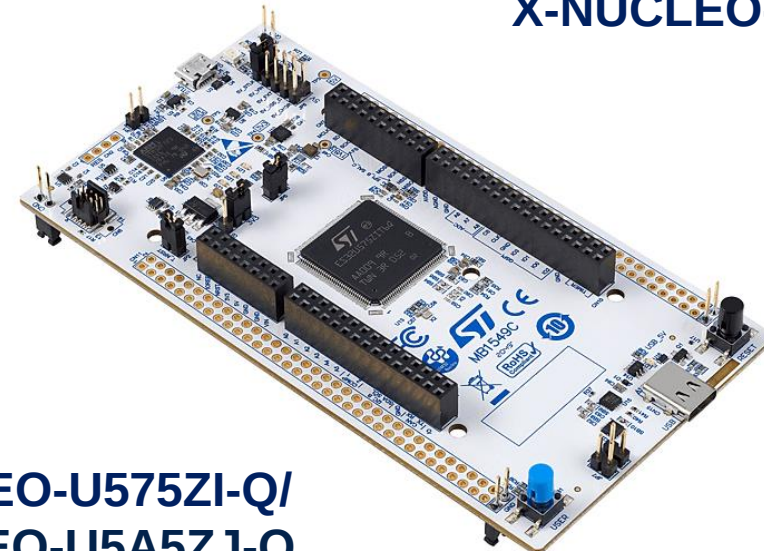
Flashing ST67 & Device Setup

STM32U5 and ST67W611M configuration

- 1x NUCLEO-U575ZI-Q/ NUCLEO-U5A5ZJ-Q board
- 1x X-NUCLEO-67W61M1 board
- Boards connected through Arduino® connectors
 - Key signals: SPI (SCK, MISO, MOSI, CS), SPI_RDY, CHIP_EN, BOOT, UART_TX/RX
- Power: MicroUSB connection to NUCLEO-U575ZI-Q

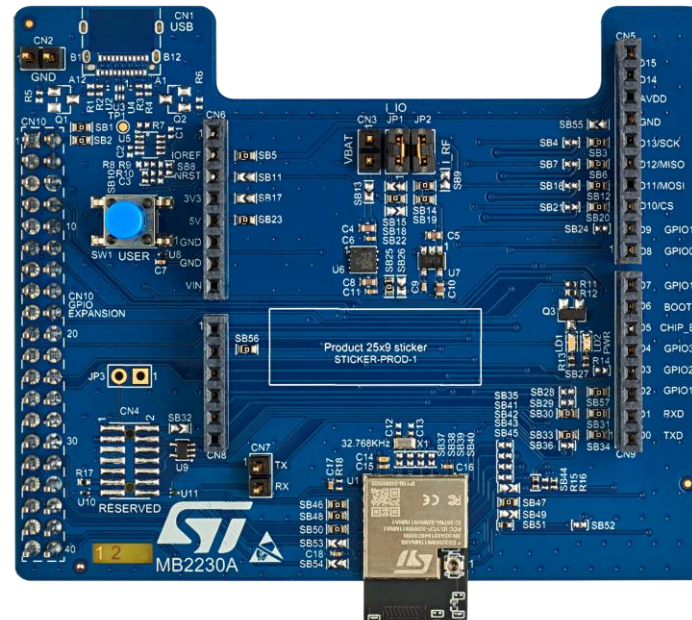


X-NUCLEO-67W61M1



**NUCLEO-U575ZI-Q/
NUCLEO-U5A5ZJ-Q**

Nucleo-U5 and X-Nucleo-67W61M1 Configuration



When stacking the Nucleo and X-Nucleo boards the Arduino connectors must be aligned.

The MicroUSB port is used when programming the STM32U575 or ST67W611M1.

The MicroUSB port is also used to read the UART messages from the STM32U575 in a serial terminal.

Flashing ST67W611M1 on NUCLEO-U575ZI-Q

Mission Mode (Functional Mode) Setup

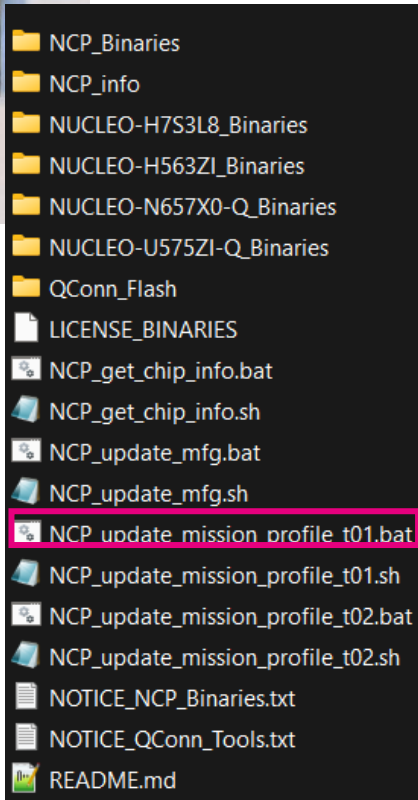
Prerequisites:

- Installed STM32CubeProgrammer
 - Installed in default directory (C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer)
- Installed X-CUBE-ST67W61
 - Installed on C: drive (C:\X_CUBE_ST67W61_V1.1.0)**

Using batch scripts located:

C:\X_CUBE_ST67W61_V1.1.0\Projects\ST67W6X_Uilities\Binaries

1. Nucleo-U575ZI-Q and X-Nucleo-67W61M1 stacked together
2. USB cable connected to microUSB port on Nucleo-U575ZI-Q and Windows PC.
3. Double-click the batch files to execute
 1. NCP_update_mission_profile_t01.bat will flash the ST67W61M with the spi_wifi binary and the host processor with CLI application.



***Note: Path length limits and paths with spaces in them for the x-cube-st67w61-v1.0.0 may cause issues. Saving on C: drive will prevent these issues.*

Flashing ST67W611M1 on NUCLEO-U5A5ZJ-Q

Mission Mode (Functional Mode) Setup

- If you are using **NUCLEO-U5A5ZJ-Q**, use the modified .bat script to flash the ST67 with the mission mode.

[ST67_Experts_Training/resources](#) at main · STMicromDEMO/ST67_Experts_Training

The screenshot shows the GitHub interface for the repository `STMicromDEMO / ST67_Experts_Training`. The left sidebar displays the file tree, with the `resources` folder expanded. The main content area shows the file list for the `resources` folder. The file `NCP_update_mission_profile_t01_U5A5ZJ-Q.bat` is highlighted with a red box, and a red arrow points to it from below. Another red arrow points to the file path `ST67_Experts_Training / resources /` in the breadcrumb navigation.

Name	Last commit message	Last commit date
..		
NCP_update_mission_profile_t01_U5A5ZJ-Q.bat	Add files via upload	now
placeholder	Create placeholder	1 minute ago

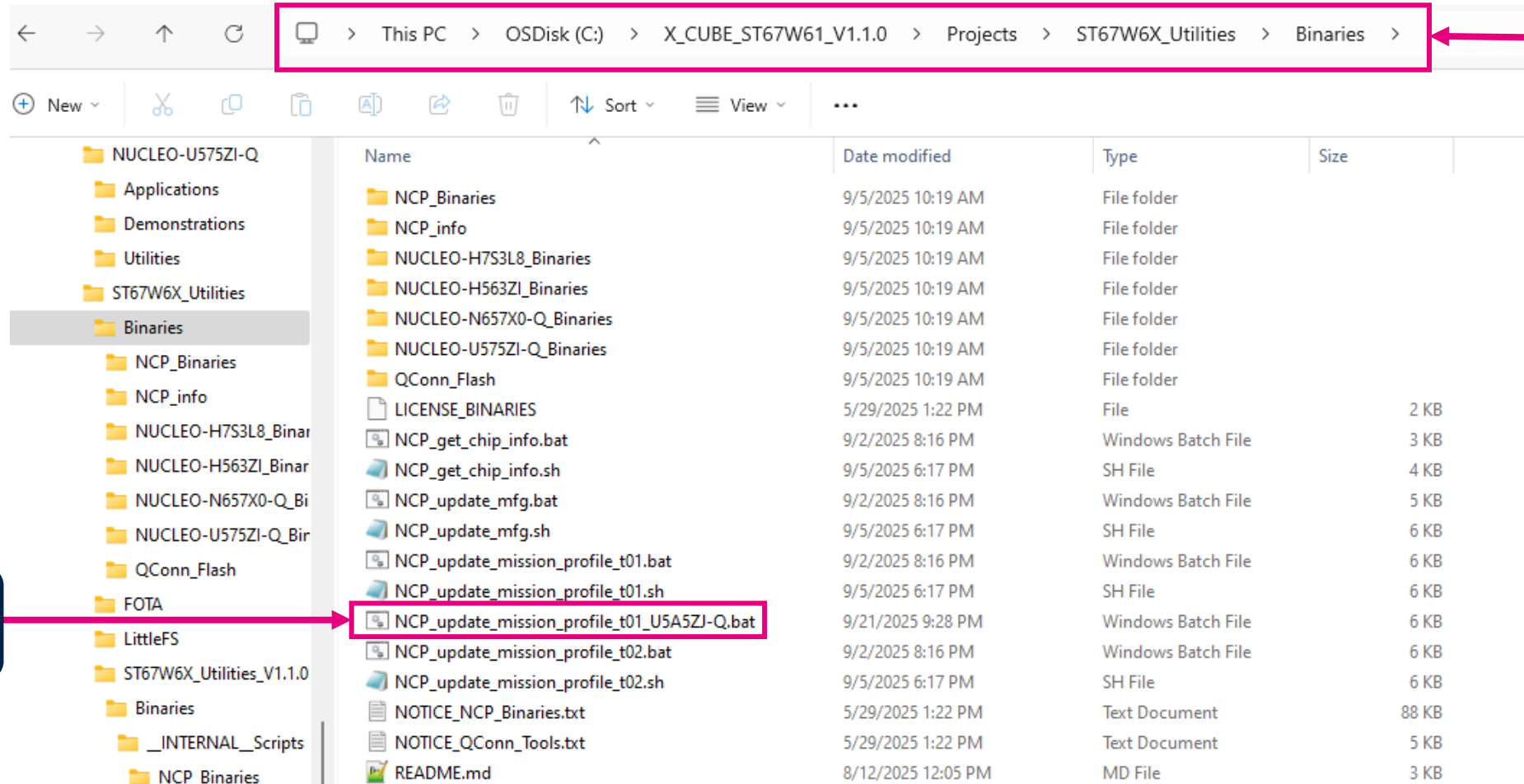


Download: `NCP_update_mission_profile_t01_U5A5ZJ-Q.bat`

Flashing ST67W611M1 on NUCLEO-U5A5ZJ-Q

Mission Mode (Functional Mode) Setup

- If you are using **NUCLEO-U5A5ZJ-Q**, use the modified .bat script to flash the ST67 with the mission mode.



```
#####  
"# You are about to load a signed binary to the NCP."  
"# This will lock the ST67W61M if not yet locked."  
#####  
Are you sure to proceed? (Y/N)  
Press Y to continue or N to abort: |
```



```
Opening and parsing file: ST67W6X_CLI.bin

Memory Programming ...
File      : ST67W6X_CLI.bin
Size     : 186.91 KB
Address  : 0x08000000

Erasing memory corresponding to segment 0:
Erasing internal memory sectors [0 23]
Download in Progress:
 100%

File download complete
Time elapsed during download operation: 00:00:02.186
```

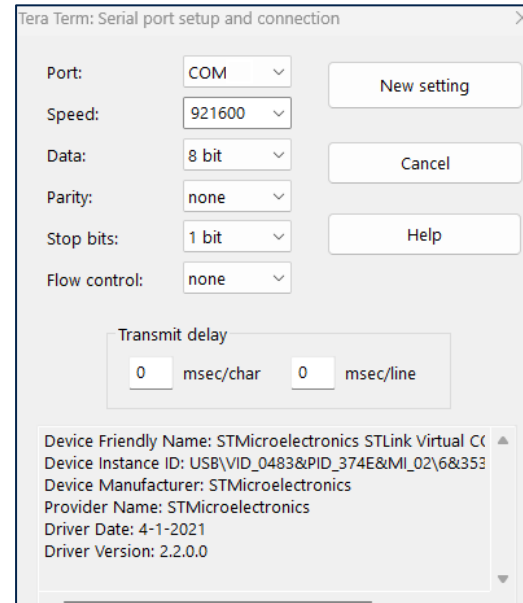
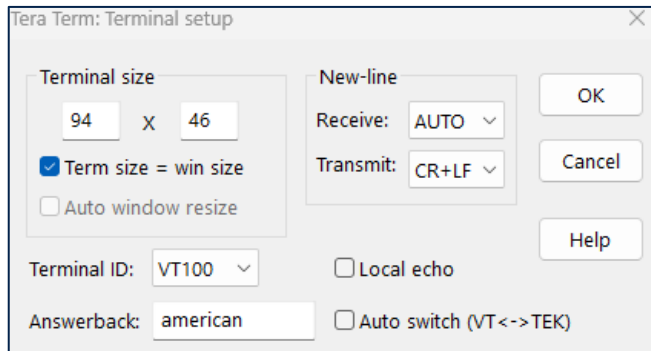
- 

Flashing ST67W611M1

Mission Mode (Functional Mode) Verification

Flashing success can be verified by connecting to the Nucleo-U575ZI-Q with a serial terminal.

Configure the serial terminal with the following parameters and connect to the corresponding COM port for the Nucleo-U575ZI-Q.



With the serial terminal program configured, press the reset button the Nucleo-U575ZI-Q and verify application and SDK version.



****Note:** “Manufacturing Year” date may be incorrect and populated with the year “2000”. This will have no effect on the functionality.

```
w61/>##### Welcome to ST67W6X CLI Application #####
# build: 15:34:46 Sep  5 2025
----- Host info -----
Host FW Version:      1.1.0
----- ST67W6X info -----
ST67W6X MW Version:   1.1.1
AT Version:           1.0.0
SDK Version:          2.0.89
Wi-Fi MAC Version:    1.6.44
BT Controller Version: 1.6.112
BT Stack Version:     1.9.56
Build Date:           Sep  4 2025 11:29:50
Module ID:            Undefined
BOM ID:               0
Manufacturing Year:   2000
Manufacturing Week:   00
Battery Voltage:      3.316 V
Trim Wi-Fi hp:        8,8,8,8,8,7,6,6,7,7,7,8,8,8
Trim Wi-Fi lp:        8,8,8,8,9,9,9,9,10,10,10,11,11,11
Trim BLE:              7,5,5,7,7
Trim XTAL:            37
MAC Address:          40:82:7b:00:0c:2d
Anti-rollback Bootloader: 0
Anti-rollback App:    0
-----
Wi-Fi init is done
Net init is done
MQTT init is done
Starting FOTA task
ready
Application started from bank 1
```




life.augmented

How to use X-CUBE-ST67W61 within STM32CubeMX

Software Development Tools

The image shows two overlapping software windows. The background window is STM32CubeIDE, labeled 'IDE' in a dark blue banner. The foreground window is STM32CubeMX, labeled 'Code Gen' in a light blue banner. A pink banner labeled 'Wi-Fi SW' points to a table of software components in the foreground window.

Component	Version	Selected
STMicroelectronics X-CUBE-ST67W61	1.0.0	
Device Applications	1.0.0	
Application		Not selected
Freertos_Tickless	1.0.0	☐
Network ST67W6X_Network_Driver	1.0.0	
ServiceShell	1.0.0	☐
ServiceAPI	1.0.0	☐
Driver / W61_at_and_bus	1.0.0	☐
> Utilities		
Debug TraceRecorder	4.10.2	
TraceRecorder	4.10.2	☐
Data Exchange cJSON	1.7.18	
cJSON	1.7.18	☐
File System LittleFS	2.10.1	
LittleFS	2.10.1	☐
Utility Utilities	1.4.2	
LPM / Tiny LPM	1.4.2	☐

“STM32CubeIDE”
development environment

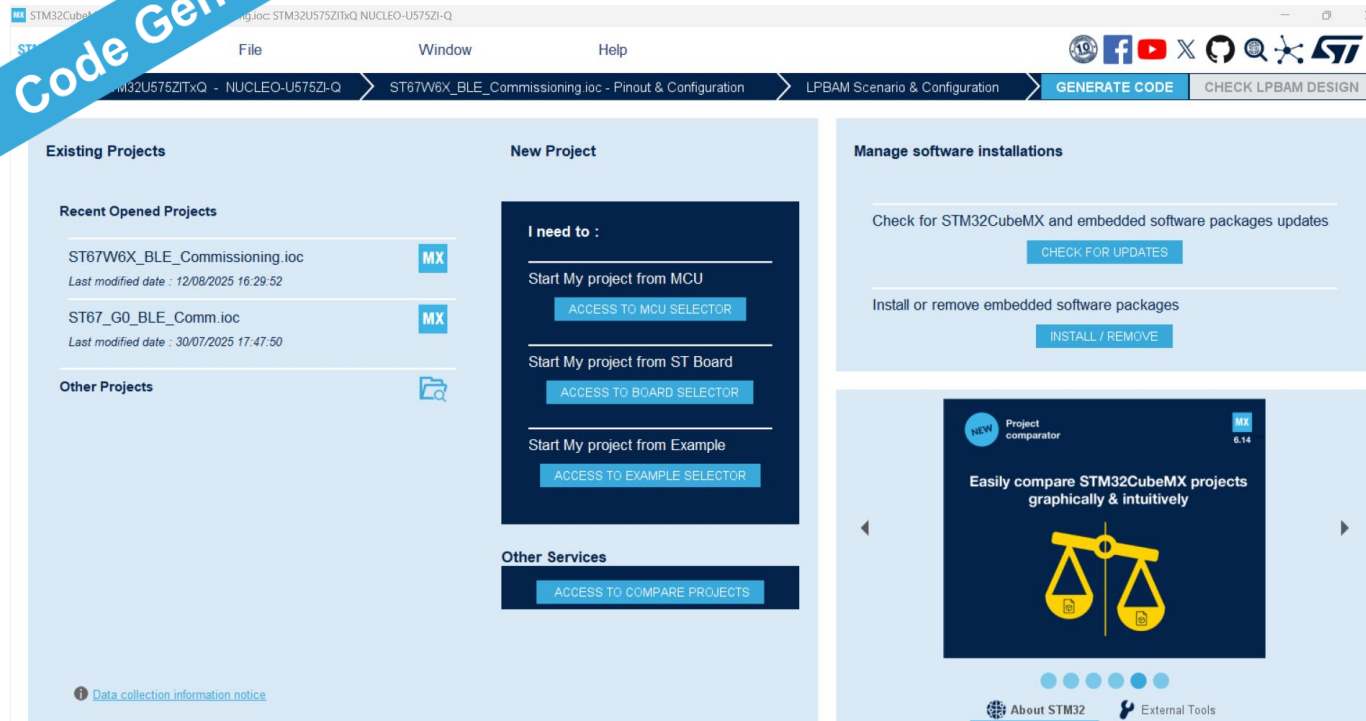
↳ **“STM32CubeMX”**
initialization code generator

↳ **“X-CUBE-ST67W61”**
application code and drivers for Wi-Fi

STM32CubeMX

an All-in-1 Development Tool

Code Gen



Very powerful configuration and code generation

Support all STM32 MCU and MPU, with integrated powerful Finder

Pinouts, clock tree, peripherals and middleware configuration.

Expandable to support wireless connectivity, sensors and much more!



Three possible cases to generate the code from the Cube MX:

- **Case 1:** How to generate a project starting from an existing .ioc
- **Case 2:** How to export a project to a pinout compatible MCU
- **Case 3:** How to generate a project starting from scratch

In this section: We would be having **hands-on** to generate **BLE Commissioning** code using the following:

- Case 1 with NUCLEO-U575ZI-Q as host processor
- Case 3 with NUCLEO-G0B1RE as host processor



BLE Commissioning App

- This application aims to demonstrate **how to provision Wi-Fi credentials via Bluetooth® LE** to establish a Wi-Fi connection to an access point.



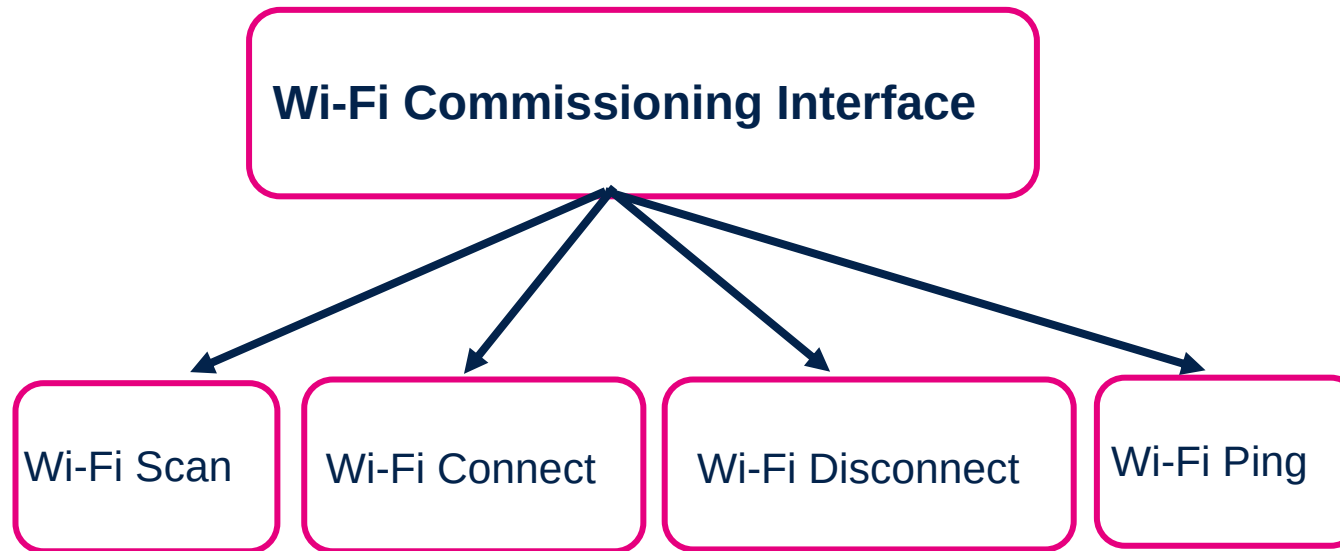
BLE Commissioning App cont..

The application acts as a peripheral device embedding the commissioning profile and its four characteristics.

At startup, the application starts to advertise.

ST Web Bluetooth® Interface must be used to scan and connect to the commissioning application: (Web-Interface available from the browser) / ST BLE Toolbox (Mobile Application)

Once connected, a Wi-Fi commissioning interface is available on web Bluetooth® page/ ST BLE Toolbox



Commissioning profile overview

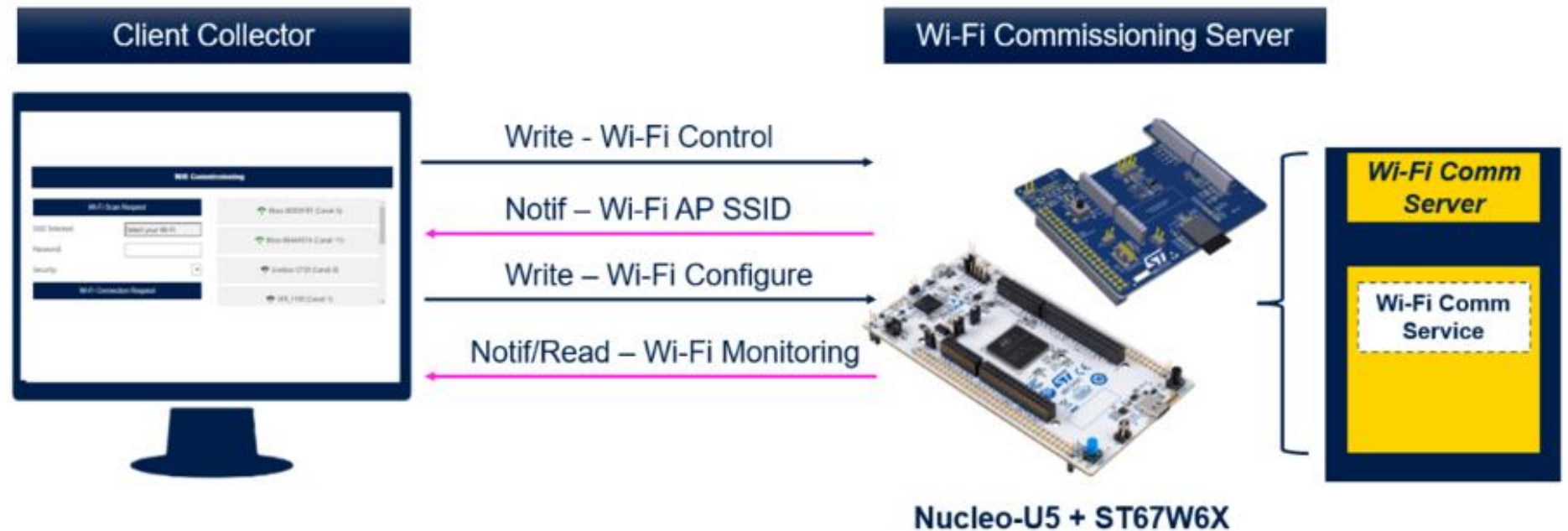
The Wi-Fi commissioning application demonstrates point to point communication using proprietary service and characteristics. It acts as a Peripheral device with one GATT service and four characteristics.

Wi-Fi commissioning server:

- Contains the Wi-Fi commissioning service, which exposes four characteristics: **Wi-Fi Control (write)**, **Wi-Fi Configure (write)**, **Wi-Fi Access Point List (notification)**, **Monitoring (Notification)** to control and monitor a Wi-Fi connection.
- Is the GATT server

Client Collector: (STBLE Toolbox/ ST Web Bluetooth page)

- Accesses the information exposed by the Wi-Fi commissioning application, **configures and controls the Wi-Fi connection** with the write characteristics, **monitors the Wi-Fi connection status** by receiving notifications from it
- Is the GATT client



The table below describes the structure of the commissioning service:

Bluetooth® LE commissioning service specification		
Service	Characteristic	Mode
Wi-Fi commissioning		
	Wi-Fi Control	Write with Response/Read
	Wi-Fi Configure	Write with Response/Read
	Wi-Fi AP List	Notify
	Monitoring	Notify

Wi-Fi commissioning - Wi-Fi control	
Byte Index	0
Name	Action
Value	0x01: Wi-Fi Start Scan 0x03: Wi-Fi Connect 0x04: Wi-Fi Disconnect 0x05: Wi-Fi Ping

Wi-Fi control characteristic:
Used to drive the Wi-Fi by launching scans, connecting or disconnecting from a network



Wi-Fi commissioning - Wi-Fi configure

Byte Index	0	1
Name	Type	Data[]
Value	0x01: AP-SSID 0x02: PWD	"ASCII" "ASCII"

Wi-Fi configure characteristic:

Used to setup the Wi-Fi parameters before establishing a connection

Wi-Fi Commissioning - Wi-Fi AP List

Byte Index	0	1	2 to 3	4 to 7	8 to 40
Name	SSID Length	Channel	Signal level	Security Flag	SSID
Value	0x00 ... 0x20	0x00 ... 0xFF	0x00 ... 0xFFFF	Security_Flag[]	Data[]

Wi-Fi access point list characteristic:

Used to notify information about scanned access points

Wi-Fi commissioning - Wi-Fi monitoring

Byte Index	0	1 up to 239
Name	Type	Data[]
Value	0x03: Connecting 0x04: Connection established 0x05: Ping response 0x06: Error	X SSID Refer to table below 0x01: Connection Timeout

Monitoring characteristic:

Used to notify information about Wi-Fi network.



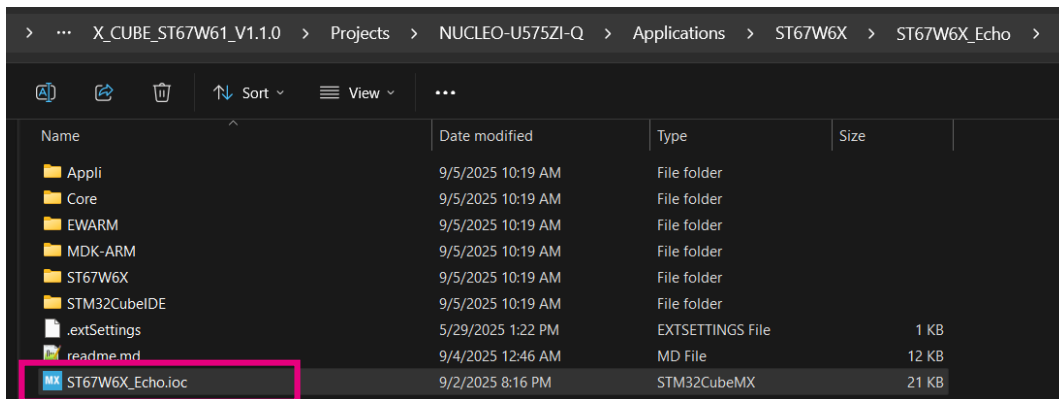
life.augmented

Generating BLE Commissioning
app using NUCLEO-U575ZI-Q as
host processor (Hands-on)

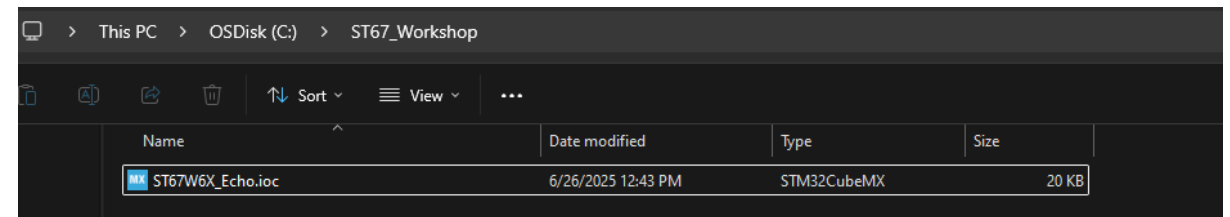
Case 1: Generating BLE Commissioning app using NUCLEO-U575ZI-Q as host processor

We would start with the ST67W6X_Echo.ioc file as a starting point and change the configuration to generate the BLE Commissioning application from the STM32CubeMX.

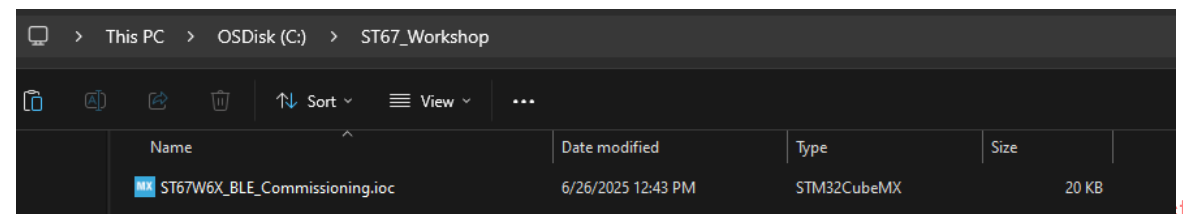
1. Browse through C:\X_CUBE_ST67W61_V1.1.0\Projects\NUCLEO-U575ZI Q\Applications\ST67W6X\ST67W6X_Echo
2. Copy ST67W6X_Echo.ioc
3. Create a new directory C:\ST67_Workshop
4. Paste and rename to ST67W6X_BLE_Commissioning.ioc



Copy
and
Paste



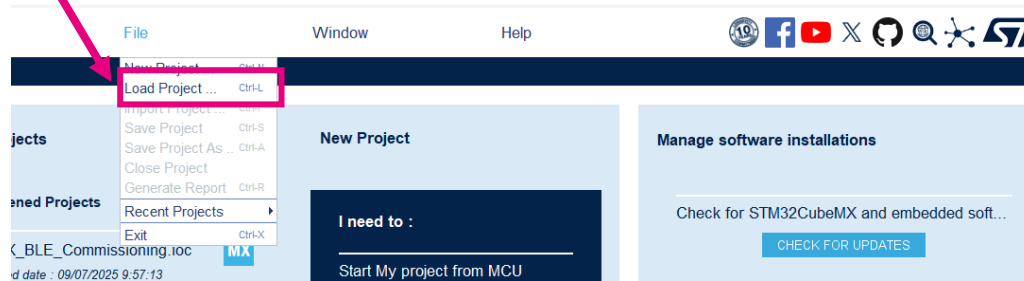
Rename



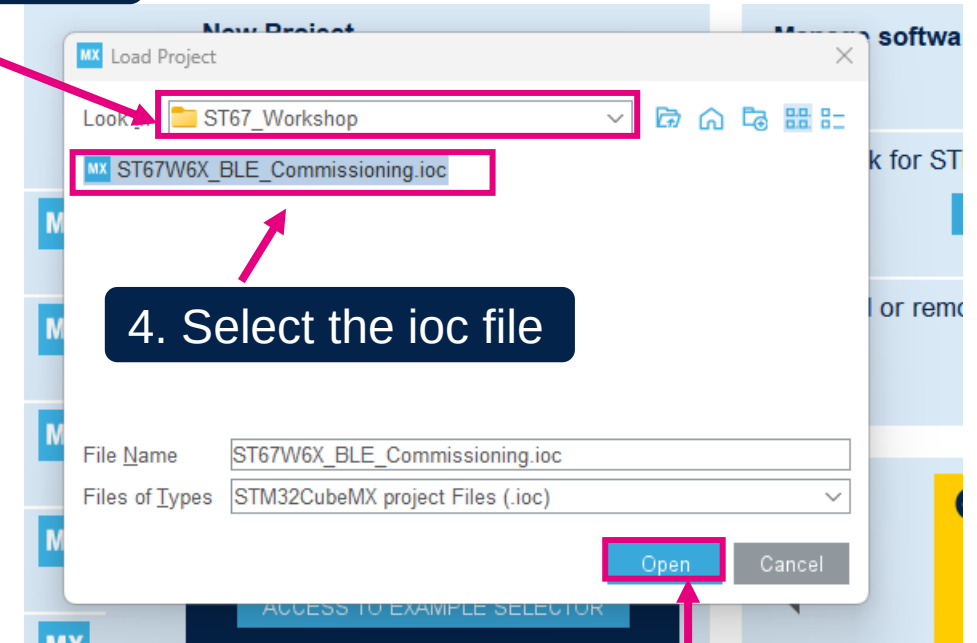
Loading ioc file in STM32CubeMX

1. Run STM32CubeMX with **admin rights**

2. Load Project



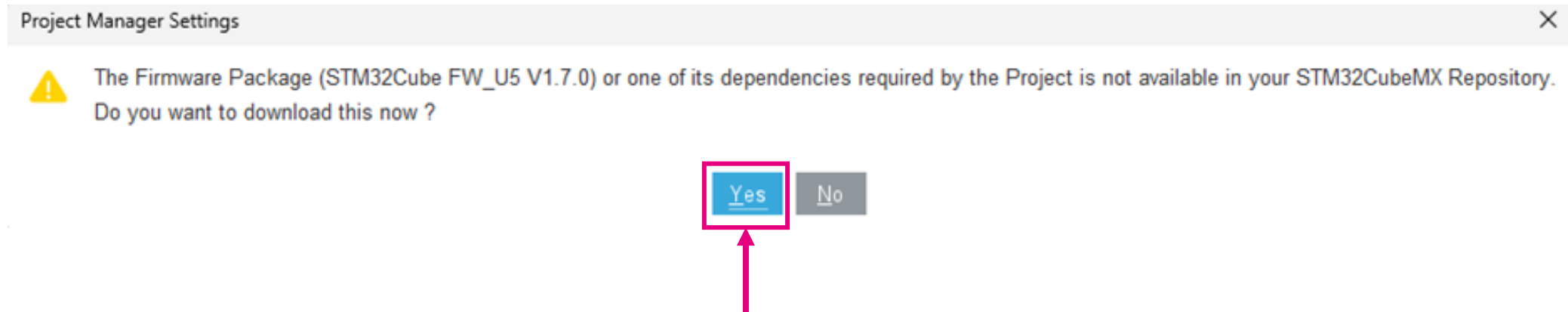
3. Go the ST67_Workshop



4. Select the ioc file

5. Click open

If you get the below warnings



Configuration

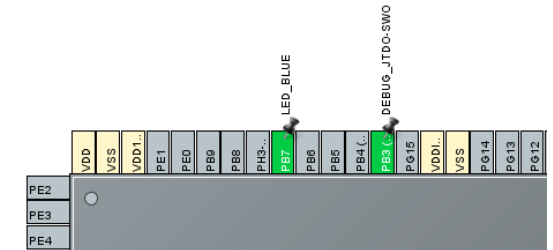
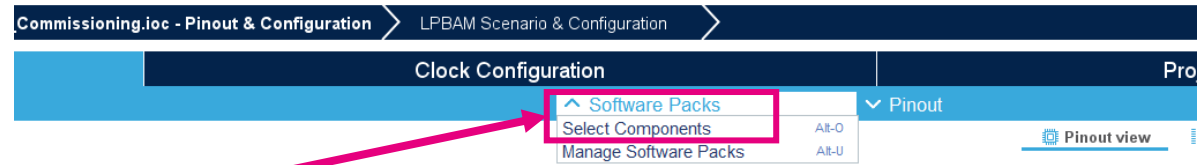
2. Select BLE
Commissioning app

1. Select Components

3. Select T01(NCP) ST67
architecture

MX Software Packs Component Selector

Pack / Bundle / Component	Status	Version	Selection
▼ STMicroelectronics.X-CUBE-ST67W61	✓	1.1.0	
▼ Device Applications	✓	1.1.0	
Application	✓	1.1.0	BLE Commissioning
FreeRTOS_Tickless	✓	1.1.0	☒
▼ Network ST67W6X_Network_Driver	✓	1.1.0	
ST67 Architecture	✓	1.1.0	T01
ATDriver	✓	1.1.0	☒
ServiceAPI	✓	1.1.0	☒
ServiceShell	✓	1.1.0	☐
▼ Utilities	✓		
Logging	✓	1.1.0	☒
Shell	✓	1.1.0	☐
Statistics	✓	1.1.0	☐
▼ Debug TraceRecorder	✓	4.10.2	
TraceRecorder	✓	4.10.2	☐
▼ Data Exchange cJSON	✓	1.7.18	
cJSON	✓	1.7.18	☐
▼ File System LittleFS	✓	2.10.1	
LittleFS	✓	2.10.1	☐
▼ Utility Utilities	✓	1.4.2	
LPM / Tiny LPM	✓	1.4.2	☒
▼ Network LwIP	✓	2.2.0	
LwIP	✓	2.2.0	☐
▼ Data Exchange MQTT-C	✓	1.1.6	
MQTT-C	✓	1.1.6	☐



4. Under Middleware and Software Packs

The screenshot shows the STM32CubeMX configuration interface. On the left, the 'Middleware and Software Packs' list is expanded, showing various packs. The 'X-CUBE-ST67W61' pack is highlighted with a red box. On the right, the 'Configuration' tab is active, showing a list of parameters to be configured. The 'W6X Init Modules' section is expanded, showing parameters like 'ST67_ARCH', 'WIFI_STATION', 'WIFI_SAP', 'BLE', 'NET', and 'MQTT'. The 'Basic Parameters' section is also expanded, showing parameters like 'DEBUGGER_ON', 'LOG_OUTPUT_MODE', 'LOW_POWER_MODE', and 'FREERTOS_TASK_NOTIF_ARRAY_LEN'. The 'Logging-Shell' section is expanded, showing the 'LOG_LEVEL' parameter. The 'W6X Parameters' section is expanded, showing parameters like 'W6X_POWER_SAVE_AUTO', 'W6X_CLOCK_MODE', 'W6X_WIFI_AUTOCONNECT', 'W6X_WIFI_COUNTRY_CODE', 'W6X_WIFI_ADAPTIVE_COUNTRY_CODE', 'W6X_BLE_HOSTNAME', 'W6X_NET_HOSTNAME', 'W6X_NET_DHCP_MODE', 'W6X_NET_RECV_TIMEOUT (ms)', 'W6X_NET_SEND_TIMEOUT (ms)', 'W6X_NET_RECV_BUFFER_SIZE', 'W6X_NET_DNS_MANUAL', 'W6X_NET_DNS_IP_1', 'W6X_NET_DNS_IP_2', and 'W6X_NET_DNS_IP_3'. The 'W61 drv Parameters' section is expanded, showing parameters like 'W61_WIFI_MAX_DETECTED_AP' and 'W61_BLE_MAX_DETECTED_PERIPHERAL'. The 'W6X_POWER_SAVE_AUTO' parameter is set to 'No'. The 'W6X_BLE_HOSTNAME' parameter is set to 'ST67W61_BLE'. The 'W61_WIFI_MAX_DETECTED_AP' parameter is set to '20'. The 'W61_BLE_MAX_DETECTED_PERIPHERAL' parameter is set to '10'.

Configuration

Bluetooth Low Energy connection with NCP in Power save mode requires accurate external low-frequency clock. To setup and use external clock, SW and HW settings must be modified.

Refer to [Wiki Bluetooth LE with low-power setup](https://wiki.st.com/stm32mcu/wiki/Connectivity:How_to_measure_ST67W611M_current_consumption#Bluetooth_h-C2-AE_LE_with_low-power_setup) page to be informed about required changes.

6. Disable W6X_POWER_SAVE_AUTO

7. Rename W6X_BLE_HOSTNAME to something unique

5. X-CUBE-ST67W61



STMicroelectronics

Home > STM32U575ZITxQ - NUCLEO-U575ZI-Q > ST67W6X_BLE_Commissioning.ioc - Project Manager > LPBAM Scenario & Configuration > GENERATE CODE

Pinout & Configuration | Clock Configuration | Project Manager

Project Settings

Project Name: ST67W6X_BLE_Commissioning

Project Location: C:\ST67_Workshop\ Browse

Application Structure: Advanced ☐ Do not generate the main()

Toolchain Folder Location: C:\ST67_Workshop\ST67W6X_BLE_Commissioning\

Toolchain / IDE: STM32CubeIDE ☒ Generate Under Root

Linker Settings

Minimum Heap Size: 0x200

Minimum Stack Size: 0x400

Thread-safe Settings

CortexM33

☐ Enable multi-threaded support

Thread-safe Locking Strategy: Default - Mapping suitable strategy depending on RTOS selection.

Mcu and Firmware Package

Mcu Reference: STM32U575ZITxQ

Firmware Package Name and Version: STM32Cube_FW_U5_V1.8.0

☒ Use Default Firmware Location

Firmware Relative Path: C:/Users/kulkarni/STM32Cube_FW_U5_V1.8.0/STM32Cube_FW_U5_V1.8.0 Browse

1. Enable "Generate Under Root"

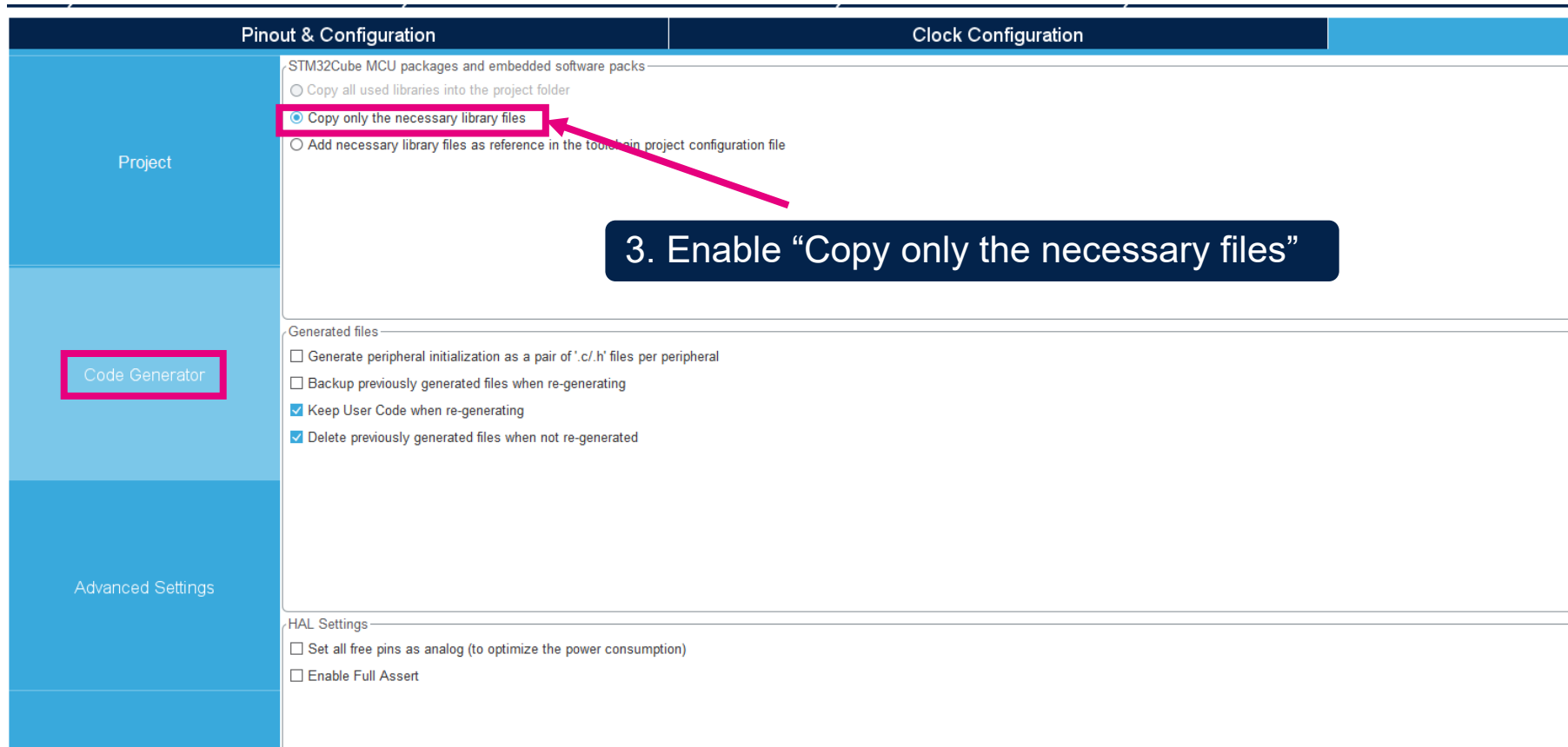
2. Enable "Use Default Firmware Location"

Project Settings

Here, we are enabling the Flat directory structure: (i.e. Driver and MW directories copied into your project directory)



Project Settings cont..



Pinout & Configuration

Clock Configuration

Project

Code Generator

Advanced Settings

STM32Cube MCU packages and embedded software packs

- ☐ Copy all used libraries into the project folder
- ☒ Copy only the necessary library files
- ☐ Add necessary library files as reference in the toolchain project configuration file

3. Enable "Copy only the necessary files"

Generated files

- ☐ Generate peripheral initialization as a pair of '.c/.h' files per peripheral
- ☐ Backup previously generated files when re-generating
- ☒ Keep User Code when re-generating
- ☒ Delete previously generated files when not re-generated

HAL Settings

- ☐ Set all free pins as analog (to optimize the power consumption)
- ☐ Enable Full Assert

Generate Code

Home > STM32U575ZITxQ - NUCLEO-U575ZI-Q > ST67W6X_BLE_Commissioning.ioc - Project Manager > LPBAM Scenario & Configuration > **GENERATE CODE** > CHECK LPE

Pinout & Configuration | Clock Configuration | **Project Manager** | T

Project

Code Generator

Advanced Settings

Driver Selector

Search (Ctrl+F)

- CORTEX_M33_NS
- PWR
- RCC
- GPIO
- > GPDMA
- ICACHE
- > SPI
- > USART
- > LPTIM
- STMicroelectronics.X-CUBE-ST67W61.1.1.0

HAL
HAL
HAL
HAL
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HAL
HAL
HAL
HAL
HAL

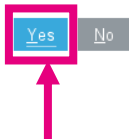
Generated Function Calls

Generate Code	Rank	Function Name	Peripheral Instance Name	☐ Do Not Generate Function Call	☐ Visibility (Static)
☒	1	SystemClock_Config	RCC	☐	☐
☒	2	MX_GPIO_Init	GPIO	☐	☒
☒	3	MX_GPDMA1_Init	GPDMA1	☐	☒
☒	4	MX_ICACHE_Init	ICACHE	☐	☒
☒	5	MX_SPI1_Init	SPI1	☐	☒
☒	6	MX_USART1_UART_Init	USART1	☐	☒
☒	7	MX_LPTIM1_Init	LPTIM1	☐	☒
☒	10	MX_ST67W6X_Init	STMicroelectronics.X-CUBE-ST67...	☐	☐

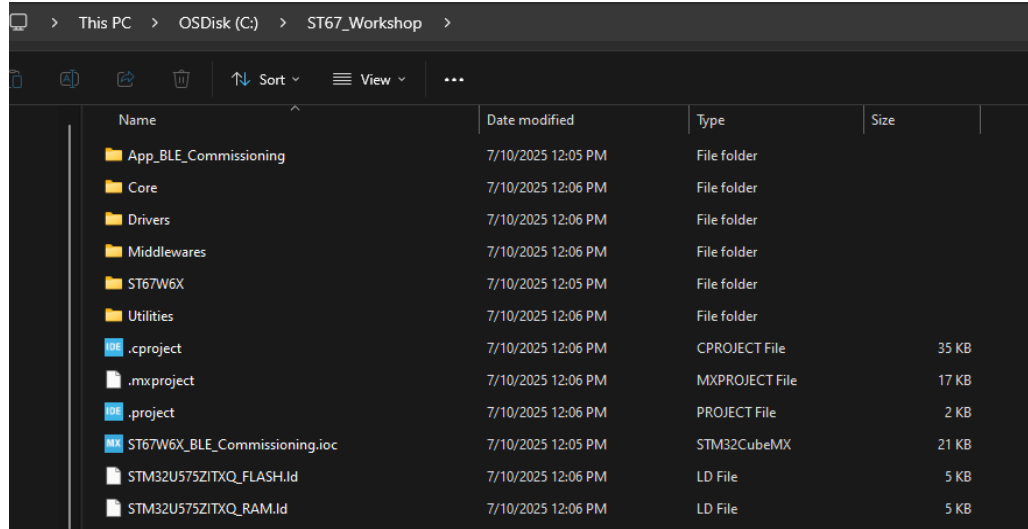
4. Generate Code

Project Manager Settings

⚠ The Firmware Package (STM32Cube FW_U5 V1.7.0) or one of its dependencies required by the Project is not available in your STM32CubeMX Repository.
Do you want to download this now ?



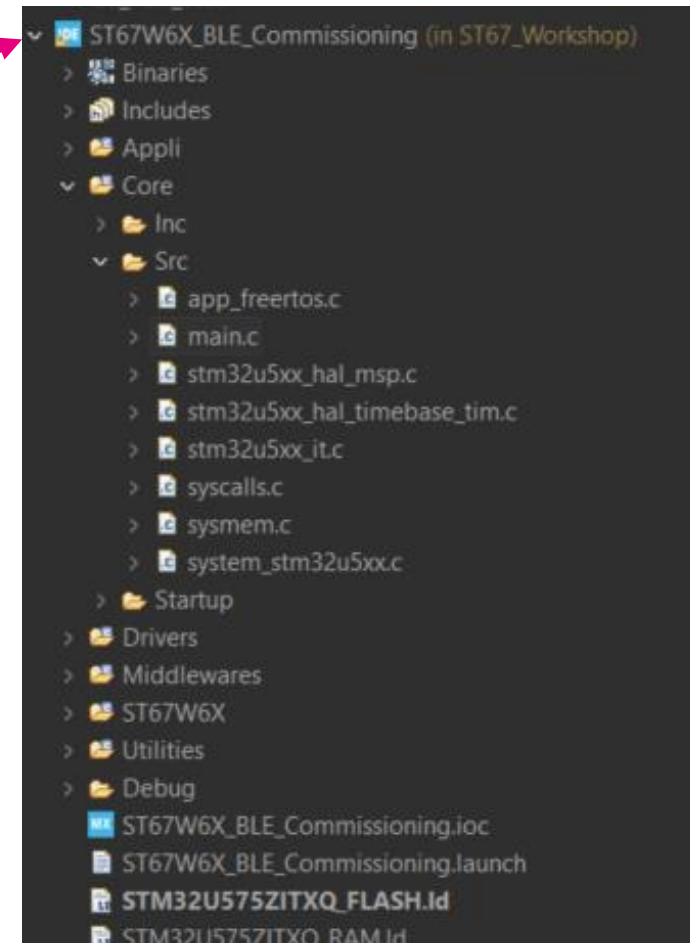
STM32CubeIDE



Name	Date modified	Type	Size
App_BLE_Commissioning	7/10/2025 12:05 PM	File folder	
Core	7/10/2025 12:06 PM	File folder	
Drivers	7/10/2025 12:06 PM	File folder	
Middlewares	7/10/2025 12:06 PM	File folder	
ST67W6X	7/10/2025 12:05 PM	File folder	
Utilities	7/10/2025 12:06 PM	File folder	
.cproject	7/10/2025 12:06 PM	CPROJECT File	35 KB
.mxproject	7/10/2025 12:06 PM	MXPROJECT File	17 KB
.project	7/10/2025 12:06 PM	PROJECT File	2 KB
ST67W6X_BLE_Commissioning.ioc	7/10/2025 12:05 PM	STM32CubeMX	21 KB
STM32U575ZITXQ_FLASH.ld	7/10/2025 12:06 PM	LD File	5 KB
STM32U575ZITXQ_RAM.ld	7/10/2025 12:06 PM	LD File	5 KB

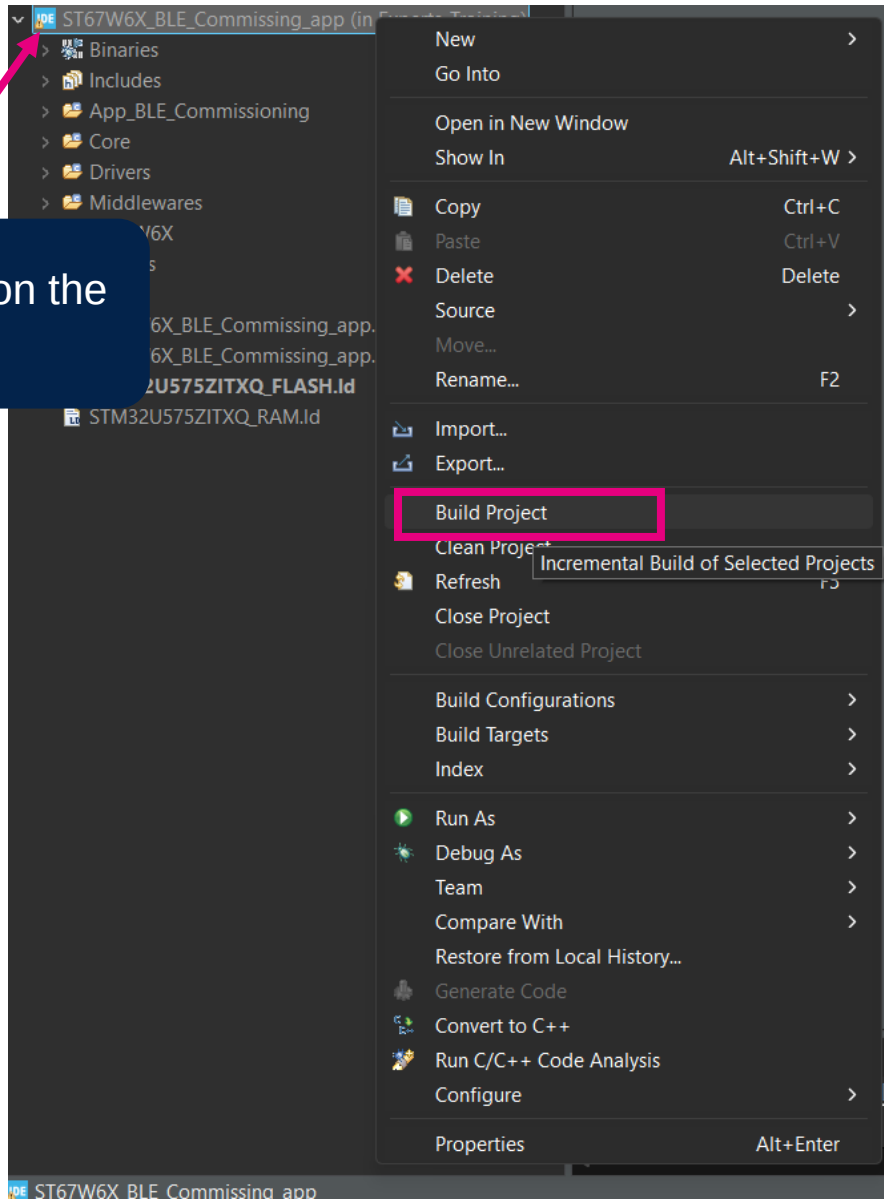
After, generating the code, you would view similar folder structure under ST67_Workshop

Load the project on the STM32CubeIDE



Time to build

Right click on the project



```

CDT Build Console [ST67W6X_BLE_Commissioning]
arm-none-eabi-gcc -mcpu=cortex-m33 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -c -I../Core/1
arm-none-eabi-gcc -I../Core/Src/app_freertos.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTI
arm-none-eabi-gcc -I../Core/Src/main.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION
arm-none-eabi-gcc -I../Core/Src/stm32u5xx_hal_msp.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASI
arm-none-eabi-gcc -I../Core/Src/stm32u5xx_hal_timebase_tim.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -Dc
arm-none-eabi-gcc -I../Core/Src/stm32u5xx_it.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTI
arm-none-eabi-gcc -I../Core/Src/syscalls.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFIC
arm-none-eabi-gcc -I../Core/Src/system.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICAT
arm-none-eabi-gcc -I../Core/Src/system_stm32u5xx.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK
arm-none-eabi-gcc -I../App_BLE_Commissioning/Target/freertos_tickless.c -mcpu=cortex-m33 -std=gnu11 -g3
arm-none-eabi-gcc -I../App_BLE_Commissioning/Target/logshell_ctrl.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDI
arm-none-eabi-gcc -I../App_BLE_Commissioning/Target/stm32_lpm_if.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEI
arm-none-eabi-gcc -I../App_BLE_Commissioning/App/main_app.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -Dc
arm-none-eabi-gcc -o "ST67W6X_BLE_Commissioning.elf" @"objects.list" -mcpu=cortex-m33 -T"C:\ST67_Work
Finished building target: ST67W6X_BLE_Commissioning.elf

arm-none-eabi-size ST67W6X_BLE_Commissioning.elf
arm-none-eabi-objdump -h -S ST67W6X_BLE_Commissioning.elf > "ST67W6X_BLE_Commissioning.list"
   text    data    bss     dec     hex filename
143952    328    208016    352296    56028 ST67W6X_BLE_Commissioning.elf
Finished building: default.size.stdout

Finished building: ST67W6X_BLE_Commissioning.list

08:12:05 Build Finished. 0 errors, 0 warnings. (took 14s.260ms)

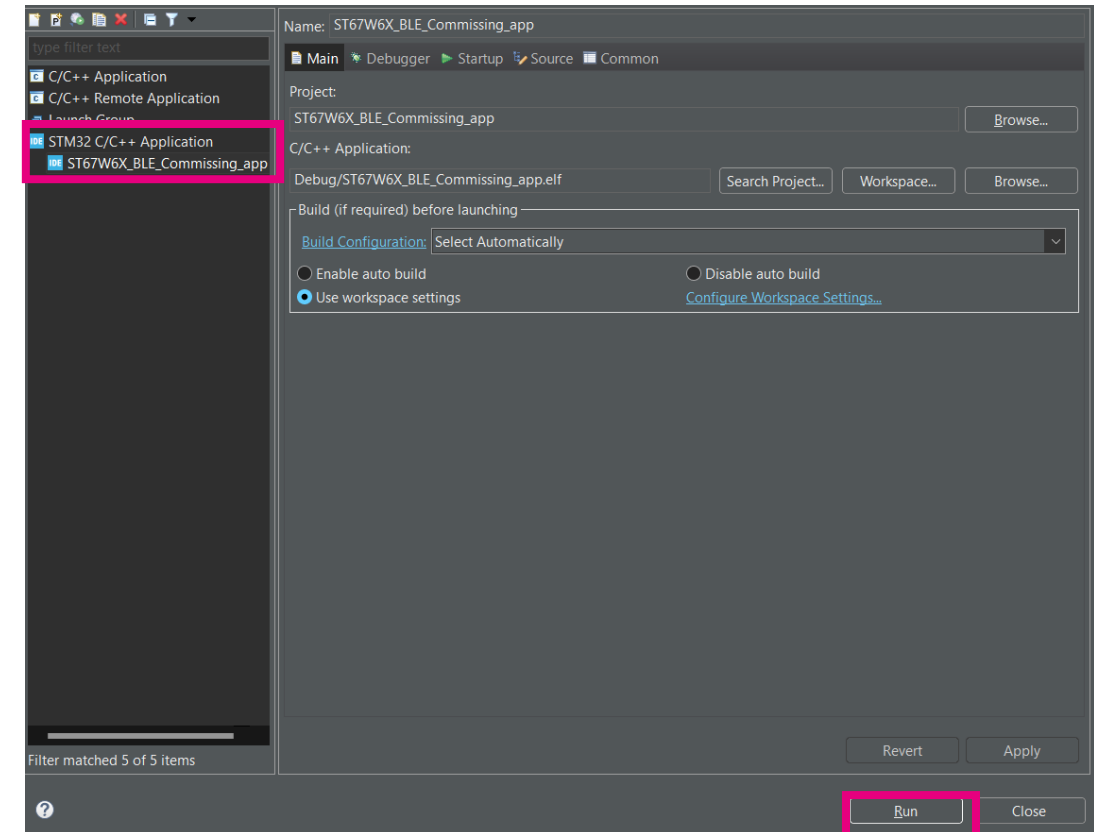
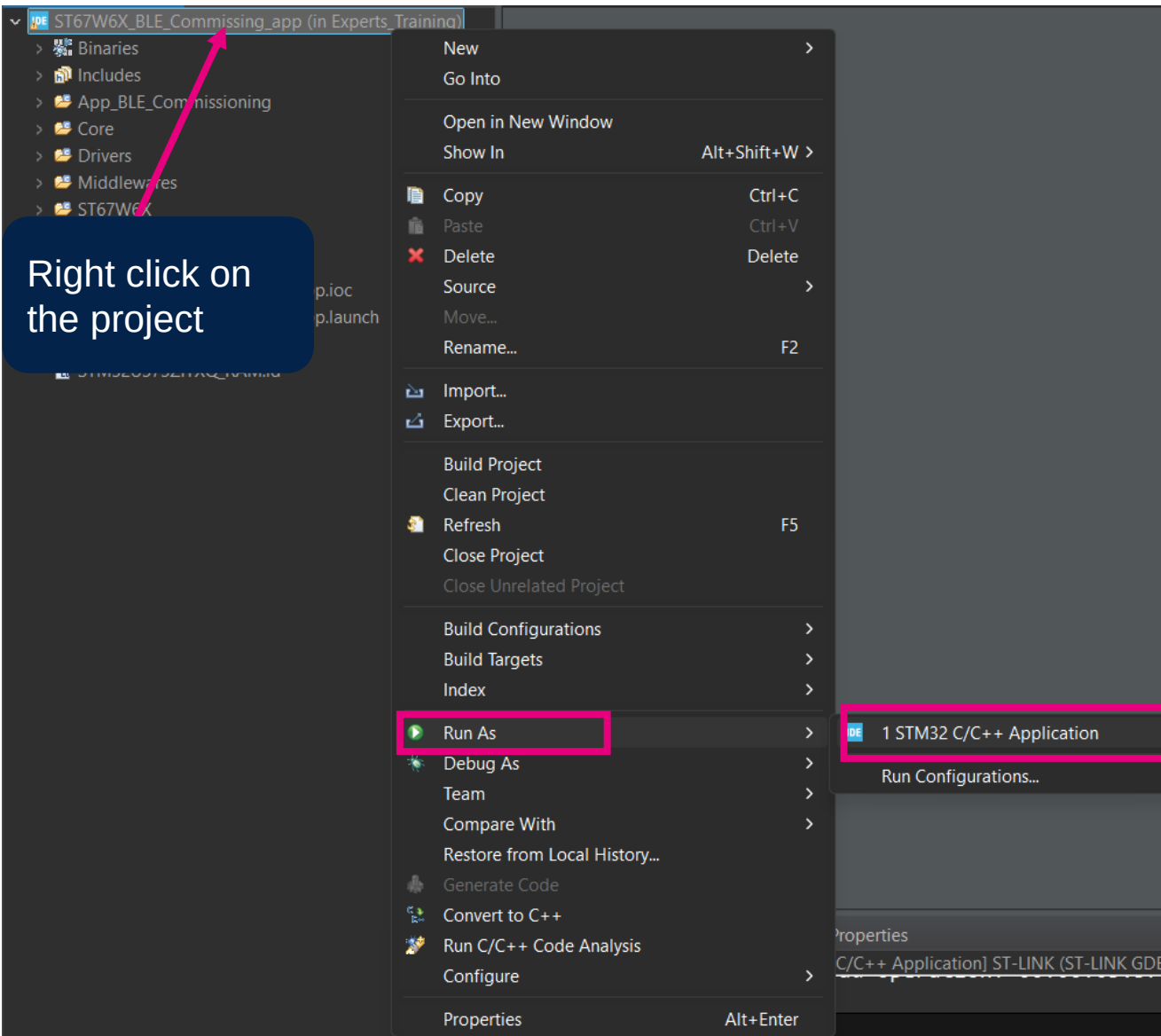
```

Console after building the application

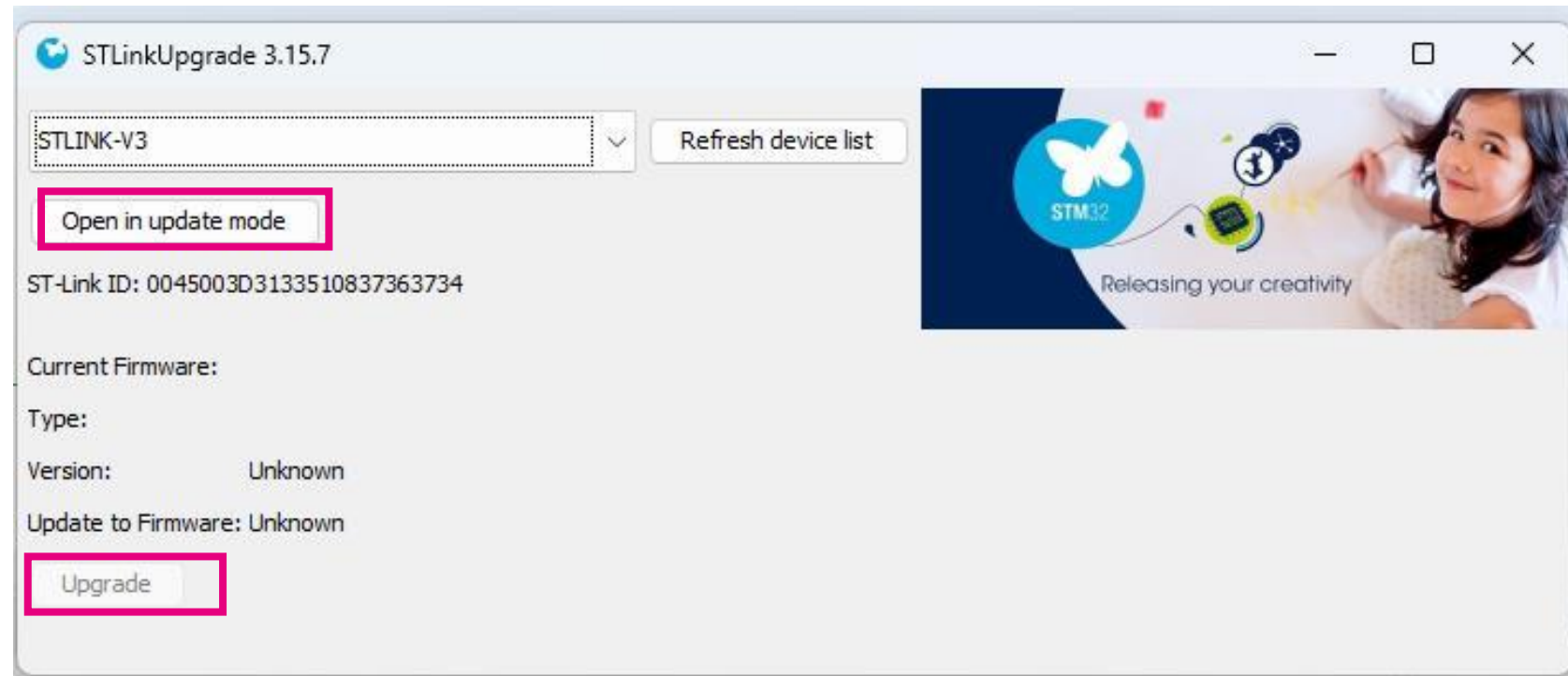
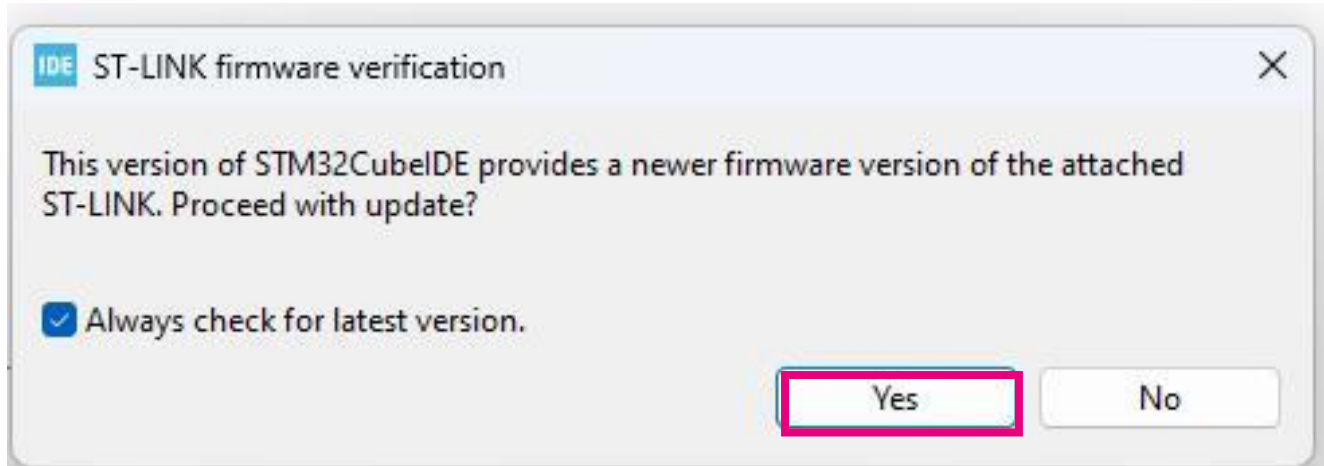


Flash the project

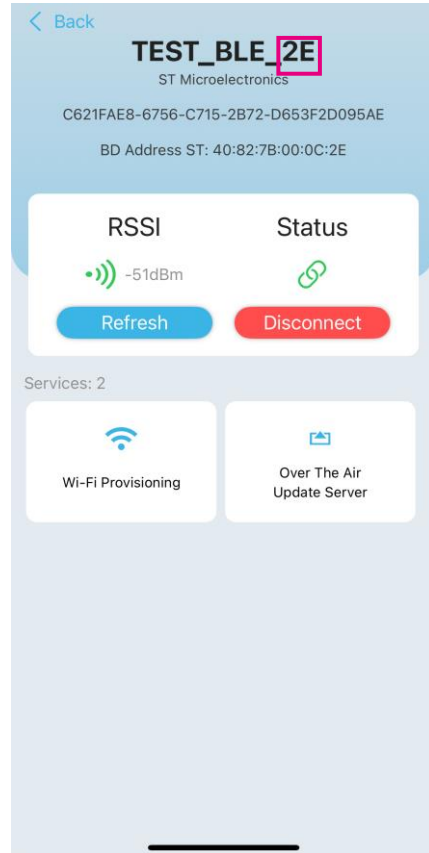
Right click on
the project



ST-LINK Update Warning



Tera Term Logs



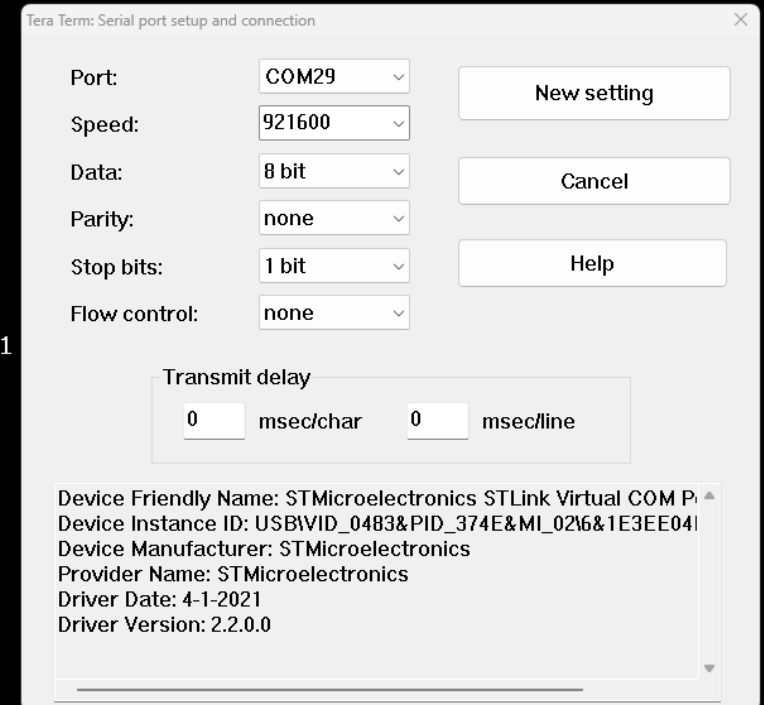
Last 1 byte of BD_Address would match last byte advertising on the ST_BLE_Toolbox

```
COM29 - Tera Term VT
File Edit Setup Control Window Help

##### Welcome to ST67W6X Wi-Fi Commissioning over BLE Application #####
# build: 19:57:12 Sep 7 2025
----- Host info -----
Host FW Version: 1.1.0
----- ST67W6X info -----
ST67W6X MW Version: 1.1.1
AT Version: 1.0.0
SDK Version: 2.0.89
Wi-Fi MAC Version: 1.6.44
BT Controller Version: 1.6.112
BT Stack Version: 1.9.56
Build Date: Sep 4 2025 11:29:50
Module ID: Undefined
BOM ID: 0
Manufacturing Year: 2000
Manufacturing Week: 00
Battery Voltage: 3.324 V
Trim Wi-Fi hp: 8,8,8,8,8,7,6,6,7,7,7,8,8,8
Trim Wi-Fi lp: 8,8,8,8,9,9,9,10,10,10,11,11,11
Trim BLE: 7,5,5,7,7
Trim XTAL: 37
MAC Address: 40:82:7b:00:0c:2d
Anti-rollback Bootloader: 0
Anti-rollback App: 0
-----
Wi-Fi init is done
Net init is done
Ble init is done
BD Address: 40:82:7b:00:0c:2e
Configure BLE
BLE configuration is done

BLE Commissioning Service Creation
- BLE service created
- BLE WIFI Control charac created
- BLE WIFI Configure charac created
- BLE WIFI Monitoring charac created
- BLE FUOTA base address charac created
- BLE FUOTA confirmation charac created
- BLE FUOTA raw data charac created
- BLE services and charac registered
- BLE service and charac creation is done

Start BLE advertising
BLE advertising is started
```



Tera Term logs after flashing the BLE Commissioning application

ST BLEToolbox/ ST BLE Webpage

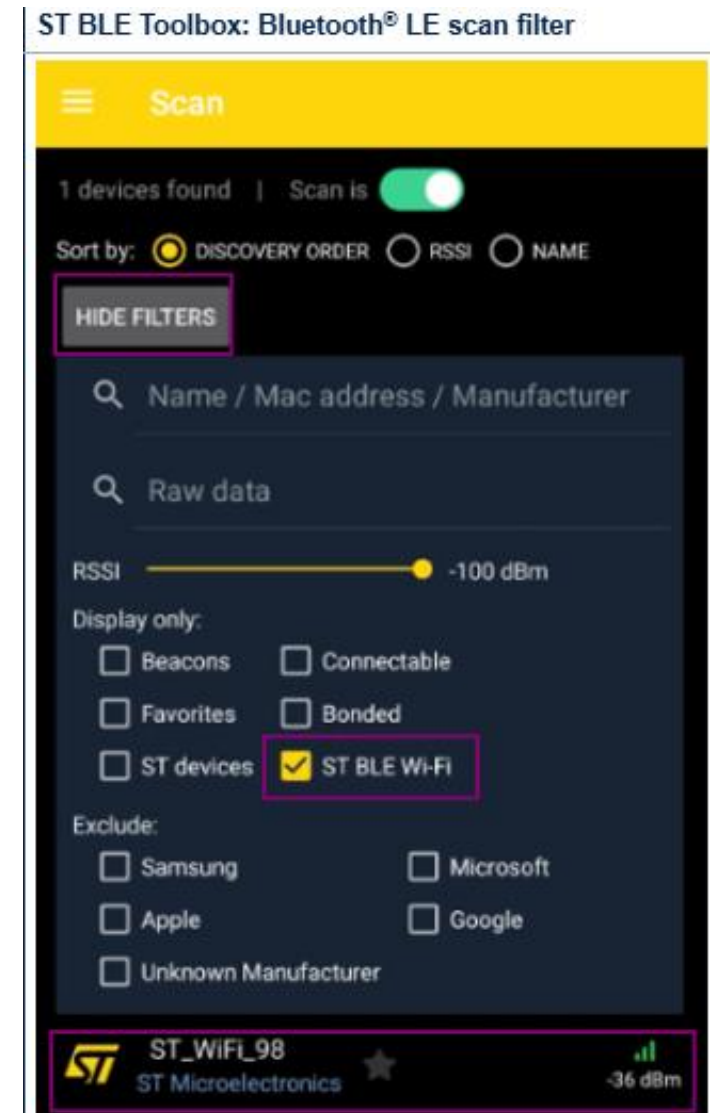
To interface with the Wi-Fi commissioning embedded application, you can use [STBLEToolbox Andoid/iOS application](#) / [ST Web Bluetooth® Interface](#)

Download [STBLEToolBox](#)

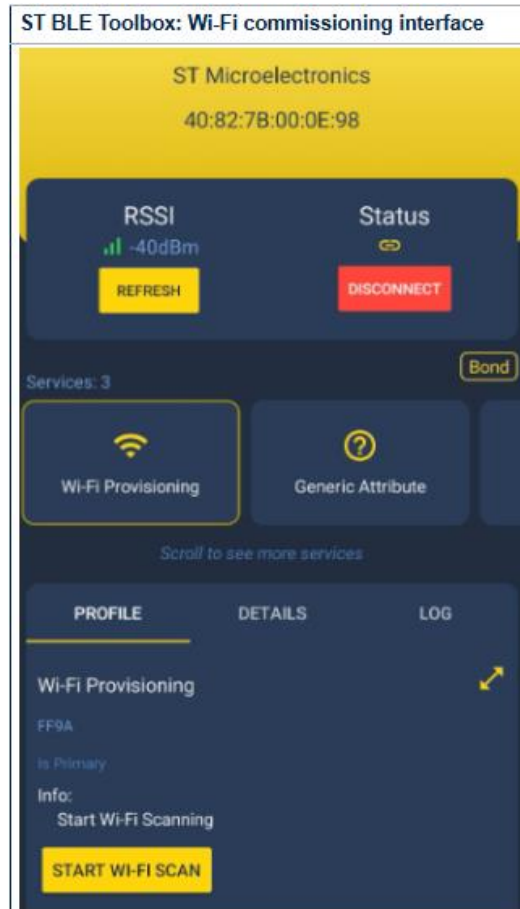
The versions supporting Wi-Fi commissioning service are **v1.4.3 for Android** and **1.2.8 for iOS**

Steps to launch and connect to Wi-Fi AP:

1. Launch the smartphone application and enable the scan button, if it is not already activated.
2. You can filter scanned devices by clicking on the SHOW FILTERS button and select ST BLE Wi-Fi to show only devices with Wi-Fi commissioning service.
3. Select your device, named ST_WiFi_XX where XX represents the last BD address digit, and connect to it.

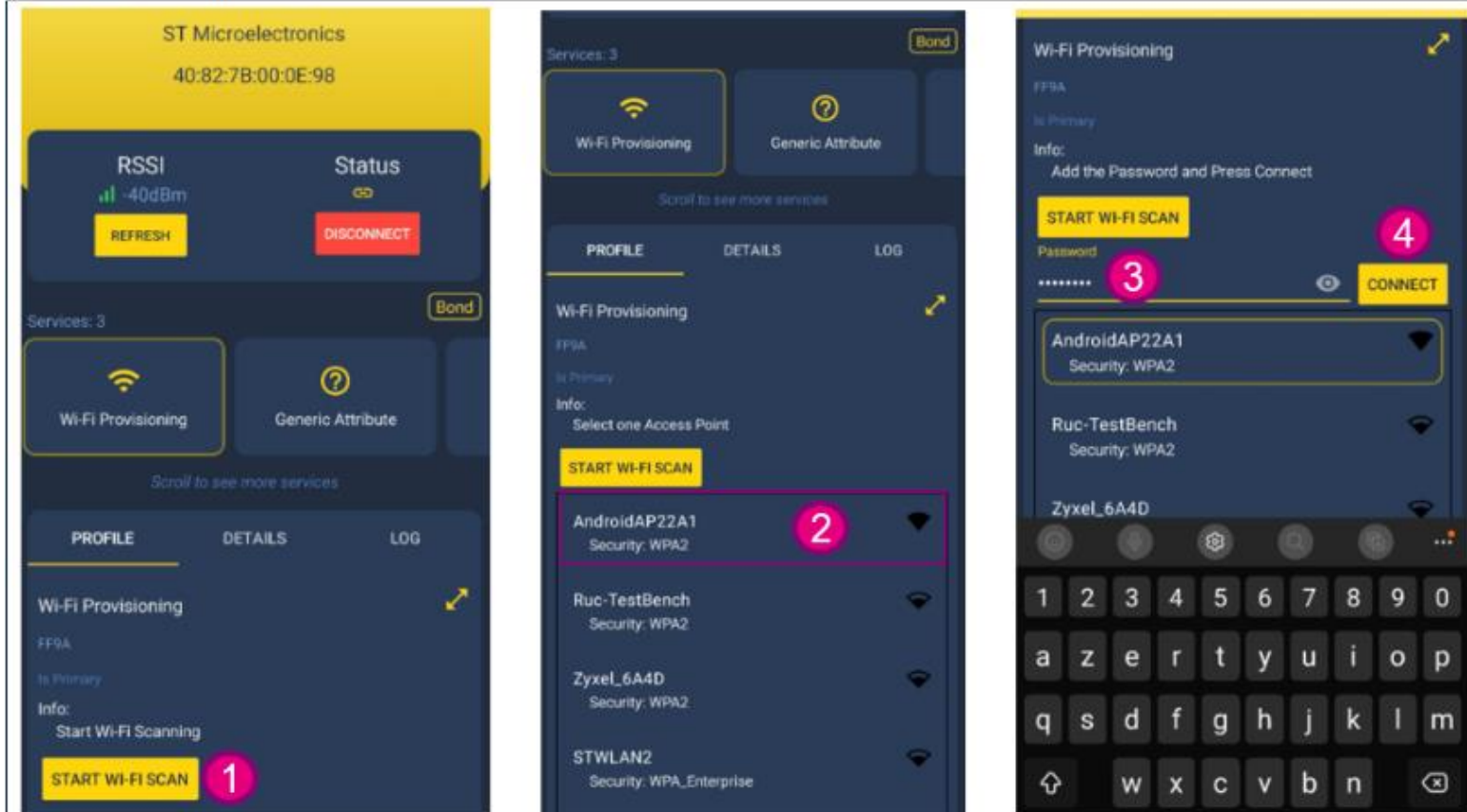


4. Once connected to the Bluetooth® LE device, click on Wi-Fi commissioning button to open the commissioning dedicated interface.

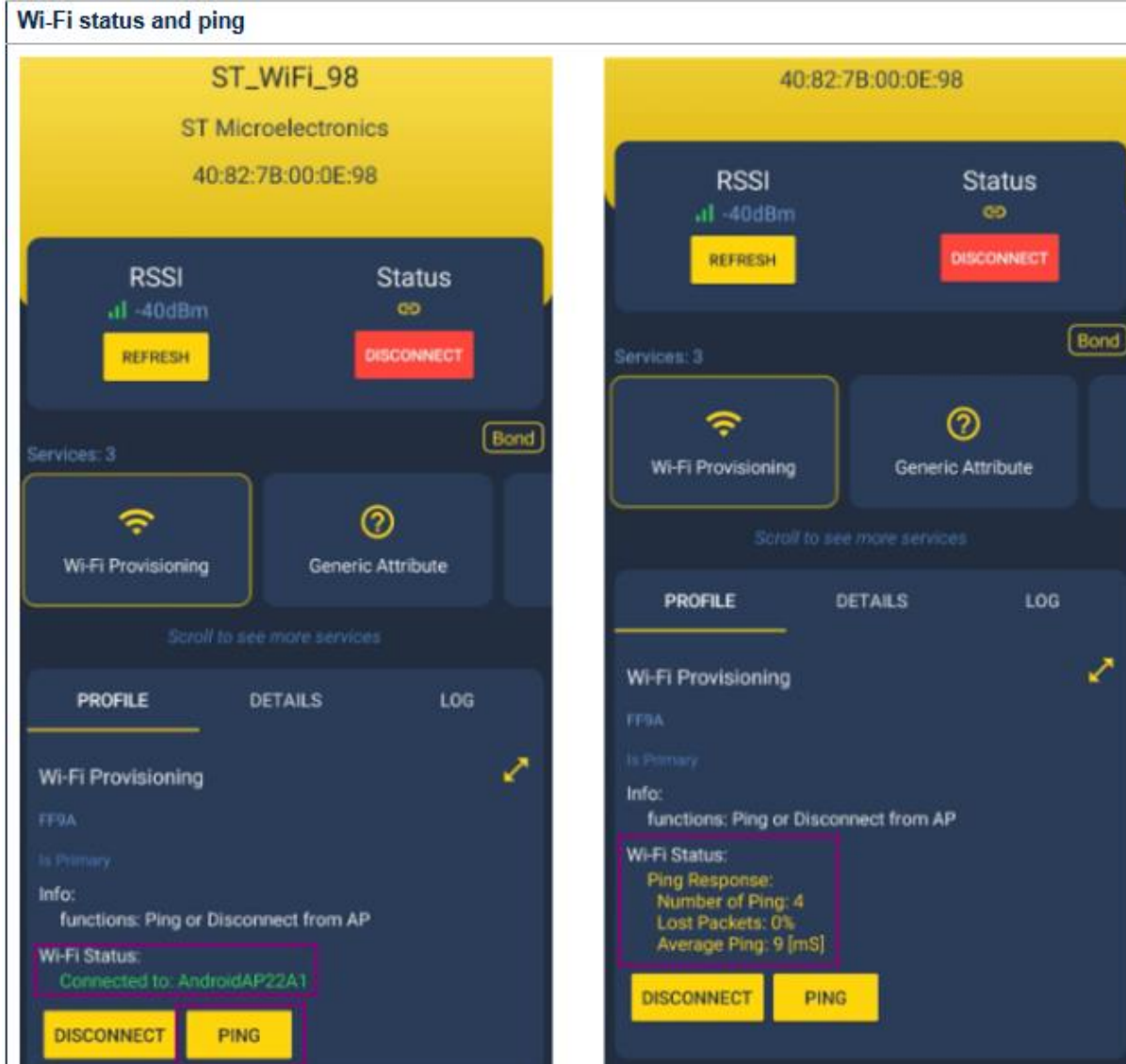


From this interface,
Launch a Wi-Fi scan (1),
To detect and list all available networks(2).
Once the available networks are displayed, select the one you want to connect to (2)
Type a password if needed (3)
Click on CONNECT button (4).

Wi-Fi connection interface

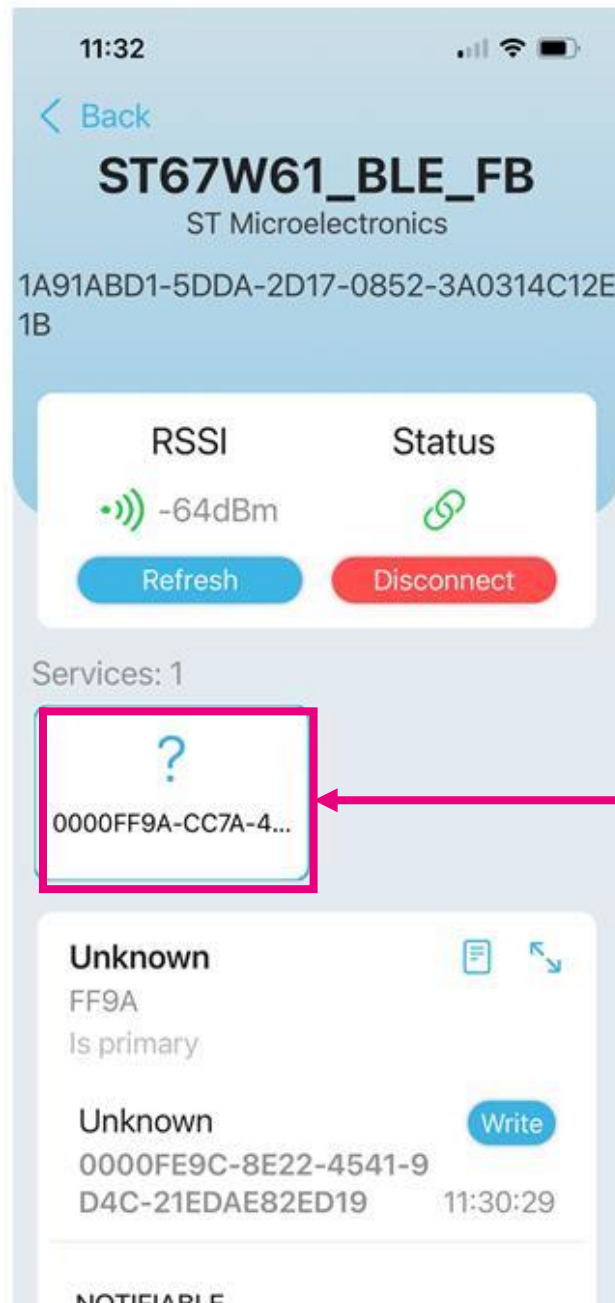


Clicking on the Ping button start a ping test with the Wi-Fi access point and returns results in the interface.

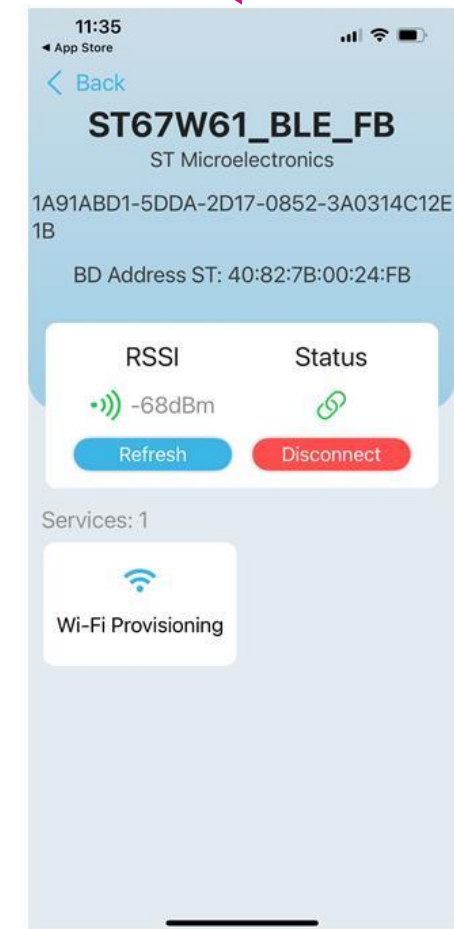
[illegible]

ST BLE Toolbox

Solution: Uninstall the app and re-install the app back



Cannot see the Wi-Fi Provisioning button





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Porting to a new host mcu using STM32CubeMX

Porting over new host mcu(1/3)

1. Select the Host MCU

Choose the target STM32 part number from the STM32CubeMX

2. Check schematics of Host MCU and ST67

Review pin mappings for SPI, GPIO, USART

3. Configure USART for Debugging

Used for debugging logs via STLink virtual port

4. Configure the SPI Interface for Host MCU and ST67 communication

Set SPI pins, SPI mode, SPI Priority

5. Configure DMA Channels for SPI

Fast and reliable SPI data transfer, reducing CPU load during transmission



Porting over new host mcu(2/3)

6. **Configure GPIO Pins** (CHIP_EN, SPI_RDY, SPI_CS, BOOT, User_Button)

Set GPIO Mode, GPIO output level, GPIO Maximum Output speed

7. **Configure SYS** (Time base Source)

8. **Configure Clock**

9. **Enable FreeRTOS Middleware**

Setup default task, heap size, stack size and required priorities

10. **Enable X-CUBE-ST67W61**

Setup Application, W61 Drivers, Utilities

11. **Configure NVIC Interrupt Settings**

SPI, DMA, USART, EXTI



Porting over new host mcu(3/3)

12. Project Creation and Code Generation

Generate project files (STM32CubeIDE/IAR)

Open in IDE for application development

13. Application Entry: Call main_app()

Inside StartDefaultTask, invoke main_app()

This initializes ST67W6X Driver, ST67W6X Wi-Fi module, ST67W6X Network module, Application specific functions

14. Build and Flash the application



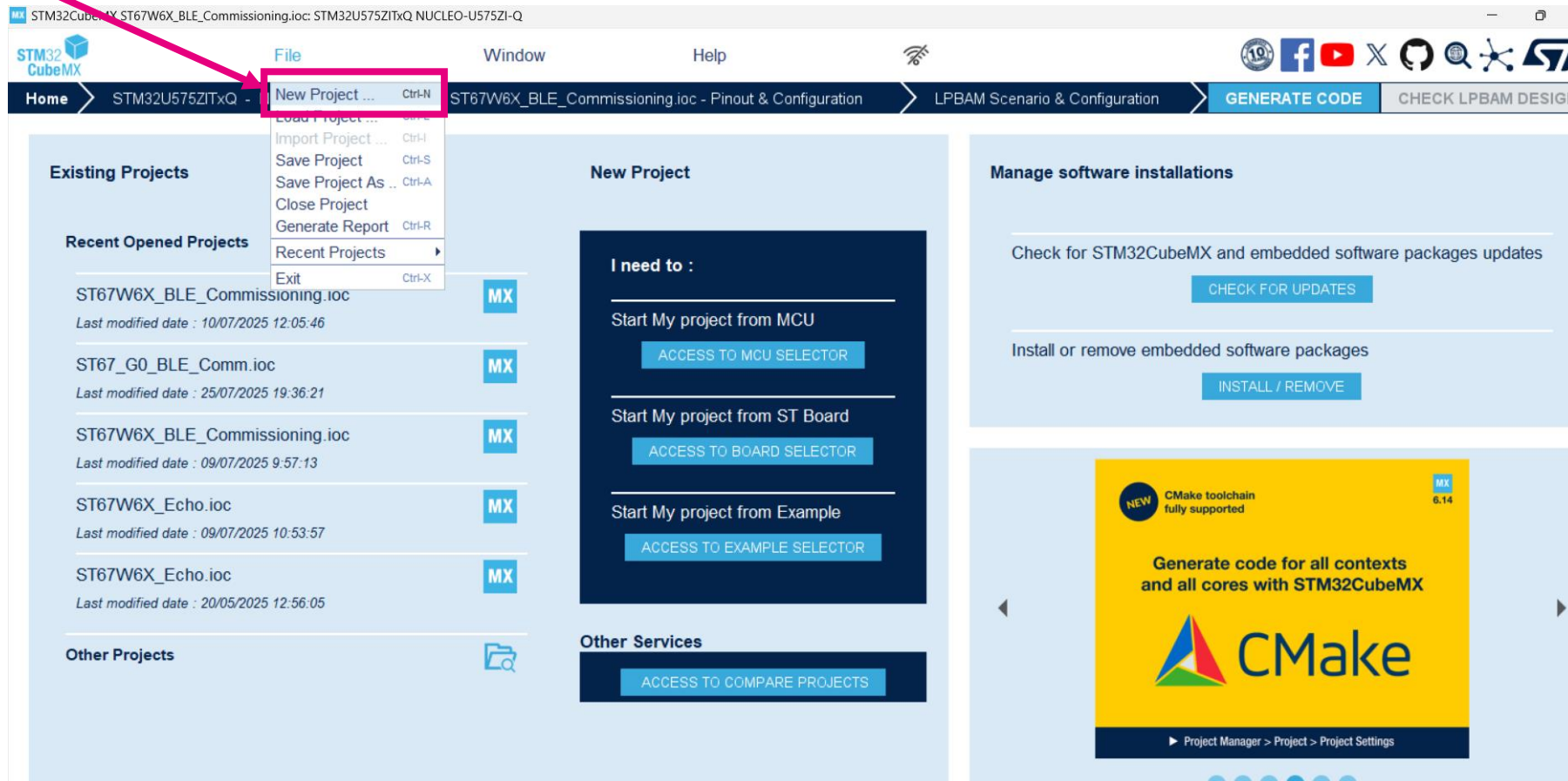


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Generating BLE Commissioning
app using NUCLEO-G0B1RE as
host processor (Hands-On)

Case 3: Generating BLE Commissioning app using NUCLEO-G0B1RE as host processor

Open STM32CubeMX and create the new project.



Select the board selector

Type STM32G0

Select STM32G0

Select NUCLEO-G0B1RE

Start Project



ST Restricted

Board Filters

Commercial Part Number

STM32G0

PRODUCT INFO

Type

Supplier

MCU / MPU Series

Check/Uncheck All (1)

STM32C0

STM32F0

STM32F1

STM32F2

STM32F3

STM32F4

STM32F7

☒ STM32G0

STM32G4

STM32H5

STM32H7

STM32L0

STM32L1

STM32L4

STM32L4+

STM32L5

Features Large Picture Docs & Resources Datasheet Buy Start Project

STM32G0 Series

NUCLEO-G0B1RE

STM32 Nucleo-64 development board with STM32G0B1RE MCU, supports Arduino and ST morpho connectivity

ACTIVE
Product is in mass production

Part Number : NUCLEO-G0B1RE
Commercial Part Number : NUCLEO-G0B1RE

Unit Price (US\$) : 10.32
Mounted Device : [STM32G0B1RET6](#)

The STM32 Nucleo-64 board provides an affordable and flexible way for users to try out new concepts and build prototypes by choosing from the various combinations of performance and power consumption features provided by the STM32 microcontroller. For the compatible boards, the internal or external SMPS significantly reduces power consumption in Run mode.

ARDUINO® Uno V3 connectivity support and the ST morpho headers allow the easy integration of the functionality of the STM32 Nucleo open development platform with a wide choice of specialized shields.

STM32 Nucleo-64 board does not require any separate probe as it integrates the

Board Project Options: NUCLEO-G0B1RE

Initialize all peripherals with their default Mode ?

Yes No

Boards List: 8 items

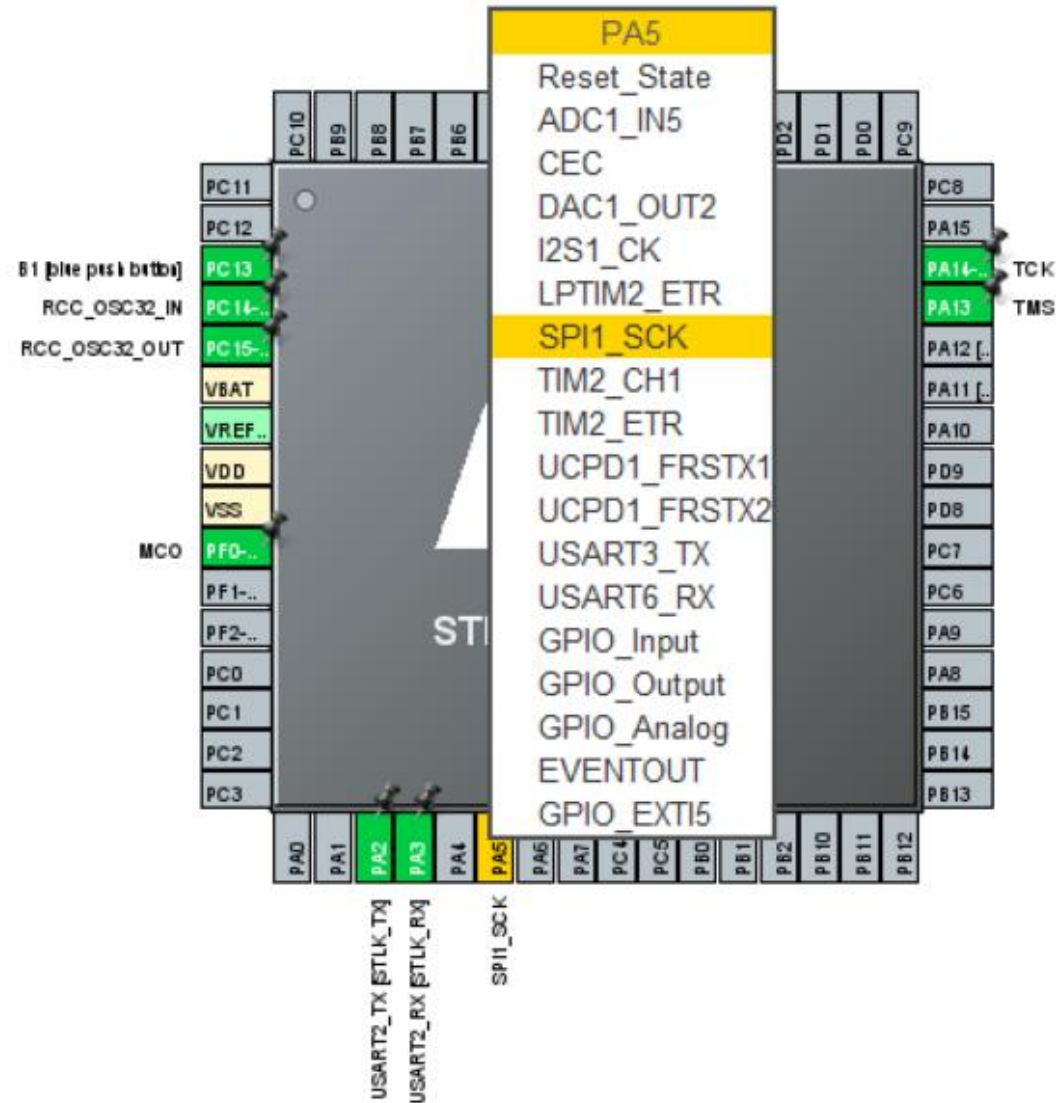
	Overview	Commercial Part No	Type	Marketing Status	Unit Price (US\$)	Mounted Device
☆		NUCLEO-G070RB	Nucleo-64	Active	10.32	STM32G070RBT6
☆		NUCLEO-G071RB	Nucleo-64	Active	10.32	STM32G071RBT6
☆		NUCLEO-G0B1RE	Nucleo-64	Active	10.32	STM32G0B1RET6
☆		STM32G0316-DISCO	Discovery Kit	Active	9.89	STM32G0316M6

Pinout

When generating an application from scratch and if X-Nucleo-67W61M1 is used, the pinout must be set in accordance with the ST67W61M1 Arduino interfaces.

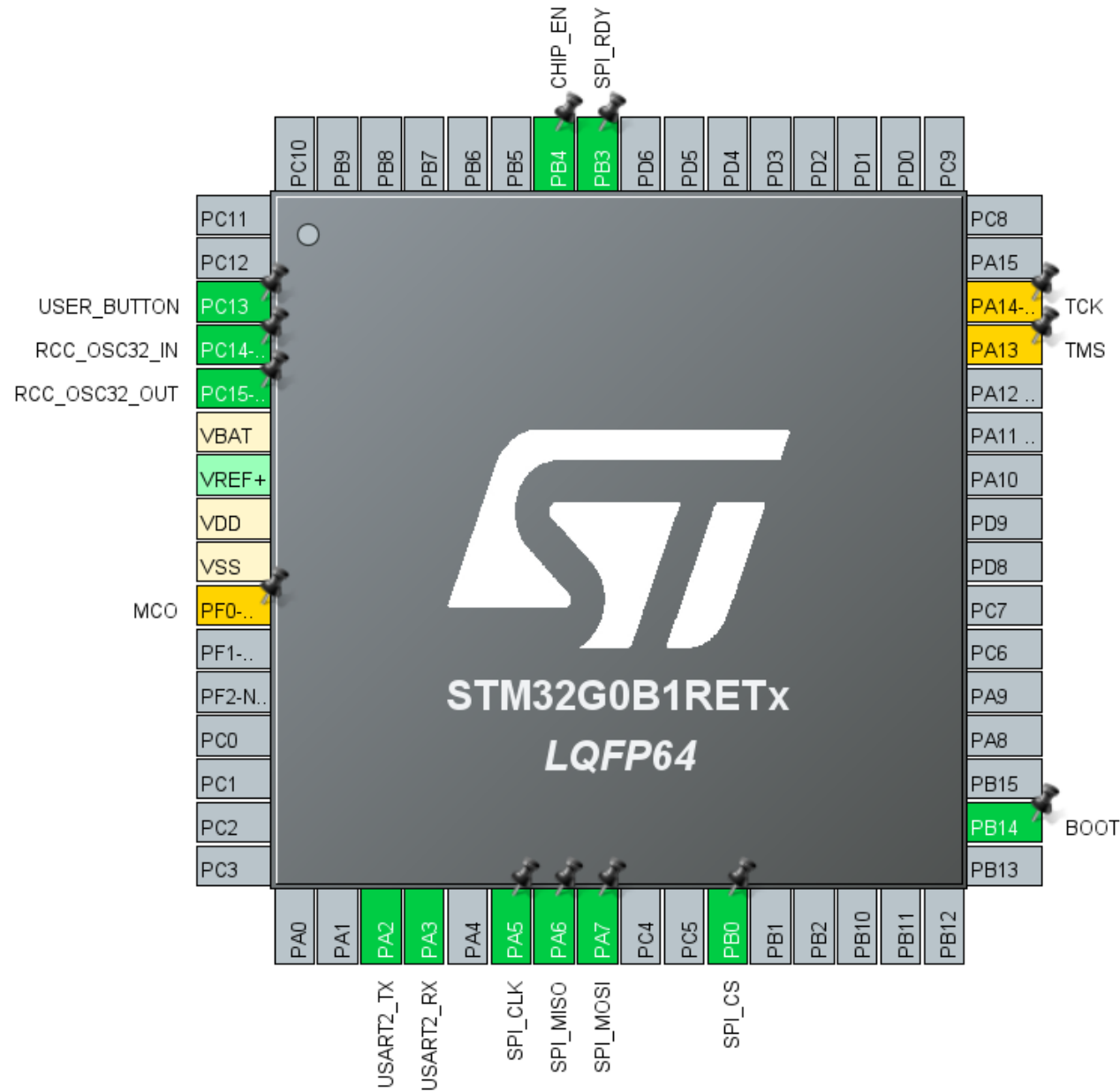
Pin function	Arduino Connector	STM32G0 Pin	GPIO mode	GPIO pull mode	GPIO speed
SPI_CLK	CN5.D13	PA5	AF PP	No Pull	Very high
SPI_MISO	CN5.D12	PA6	AF PP	No Pull	Very high
SPI_MOSI	CN5.D11	PA7	AF PP	No Pull	Very high
SPI_CS	CN5.D10	PB0	Output PP	No Pull	high
USER_BUTTON	-	PC13	EXTI Falling	Pull-up	N/A
CHIP_EN	CN9.D5	PB4	Output PP	No Pull	High
BOOT	CN9.D6	PB14	Output PP	No Pull	Low
SPI_RDY	CN9.D3	PB3	EXTI Falling/Rising	No Pull	N/A

Assigning functionality



For assigning, click on the pin and select functionality.
(e.g) PA5->SPI_SCK

Pinout View



After, assigning all the pins mentioned from the pin out table, this would be overall pinout view

UART GPIO(USART 2)

PA2: USART2_TX
PA3: USART2_RX

Select USART2

Parameter Settings

Mode: Asynchronous

Hardware Flow Control (RS232): Disable

Hardware Flow Control (RS485): ☐

Slave Select(NSS) Management: Disable

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

Basic Parameters

Baud Rate: 115200 Bits/s

Word Length: 8 Bits (including Parity)

Parity: None

Stop Bits: 1

Advanced Parameters

Data Direction: Receive and Transmit

Over Sampling: 16 Samples

Single Sample: Disable

ClockPrescaler: 1

Fifo Mode: Disable

Txfifo Threshold: 1 eighth full configuration

Rxfifo Threshold: 1 eighth full configuration

Advanced Features

Auto Baudrate: Disable

TX Pin Active Level Inversion: Disable

RX Pin Active Level Inversion: Disable

Data Inversion: Disable

TX and RX Pins Swapping: Disable

Overrun: Enable

DMA on RX Error: Enable

MSB First: Disable

GPIO Settings

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Search Signals

Search (Ctrl+F)

☐ Show only Modified Pins

Pin Name	Signal on Pin	GPIO output level	GPIO mode	GPIO Pull-up/Pull-down	Maximum out...	Fast Mode	User Label	Modified
PA2	USART2_TX	Low	Alternate Functio...	No pull-up and no pull-down	Low	n/a		☐
PA3	USART2_RX	Low	Alternate Functio...	No pull-up and no pull-down	Low	n/a		☐

SPI GPIO(SPI1)

PA5: SPI_SCK
PA6: SPI_MISO
PA7: SPI_MOSI

Home

STM32G0B1RETx - NUCLEO-G0B1RE

ST67_G0_BLE_Comm.ioc - Pinout & Configuration

Pinout & Configuration

Clock Configuration

Project Manager

Software Packs

Pinout

SPI1 Mode and Configuration

Mode

Full-Duplex Master

Hardware NSS Signal

Disable

Configuration

Reset Configuration

Parameter Settings

User Constants

NVIC Settings

DMA Settings

GPIO Settings

Search Signals

Search (Ctrl+F)

Show only Modified Pins

Pin Name	Signal on Pin	GPIO output level	GPIO mode	GPIO Pull-up/Pull-down	Maximum output current	Fast Mode	User Label	Modified
PA5	SPI1_SCK	Low	Alternate Function	No pull-up and no pull-down	Very High	n/a	SPI_SCK	✓
PA6	SPI1_MISO	Low	Alternate Function	No pull-up and no pull-down	Very High	n/a	SPI_MISO	✓
PA7	SPI1_MOSI	Low	Alternate Function	No pull-up and no pull-down	Very High	n/a	SPI_MOSI	✓

Select SPI1

Set the Maximum output to Very High

Set the User Labels



SPI GPIO(SPI1)

Parameter
Settings

The screenshot displays the 'SPI1 Mode and Configuration' window. On the left, a tree view lists various components, with 'SPI1' highlighted. The main panel shows the 'Parameter Settings' tab selected, which includes sections for 'Basic Parameters', 'Clock Parameters', and 'Advanced Parameters'. Two callouts with arrows point to specific settings: one points to 'Data Size' set to '8 Bits' under 'Basic Parameters', and another points to 'Prescaler' set to '2' under 'Clock Parameters'.

Mode

Mode: Full-Duplex Master

Hardware NSS Signal: Disable

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

Basic Parameters

Frame Format

Data Size: 8 Bits

First Bit

MSB First

Clock Parameters

Prescaler (for Baud Rate): 2

Baud Rate: 8.0 MBits/s

Clock Polarity (CPOL): Low

Clock Phase (CPHA): 1 Edge

Advanced Parameters

CRC Calculation: Disabled

NSSP Mode: Enabled

NSS Signal Type: Software

Set the Data size to 8

Set the Prescaler to 2

DMA Channels

Select DMA

The screenshot shows the STM32CubeMX Pinout & Configuration tab. On the left, the 'System Core' section has 'DMA' selected and highlighted with a red box. A red arrow points from the 'Select DMA' text to this box. The main area shows the 'DMA Mode and Configuration' section. Under 'Mode', 'DMA1_DMA2' and 'MemToMem' are selected. Below this is a table with the following data:

DMA Request	Channel	Direction	Priority
SPI1_TX	DMA1 Channel 1	Memory To Peripheral	High
SPI1_RX	DMA1 Channel 2	Peripheral To Memory	Low
			Medium
			High
			Very High

The 'High' priority for the SPI1_TX row is highlighted with a red box. Below the table are 'Add' and 'Delete' buttons. The 'DMA Request Settings' section shows 'Mode' set to 'Normal', 'Increment Address' unchecked, and 'Data Width' set to 'Byte'. The 'DMA Request Synchronization Settings' section has 'Enable synchronization' unchecked, 'Synchronization signal' and 'Signal polarity' set to default, 'Enable event' unchecked, and 'Request number' set to default.

SPI1_TX -> DMA1 CHANNEL 1->High
SPI1_RX -> DMA1 CHANNEL 2-> High

GPIO

Select GPIO

Categories A->Z

System Core

- ✓ DMA
- ✓ **GPIO**
- ✓ IWDG
- ✓ NVIC
- ⚠ RCC
- ⚠ SYS
- WWDG

Analog >

Timers >

Connectivity >

Multimedia >

GPIO Mode and Configuration

Mode

Configuration

Group By Peripherals

✓ GPIO ✓ Single Mapped Signals ✓ RCC ✓ SPI ✓ USART ✓ NVIC

Search Signals
Search (Ctrl+F)

☐ Show only Modified Pins

Pin Name	Signal on Pin	GPIO output level	GPIO mode	GPIO Pul...	Maximu...	Fast Mode	User Label	Modified
PB0	n/a	Low	Output Push Pull	No pull-u...	High	n/a	SPI_CS	✓
PB3	n/a	n/a	External Interrupt Mode with Rising/Falling edge trigger detection	No pull-u...	n/a	n/a	SPI_RDY	✓
PB4	n/a	Low	Output Push Pull	No pull-u...	High	n/a	CHIP_EN	✓
PB14	n/a	Low	Output Push Pull	No pull-u...	Low	n/a	BOOT	✓
PC13	n/a	n/a	External Interrupt Mode with Falling edge trigger detection	No pull-u...	n/a	n/a	USER_BU...	✓

GPIO pin names

GPIO output level

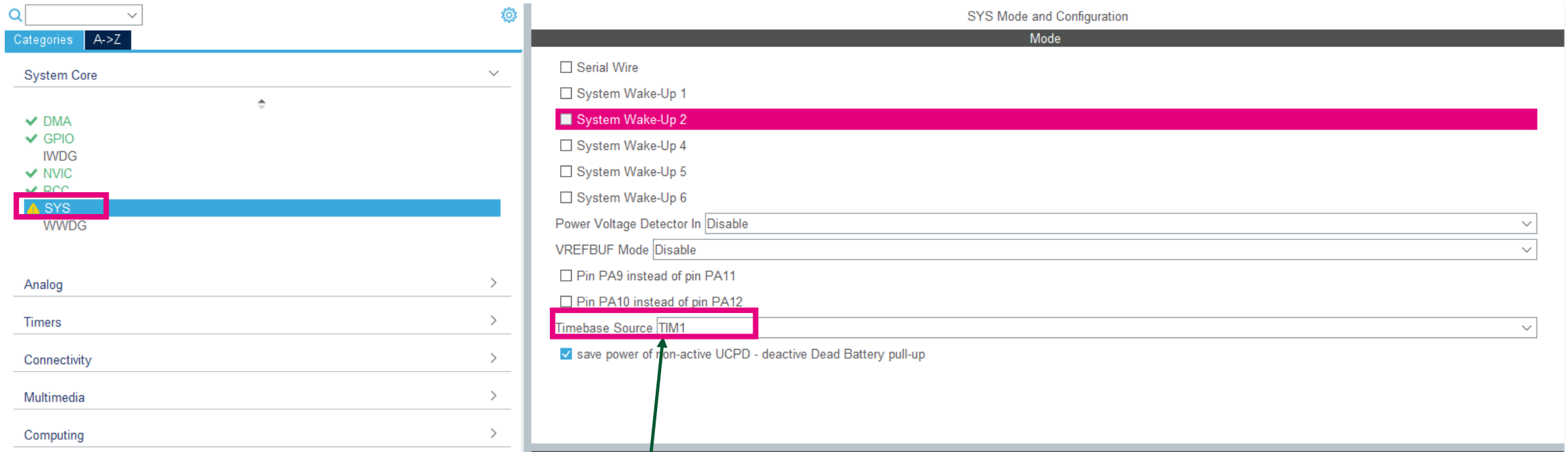
GPIO mode

GPIO maximum speed

GPIO user label



Timer



Sys mode configuration: TIM1 in place of SysTick: to avoid RTOS conflicts, enable low power operation.

RCC

Home > STM32G0B1RETx - NUCLEO-G0B1RE > Untitled - Pinout & Configuration

Pinout & Configuration | Clock Configuration | Project Manager

Software Packs | Pinout

RCC Mode and Configuration

Mode

High Speed Clock (HSE) **Disable**

☐ OSC enable

Low Speed Clock (LSE) Crystal/Ceramic Resonator

☐ OSC32 enable

☐ Master Clock Output

☐ Master Clock Output 2

☒ LSCO Clock Output

☐ Audio Clock Input (I2S_CKIN)

CRS SYNC **Disable**

HSE disabled

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

System Parameters

VDD voltage (V)	3.3 V
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	2 WS (3 CPU cycle)

RCC Parameters

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters

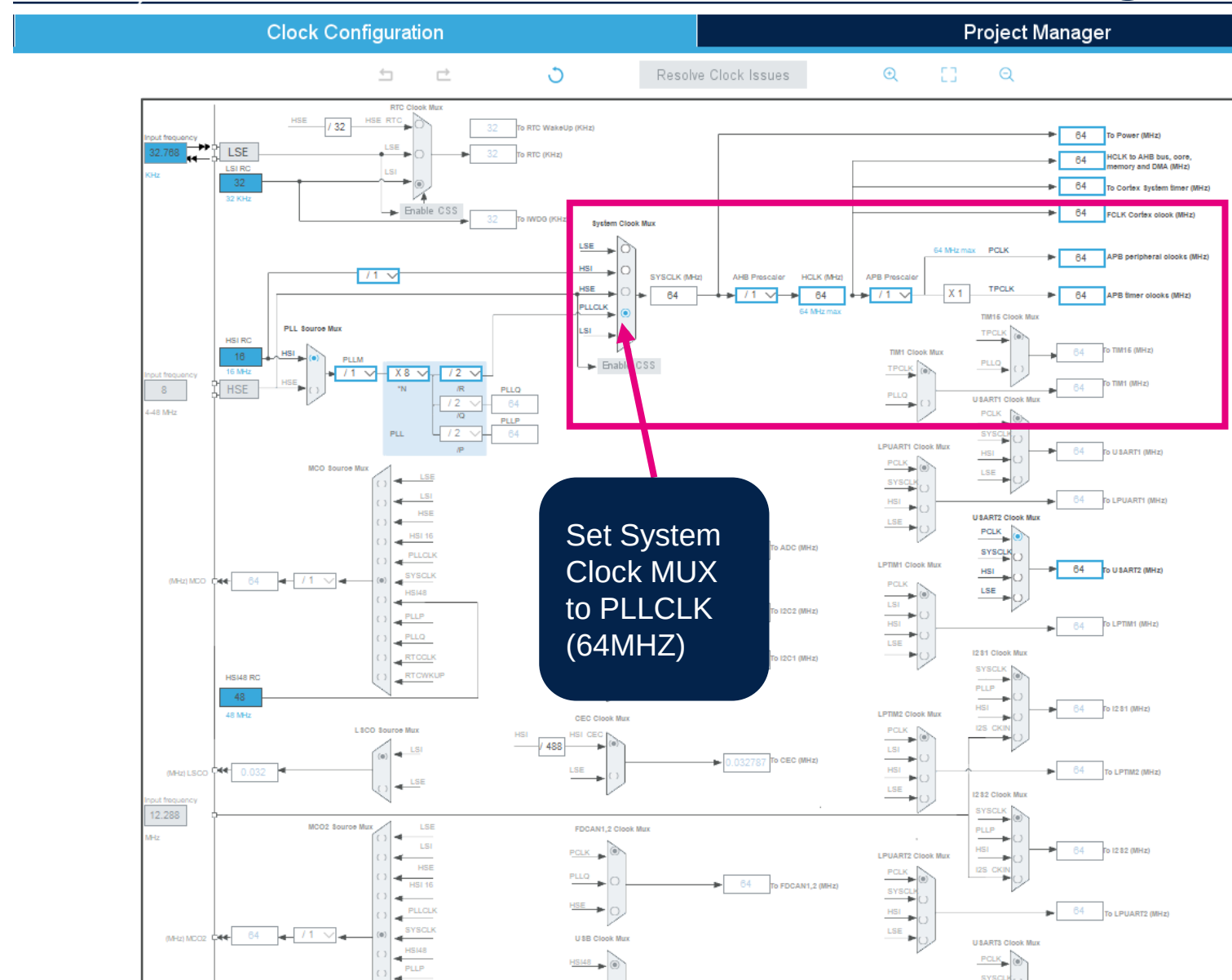
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
-------------------------------	---------------------------------

Peripherals Clock Configuration

Generate the peripherals clock configuration	TRUE
--	------

Clock Configuration

- In the "Clock Configuration" panel, system clock is set to the maximum frequency of 64 MHz, which increases the SPI clock speed to 32 MHz



Middleware & Software Packs

FreeRTOS

- FreeRTOS component can be either be selected from Middleware (FREERTOS) or from the Software Pack (X-CUBE-FREERTOS) depending on the chosen STM32.
- By default, the stack size of the default task defined by STM32CubeMx tool is equal to 128 words. The application default task required **512 words**.

The screenshot displays the STM32CubeMX interface for configuring FreeRTOS. On the left, the 'Middleware and Software Packs' list shows 'FREERTOS' selected. The main window, titled 'FREERTOS Mode and Configuration', has the 'Tasks and Queues' tab active. It shows a table of tasks with the following data:

Task Name	Priority	Stack Size (Words)	Entry Function	Code Generation	Parameter	Allocation	Buffer Name	Control Block Name
defaultTask	osPriorityNormal	512	StartDefaultTask	Default	NULL	Dynamic	NULL	NULL

An 'Edit Task' dialog box is open, showing the configuration for 'defaultTask'. The 'Stack Size (Words)' field is highlighted with a pink box and contains the value '512'.

Middleware & Software Packs

FreeRTOS

- Adjust heap :The size heap value should be set at least to 40Kbytes. To optimize the memory, this value could be adjusted during integration and validation tests of the application

The screenshot displays the STM32CubeIDE interface for configuring FreeRTOS. On the left, the 'Middleware and Software Packs' list includes various packs, with 'FREERTOS' highlighted. On the right, the 'FREERTOS Mode and Configuration' dialog is open, showing the 'Config parameters' tab. The 'TOTAL_HEAP_SIZE' is set to '40000 Bytes'.

Parameter	Value
FreeRTOS version	10.3.1
CMSIS-RTOS version	2.00
MPU/FPU	
ENABLE_MPU	Disabled
ENABLE_FPU	Disabled
Kernel settings	
USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128 Words
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled
Memory management settings	
Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	40000 Bytes
Memory management scheme	heap_4
Hook function related definitions	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled

Middleware & Software Packs

FreeRTOS

Pinout & Configuration

Clock Configuration

Project Manager

Software Packs

Pinout

FREERTOS Mode and Configuration

Mode

Interface: CMSIS_V2

Configuration

Reset Configuration

Config parameters

Include parameters

Advanced settings

User Constants

Tasks and Queues

Timers and Semaphores

Mutexes

Events

FreeRTOS Heap Usage

Configure the below parameters :

Search (Ctrl+F)

Newlib settings (see parameter description first)

USE_NEWLIB_REENTRANT

Enabled

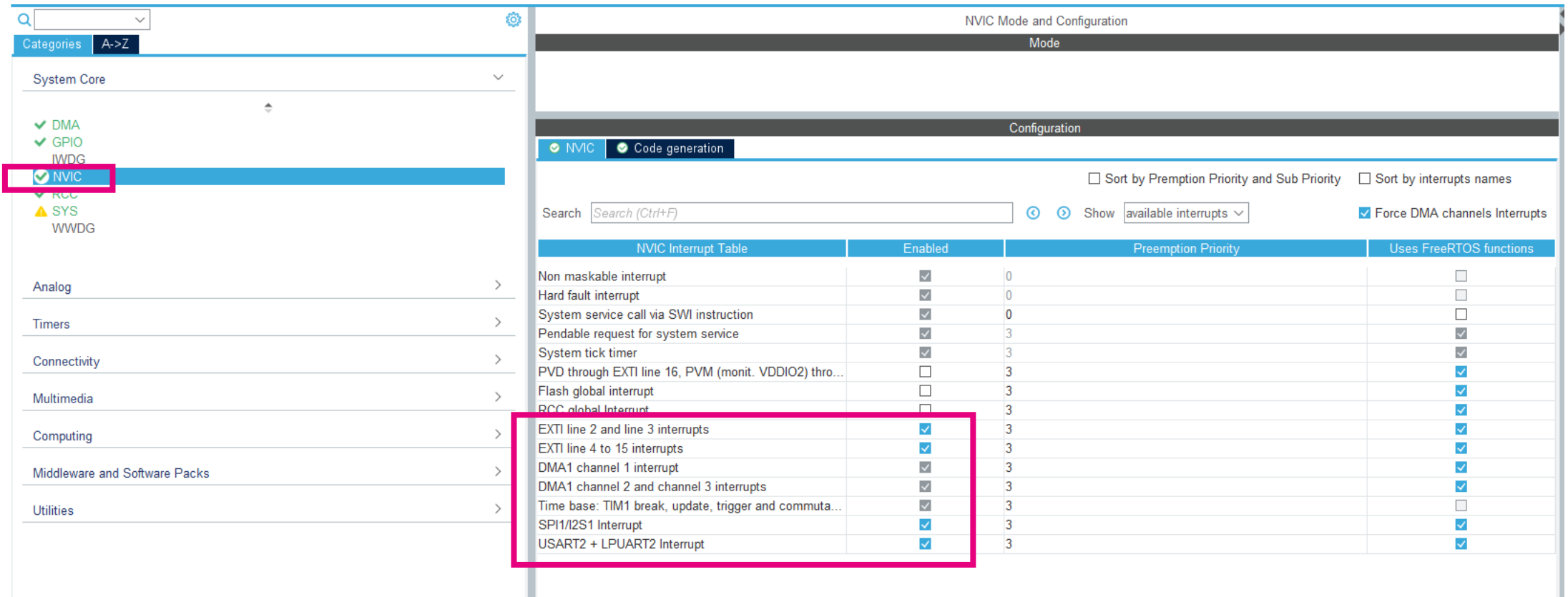
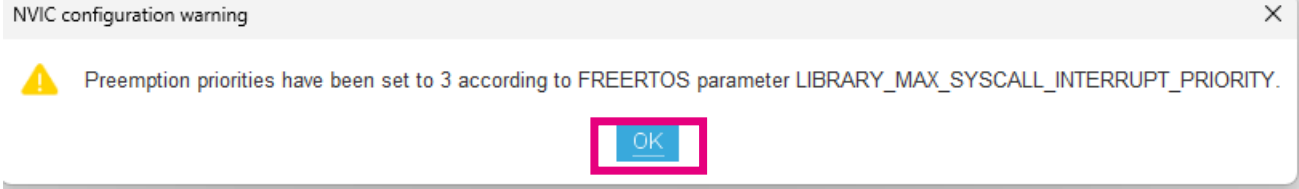
Project settings (see parameter description first)

Use FW pack heap file

Enabled



Enable
USE_NEWLIB_REENTRANT



The image shows the 'NVIC Mode and Configuration' window in a development tool. On the left, a tree view under 'System Core' lists various components: DMA, GPIO, IWDG, **NVIC** (highlighted with a red box), RCC, SYS, and WWDG. The main panel is titled 'NVIC Mode and Configuration' and has two tabs: 'NVIC' (selected) and 'Code generation'. Below the tabs, there are search and filter options: a search bar with 'Search (Ctrl+F)', a 'Show' button, a dropdown menu set to 'available interrupts', and two checkboxes: 'Sort by Preemption Priority and Sub Priority' (unchecked) and 'Sort by interrupts names' (unchecked). A checkbox 'Force DMA channels Interrupts' is checked. Below these options is a table with four columns: 'NVIC Interrupt Table', 'Enabled', 'Preemption Priority', and 'Uses FreeRTOS functions'. The table lists various interrupts, with a red box highlighting the bottom section of the table.

NVIC Interrupt Table	Enabled	Preemption Priority	Uses FreeRTOS functions
Non maskable interrupt	☒	0	☐
Hard fault interrupt	☒	0	☐
System service call via SWI instruction	☒	0	☐
Pendable request for system service	☒	3	☒
System tick timer	☒	3	☒
PVD through EXTI line 16, PVM (monit. VDDIO2) thro...	☐	3	☒
Flash global interrupt	☐	3	☒
RCC global interrupt	☐	3	☒
EXTI line 2 and line 3 interrupts	☒	3	☒
EXTI line 4 to 15 interrupts	☒	3	☒
DMA1 channel 1 interrupt	☒	3	☒
DMA1 channel 2 and channel 3 interrupts	☒	3	☒
Time base: TIM1 break, update, trigger and commuta...	☒	3	☐
SPI1/I2S1 Interrupt	☒	3	☒
USART2 + LPUART2 Interrupt	☒	3	☒

Middleware & Software Packs

X-CUBE-ST67W61 Software package

Pack / Bundle / Component	Status	Version	Selection
▼ STMicroelectronics.X-CUBE-ST67W61	✓	1.1.0	▼
▼ Device Applications	✓	1.1.0	
Application	✓	1.1.0	BLE_Commissioning ▼
FreeRTOS_Tickless		1.1.0	☐
▼ Network ST67W6X_Network_Driver	✓	1.1.0	
ST67 Architecture	✓	1.1.0	T01 ▼
ATDriver	✓	1.1.0	☒
ServiceAPI	✓	1.1.0	☒
ServiceShell		1.1.0	☐
▼ Utilities	✓		
Logging	✓	1.1.0	☒
Shell		1.1.0	☐
Statistics		1.1.0	☐
▼ Debug TraceRecorder		4.10.2	
TraceRecorder		4.10.2	☐
▼ Data Exchange cJSON		1.7.18	
cJSON		1.7.18	☐
▼ File System LittleFS		2.10.1	
LittleFS		2.10.1	☐
▼ Utility Utilities		1.4.2	
LPM / Tiny LPM		1.4.2	☐
▼ Network LwIP		2.2.0	
LwIP		2.2.0	☐
▼ Data Exchange MQTT-C		1.1.6	
MQTT-C		1.1.6	☐

Select BLE_Commissioning

Select T01 ST67
Architecture

Enable AT Driver

Enable
ServiceAPI

Enable Logging

Middleware & Software Packs

X-CUBE-ST67W61 Software package

The screenshot displays the STM32CubeIDE interface with the 'Pinout & Configuration' tab selected. The left sidebar shows a list of software packs, with 'X-CUBE-ST67W61' highlighted. The main area shows the 'Mode and Configuration' for 'STMicroelectronics.X-CUBE-ST67W61.1.0.0'. The 'Mode' section has checkboxes for 'Device Applications' and 'Network ST67W6X Network Driver'. The 'Configuration' section has a 'Reset Configuration' button and a tabbed interface with 'Platform Settings' selected. A pink arrow points from a 'Platform settings' label to the 'Platform Settings' tab. The 'Platform proposal' section shows a table with columns 'Name', 'IPs or Components', 'Found Solutions', and 'BSP API'.

Name	IPs or Components	Found Solutions	BSP API
LogSh_UART	USART:Asynchronous	USART2	Unknown
NCP_SPI	SPI:Full-Duplex Master	SPI1	Unknown

Middleware & Software Packs

X-CUBE-ST67W61 Software package

STMicroelectronics.X-CUBE-ST67W61.1.1.0 Mode and Configuration

Mode

- ☒ Device Applications
- ☒ Network ST67W6X Network Driver

Configuration

Reset Configuration

Parameter Settings | User Constants | Platform Settings

Configure the below parameters :

Search (Ctrl+F)

Parameter	Value
W6X Init Modules	
ST67_ARCH	W6X_ARCH_T01
WIFI_STATION	Yes
WIFI_SAP	No
BLE	Yes
NET	Yes
MQTT	No
Basic Parameters	
LOG_OUTPUT_MODE	LOG_OUTPUT_UART
FREERTOS_TASK_NOTIF_ARRAY_LEN	8
Logging-Shell	
LOG_LEVEL	LOG_DEBUG
W6X Parameters	
W6X_POWER_SAVE_AUTO	No
W6X_CLOCK_MODE	Internal RC oscillator
W6X_WIFI_AUTOCONNECT	No
W6X_WIFI_COUNTRY_CODE	00
W6X_WIFI_ADAPTIVE_COUNTRY_CODE	No
W6X_BLE_HOSTNAME	ST67W61_BLE
W6X_NET_HOSTNAME	ST67W61_WIFI
W6X_NET_DHCP_MODE	DHCP STA+SAP
W6X_NET_RECV_TIMEOUT (ms)	10000
W6X_NET_SEND_TIMEOUT (ms)	10000
W6X_NET_RECV_BUFFER_SIZE	9216
W6X_NET_DNS_MANUAL	No
W6X_NET_DNS_IP_1	{208, 67, 222, 222}
W6X_NET_DNS_IP_2	{8, 8, 8, 8}
W6X_NET_DNS_IP_3	{0, 0, 0, 0}

Disable
W6X_POWER_SAVE_AUTO

Rename W6X_BLE_HOSTNAME to
something unique



Project Creation

Home > STM32G0B1RETx - NUCLEO-G0B1RE > ST67_G0_BLE_Comm.ioc - Project Manager

Pinout & Configuration | Clock Configuration | **Project Manager**

Project

Code Generator

Advanced Settings

Project Settings

Project Name: ST67_G0_BLE_Comm

Project Location: C:\ST67_Test Browse

Application Structure: Advanced ☐ Do not generate the main()

Toolchain Folder Location: C:\ST67_Test\ST67_G0_BLE_Comm\

Toolchain / IDE: STM32CubeIDE ☒ Generate Under Root

Linker Settings

Minimum Heap Size: 0x200

Minimum Stack Size: 0x400

Thread-safe Settings

Cortex-M0+NS

☐ Enable multi-threaded support

Thread-safe Locking Strategy: Default - Mapping suitable strategy depending on RTOS selection.

Mcu and Firmware Package

Mcu Reference: STM32G0B1RETx

Firmware Package Name and Version: STM32Cube_FW_G0_V1.6.2 ☒ Use latest available version

☒ Use Default Firmware Location

Firmware Relative Path: C:/Users/kulkams/STM32Cube/Repository/STM32Cube_FW_G0_V1.6.2 Browse

1. Enable “Generate Under Root”

Here, we are enabling the Flat directory structure: (i.e. Driver and MW directories copied into your project directory)

2. Enable “ Use Default Firmware Location”

Code Generator

Home > STM32G0B1RETx - NUCLEO-G0B1RE > ST67_G0_BLE_Comm.ioc - Project Manager

Pinout & Configuration | Clock Configuration | Project Manager

Project

Code Generator

Advanced Settings

STM32Cube MCU packages and embedded software packs

- ☐ Copy all used libraries into the project folder
- ☒ Copy only the necessary library files
- ☐ Add necessary library files as reference in the toolchain project configuration file

Generated files

- ☐ Generate peripheral initialization as a pair of '.c/.h' files per peripheral
- ☐ Backup previously generated files when re-generating
- ☒ Keep User Code when re-generating
- ☒ Delete previously generated files when not re-generated

HAL Settings

- ☐ Set all free pins as analog (to optimize the power consumption)
- ☐ Enable Full Assert

3. Enable "Copy only the necessary files"

Advanced settings

Home > STM32G0B1RETx - NUCLEO-G0B1RE > Experts_Training_G0.ioc - Project Manager > GENE

Pinout & Configuration | Clock Configuration | Project Manager

Project

Code Generator

Advanced Settings

Driver Selector

Search (Ctrl+F)

RCC HAL
GPIO HAL
DMA HAL
> SPI HAL
> USART HAL
STMicroelectronics.X-CUBE-ST67W61.1.1.0 HAL

Generated Function Calls

Generate Code	Rank	Function Name	Peripheral Instance Name	☐ Do Not Generate Function Call	☐ Visibility (Static)
☒	1	SystemClock_Config	RCC	☐	☐
☒	2	MX_GPIO_Init	GPIO	☐	☒
☒	3	MX_DMA_Init	DMA	☐	☒
☒	4	MX_SPI1_Init	SPI1	☐	☒
☒	5	MX_USART2_UART_Init	USART2	☐	☒
☒	6	MX_ST67W6X_Init	STMicroelectronics.X-C...	☐	☐

4. Uncheck this option for SPI1 and USART2

5. Check Visibility for SPI1 and USART2

Generate Code

5. Generate Code

Home > STM32G0B1RETx - NUCLEO-G0B1RE > Experts_Training_G0.ioc - Project Manager > **GENERATE CODE**

Pinout & Configuration | **Clock Configuration** | **Project Manager** | **Tools**

Project

Code Generator

Advanced Settings

Driver Selector

Search (Ctrl+F)

- RCC HAL
- GPIO HAL
- DMA HAL
- > SPI HAL
- > USART HAL
- STMicroelectronics.X-CUBE-ST67W61.1.1.0 HAL

Generated Function Calls

Generate Code	Rank	Function Name	Peripheral Instance Name	☐ Do Not Generate Function Call	☐ Visibility (Static)
☒	1	SystemClock_Config	RCC	☐	☐
☒	2	MX_GPIO_Init	GPIO	☐	☒
☒	3	MX_DMA_Init	DMA	☐	☒
☒	4	MX_SPI1_Init	SPI1	☐	☒
☒	5	MX_USART2_UART_Init	USART2	☐	☒
☒	6	MX_ST67W6X_Init	STMicroelectronics.X-C...	☐	☐

Register CallBack

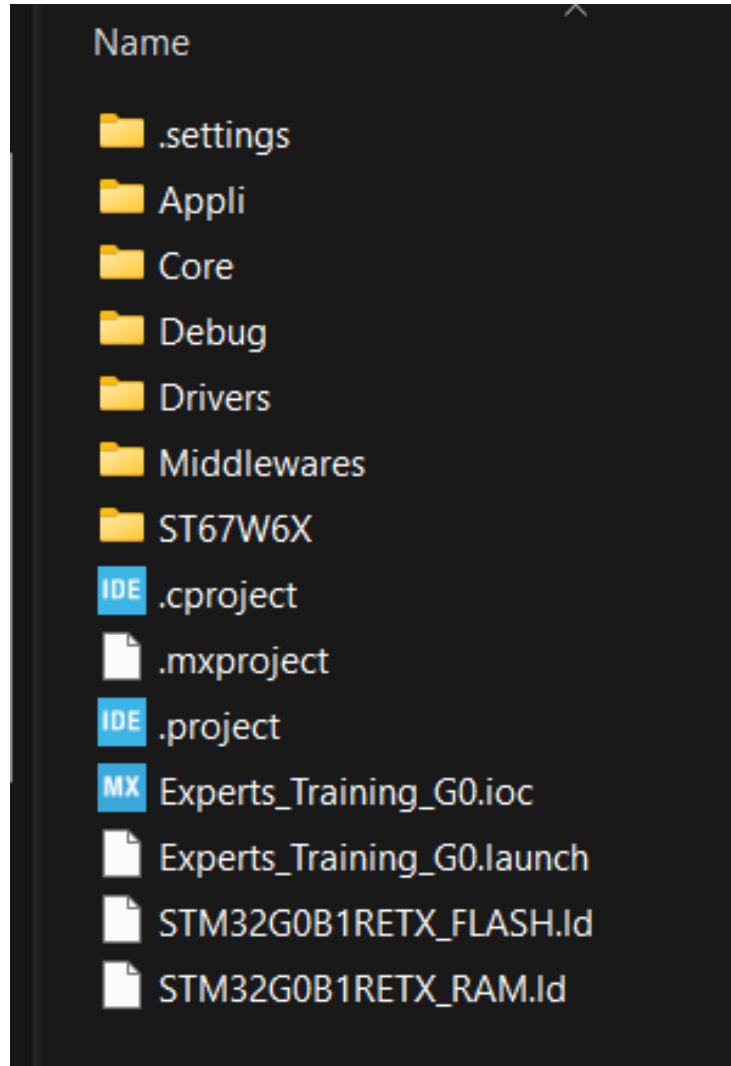
Search (Ctrl+F)

- ADC DISABLE
- CEC DISABLE
- COMP DISABLE
- CRYP DISABLE
- DAC DISABLE
- FDCAN DISABLE
- HCD DISABLE
- I2C DISABLE
- I2S DISABLE
- IRDA DISABLE
- LPTIM DISABLE
- PCD DISABLE
- RNG DISABLE
- RTC DISABLE
- SMBUS DISABLE
- SPI DISABLE
- TIM DISABLE
- UART DISABLE
- USART DISABLE
- WWDG DISABLE

Note: While generating the project, make sure the MX_SPI1_Init and MX_USART2_UART_Init are unchecked (Do Not Generate Function Call) or if they are getting enabled when you hit the Generate Code: try to close CubeMX and try again



Folder Structure

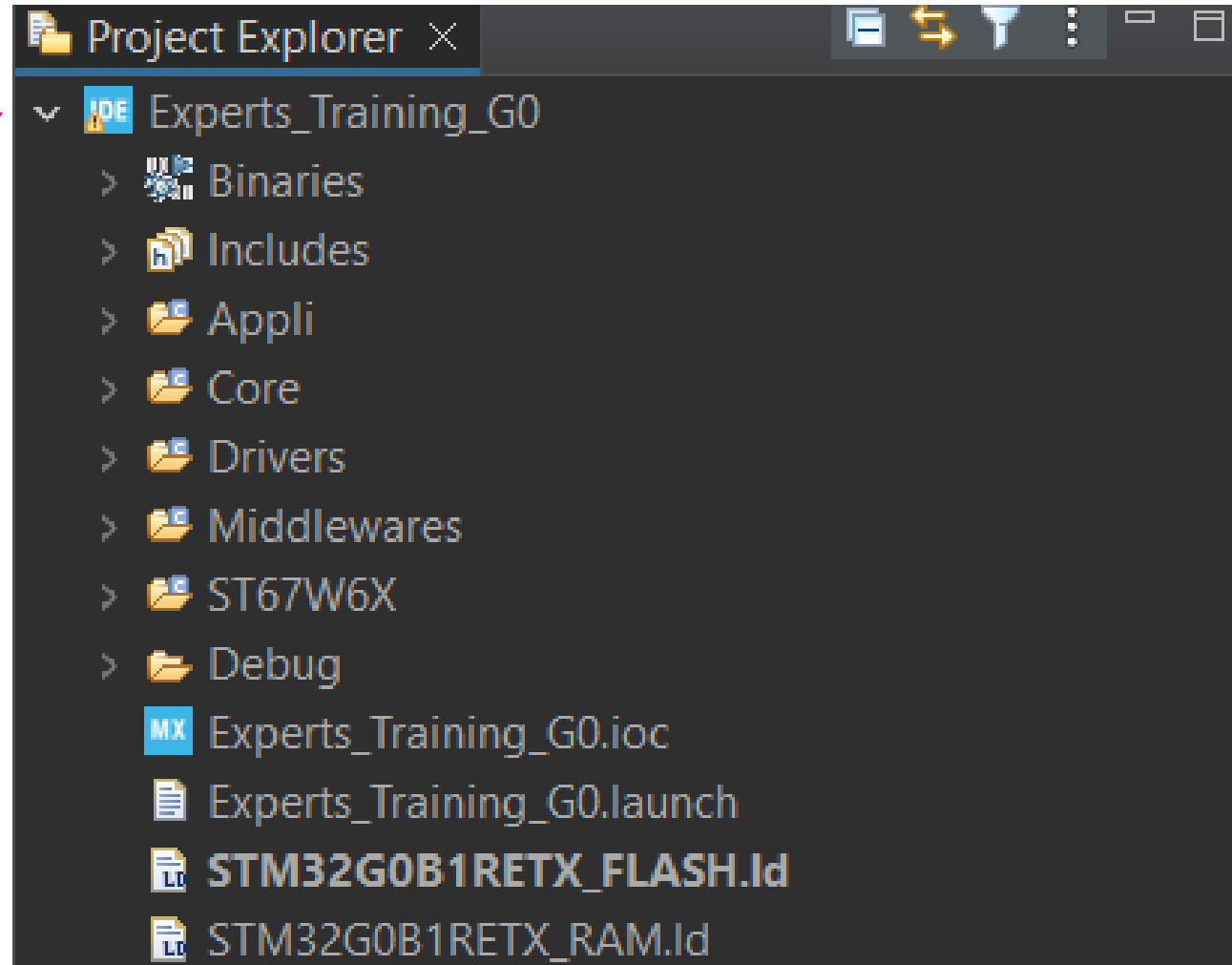


Folder structure view after generating the code from CubeMX

- App code
- Core
- Drivers
- Middlewares
- ST67W6X

STM32CubeIDE

Load the project on
the STM32CubeIDE



xPortIsInsideInterrupt()

- If the used device embeds an Arm® Cortex®M0+ (e.g. STM32G0) and its FreeRTOS version (v10.3.1) which is older than of V10.6, definition below **must be added manually after project has been generated**.
- The below function definition must be added in the **portmacro.h** file (Project/Middlewares/Third_Party/FreeRTOS/Source/portable/GCC/ARM_CM0)

```
#define portINLINE __inline
#ifndef portFORCE_INLINE
#define portFORCE_INLINE inline __attribute__((always_inline))
#endif

/*-----*/

portFORCE_INLINE static BaseType_t xPortIsInsideInterrupt( void )
{
    uint32_t ulCurrentInterrupt;
    BaseType_t xReturn;

    /* Obtain the number of the currently executing interrupt. */
    __asm volatile ( "mrs %0, ipsr" : "=r" ( ulCurrentInterrupt )::"memory" );

    if( ulCurrentInterrupt == 0 )
    {
        xReturn = pdFALSE;
    }
    else
    {
        xReturn = pdTRUE;
    }

    return xReturn;
}
```



STM32CubeIDE

Call xPortIsInsideInterrupt() function
under portmacro.h file

IMP: As we are using G0 (CortexM0) -> older
version of FreeRTOS, so we need to add this
function definition manually.

```
#define portINLINE __inline
#ifndef portFORCE_INLINE
#define portFORCE_INLINE inline
__attribute__((always_inline))
#endif

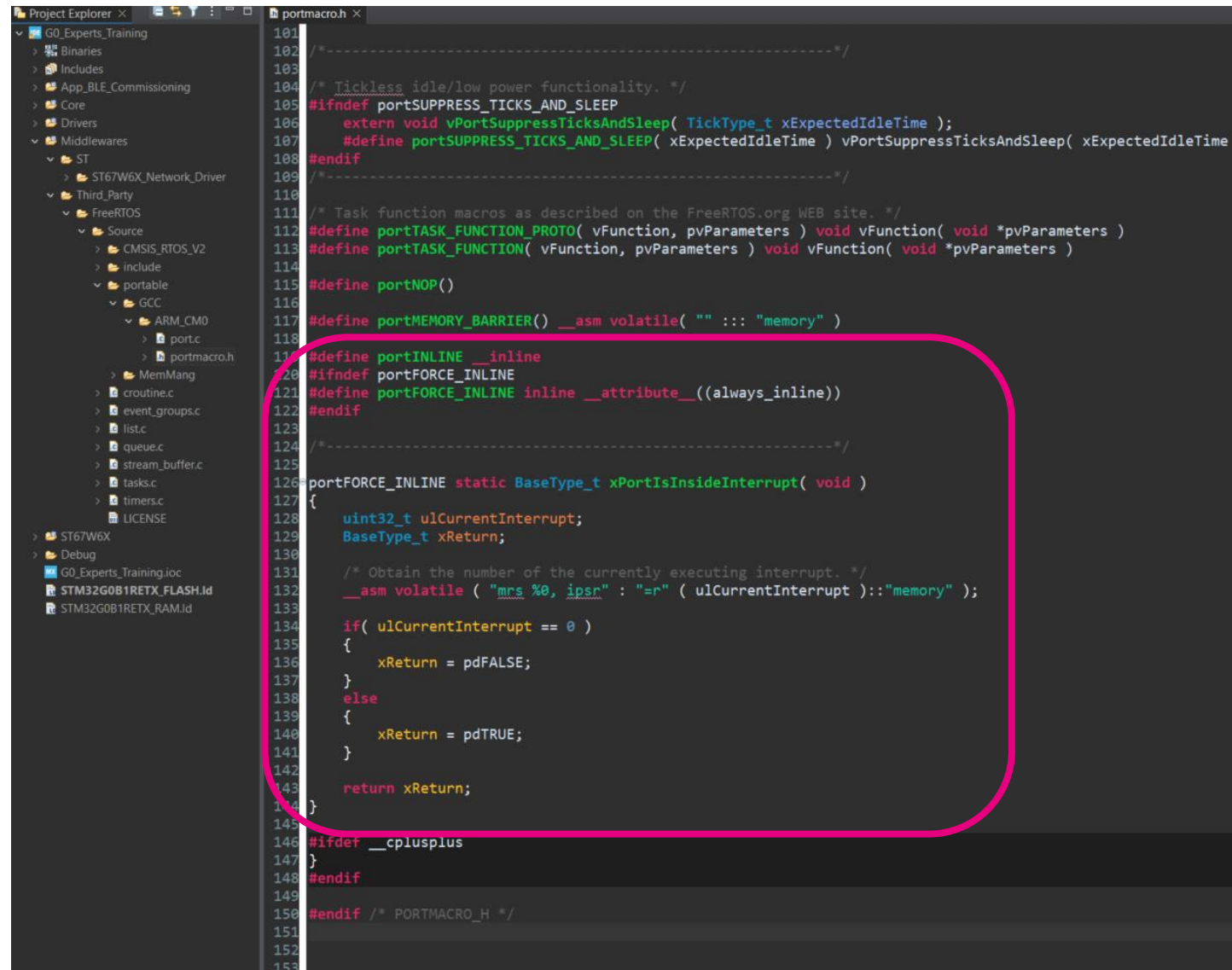
/*-----*/

portFORCE_INLINE static BaseType_t xPortIsInsideInterrupt(
void )
{
    uint32_t ulCurrentInterrupt;
    BaseType_t xReturn;

    /* Obtain the number of the currently executing interrupt. */
    __asm volatile ( "mrs %0, ipsr" : "=r" ( ulCurrentInterrupt
)::"memory" );

    if( ulCurrentInterrupt == 0 )
    {
        xReturn = pdFALSE;
    }
    else
    {
        xReturn = pdTRUE;
    }

    return xReturn;
}
```



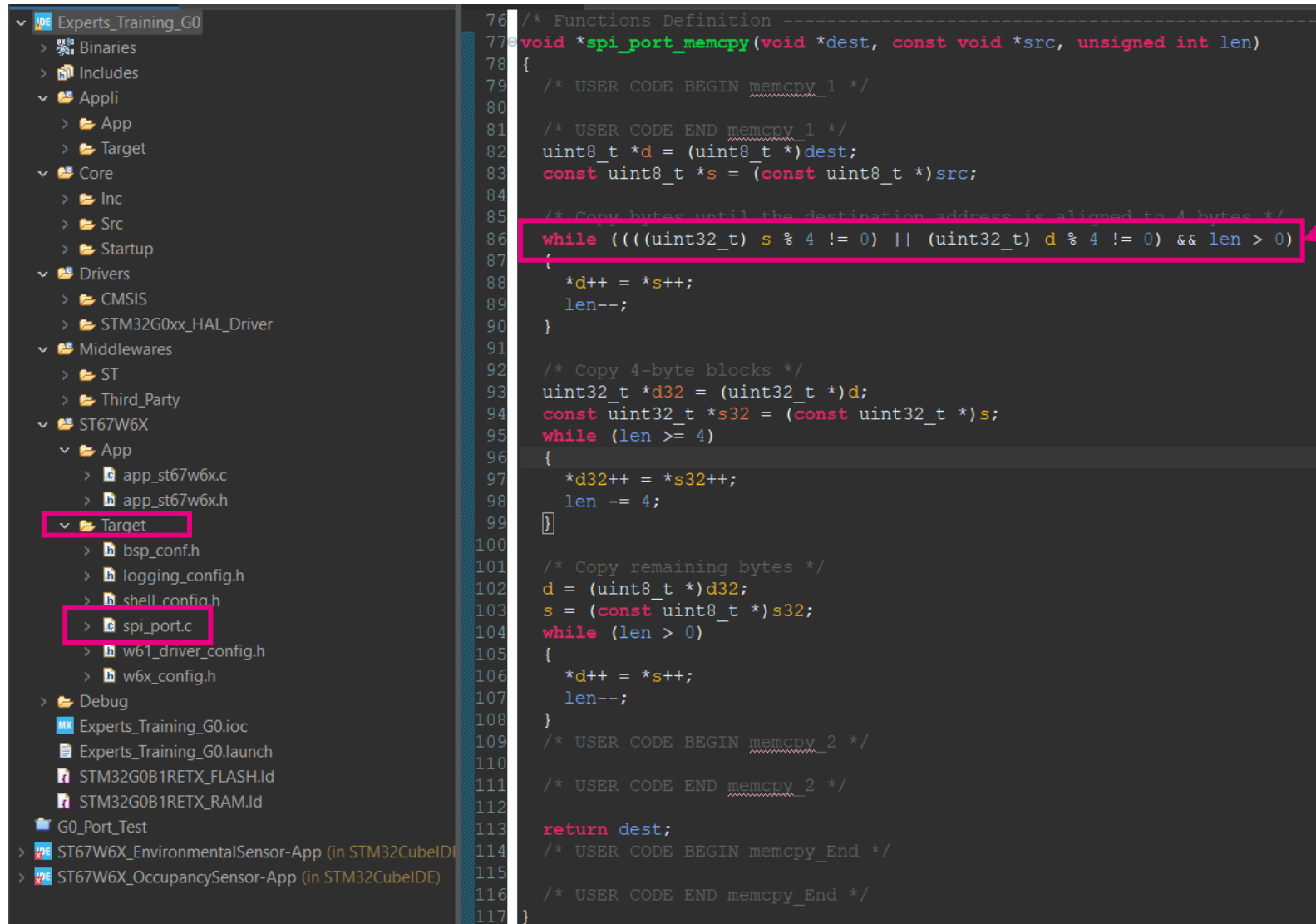
Note: For the Cortex M0+, there is a memory alignment to be followed. Introducing a 4-byte alignment check section to see if source or destination are not aligned.

If not, it handles copy byte by byte, until alignment is reached.

STM32CubeIDE

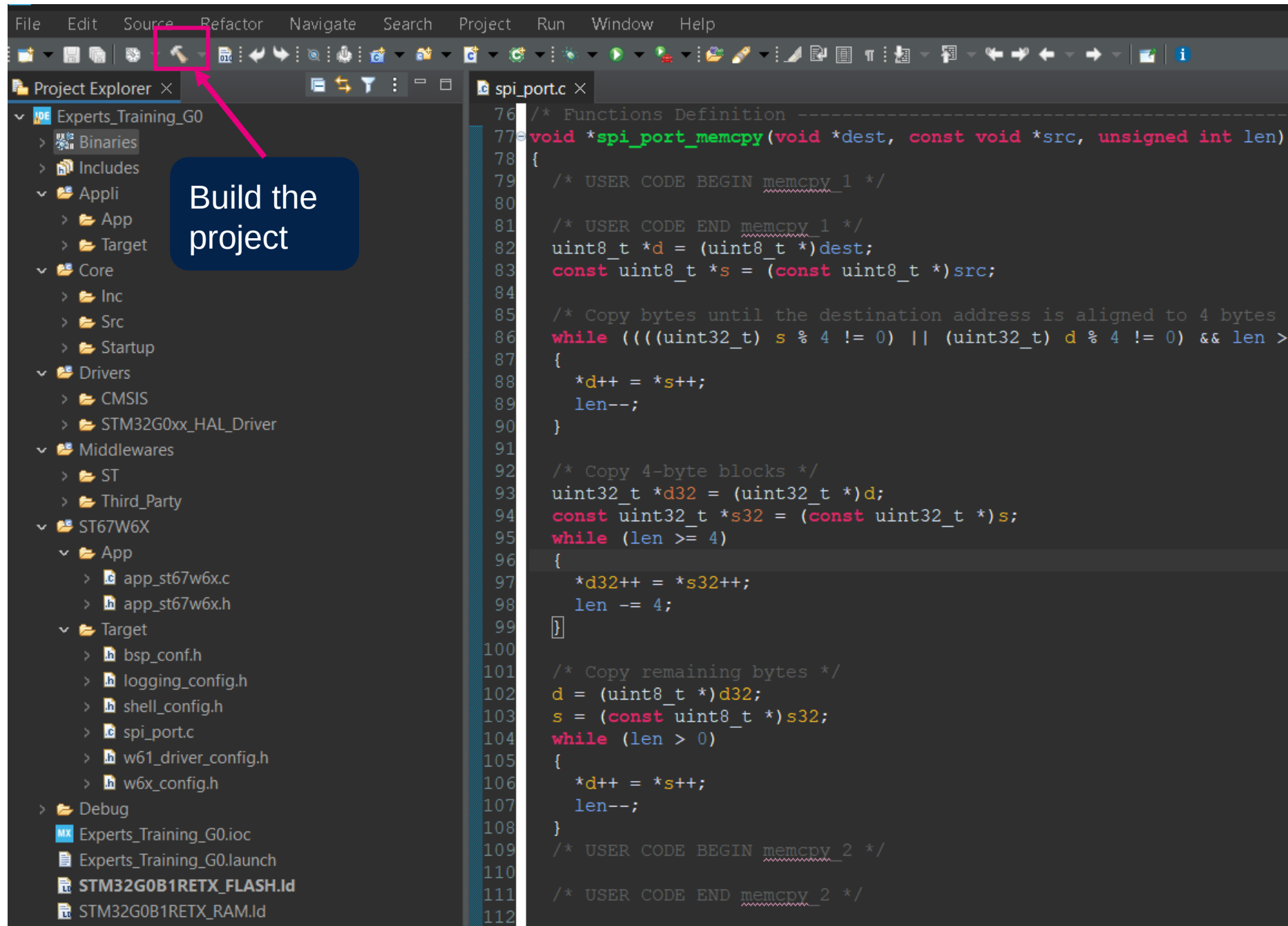
In Project/ST67W6X/spi_port.c
Add the below line of code to
check for the source as well.

while (((uint32_t) s % 4 != 0) || (uint32_t) d % 4 != 0) && len > 0)



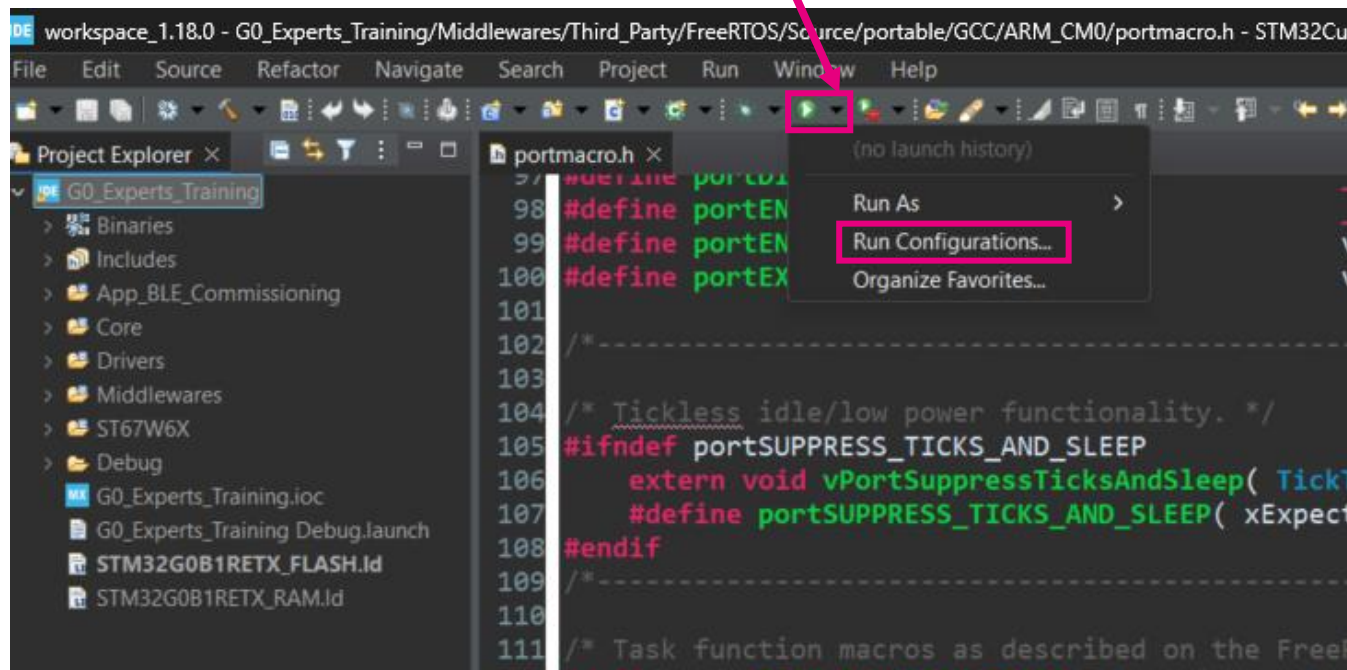
```
76 /* Functions Definition -----
77 void *spi_port_memcpy(void *dest, const void *src, unsigned int len)
78 {
79     /* USER CODE BEGIN memcpy_1 */
80
81     /* USER CODE END memcpy_1 */
82     uint8_t *d = (uint8_t *)dest;
83     const uint8_t *s = (const uint8_t *)src;
84
85     /* Copy bytes until the destination address is aligned to 4 bytes */
86     while (((uint32_t) s % 4 != 0) || (uint32_t) d % 4 != 0) && len > 0)
87     {
88         *d++ = *s++;
89         len--;
90     }
91
92     /* Copy 4-byte blocks */
93     uint32_t *d32 = (uint32_t *)d;
94     const uint32_t *s32 = (const uint32_t *)s;
95     while (len >= 4)
96     {
97         *d32++ = *s32++;
98         len -= 4;
99     }
100
101     /* Copy remaining bytes */
102     d = (uint8_t *)d32;
103     s = (const uint8_t *)s32;
104     while (len > 0)
105     {
106         *d++ = *s++;
107         len--;
108     }
109     /* USER CODE BEGIN memcpy_2 */
110
111     /* USER CODE END memcpy_2 */
112
113     return dest;
114     /* USER CODE BEGIN memcpy_End */
115
116     /* USER CODE END memcpy_End */
117 }
```

Build!

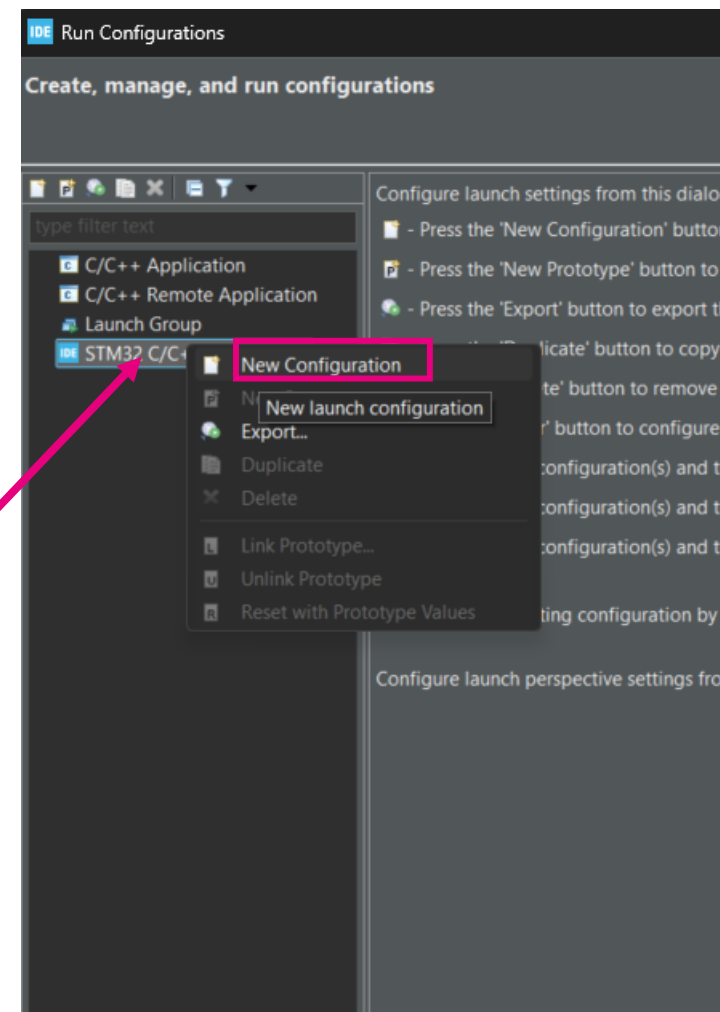


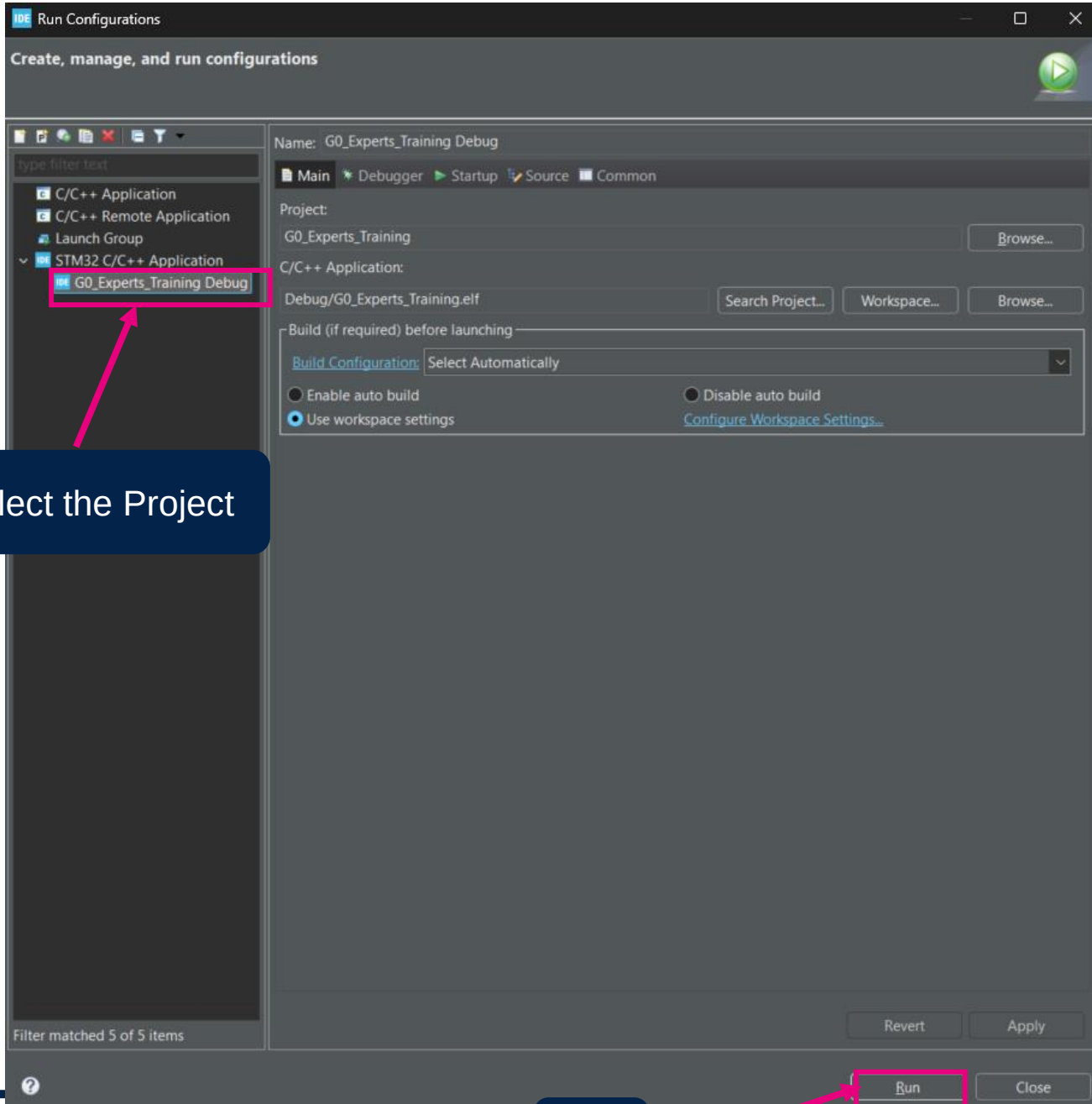
Flash!

Flash the project



Right-Click on STM32 C/C++ application





Select the Project

Run

Console logs after
flashing the project

Flash!

```
Problems Tasks Console Properties
<terminated> G0_Experts_Training Debug [STM32 C/C++ Application] ST-LINK (ST-LINK GDB server) (Terminated Aug 20, 2025, 3:14:43 PM) [pid: ...]

Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
-----
STM32CubeProgrammer v2.19.0
-----

Log output file: C:\Users\kulkarns\AppData\Local\Temp\STM32CubeProgrammer_a15436.log
ST-LINK SN : 0670FF485570854967113128
ST-LINK FW : V2J46M31
Board : NUCLEO-G0B1RE
Voltage : 3.23V
SWD freq : 4000 KHz
Connect mode: Under Reset
Reset mode : Hardware reset
Device ID : 0x467
Revision ID : Rev Z
Device name : STM32G0B0xx/B1xx/C1xx
Flash size : 512 KBytes
Device type : MCU
Device CPU : Cortex-M0+
BL Version : 0x92

Opening and parsing file: ST-LINK_GDB_server_a15436.srec

Memory Programming ...
File : ST-LINK_GDB_server_a15436.srec
Size : 122.02 KB
Address : 0x08000000

Erasing memory corresponding to sector 0:
Erasing internal memory sectors [0 61]
Download in Progress:

File download complete
Time elapsed during download operation: 00:00:03.410

Verifying ...

Download verified successfully

Shutting down...
Exit.
```

Tera Term Logs after flashing

Tera Term Logs

```
#### Welcome to ST67W6X Wi-Fi Commissioning over BLE Application ####
# build: 23:14:47 Sep  7 2025
----- Host info -----
Host FW Version:      1.1.0
----- ST67W6X info -----
ST67W6X MW Version:   1.1.1
AT Version:           1.0.0
SDK Version:          2.0.89
Wi-Fi MAC Version:    1.6.44
BT Controller Version: 1.6.112
BT Stack Version:     1.9.56
Build Date:           Sep  4 2025 11:29:50
Module ID:            Undefined
BOM ID:               0
Manufacturing Year:   2000
Manufacturing Week:   00
Battery Voltage:      3.316 V
Trim Wi-Fi hp:        8,8,8,8,8,7,6,6,7,7,7,8,8,8
Trim Wi-Fi lp:        8,8,8,8,9,9,9,10,10,10,11,11,11
Trim BLE:              7,5,5,7,7
Trim XTAL:            37
MAC Address:          40:82:7b:00:0c:2d
Anti-rollback Bootloader: 0
Anti-rollback App:    0

-----
Wi-Fi init is done
Net init is done
Ble init is done
BD Address: 40:82:7b:00:0c:2e
Configure BLE
BLE configuration is done

BLE Commissioning Service Creation
- BLE service created
- BLE WIFI Control charac created
- BLE WIFI Configure charac created
- BLE WIFI Monitoring charac created
- BLE FUOTA base address charac created
- BLE FUOTA confirmation charac created
- BLE FUOTA raw data charac created
- BLE services and charac registered
BLE service and charac creation is done

Start BLE advertising
BLE advertising is started
█
```

Tera Term: Serial port setup and connection

Port:	COM21	New setting
Speed:	115200	
Data:	8 bit	Cancel
Parity:	none	
Stop bits:	1 bit	Help
Flow control:	none	

Transmit delay

0	msec/char	0	msec/line
---	-----------	---	-----------

Device Friendly Name: STMicroelectronics STLink Virtual COM P
Device Instance ID: USB\VID_0483&PID_374B&MI_02\6&38AF34B
Device Manufacturer: STMicroelectronics
Provider Name: STMicroelectronics
Driver Date: 4-1-2021
Driver Version: 2.2.0.0



life.augmented

Documentation



Access:

The wifi wiki is opened to all



Wiki address of the main page

https://wiki.st.com/stm32mcu/wiki/Connectivity:Introduction_to_Wi-Fi



We will be happy to get your feedback!

Wiki Wi-Fi®: pages breakdown

About Wi-Fi

1 Wi-Fi® overview

1.1 What is Wi-Fi®

1.2 Wi-Fi® network architecture

1.3 Wi-Fi® features details

Details about
ST67W611M1

2 ST67W611M1

2.1 ST67W611M1 overview

2.2 X-CUBE-ST67W61

2.3 ST67W611M1: Hardware and board description

2.4 ST67W611M1 Security overview

2.5 ST67W611M1-based customer end-Product

2.6 ST67W611M1 evaluation

X-CUBE-ST67W61

3 X-CUBE-ST67W61 software application notes and user manuals

3.1 Getting started with ST67W611M1

Host boards
supported

4 ST67W611M1 associated hosts boards

5 Specific tools

Tools & References

6 Terms and definitions

7 References



Other documentation

- Getting started with X-CUBE-ST67W61 [here](#)

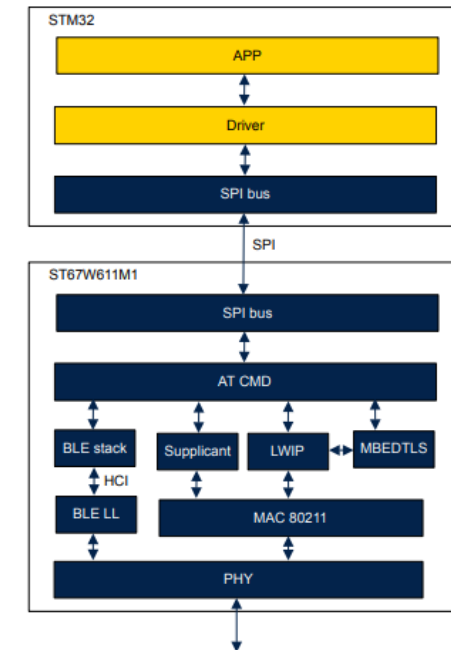
2

X-CUBE-ST67W61 architecture overview

The X-CUBE-ST67W61 is a set of software components implementing host applications driving a Wi-Fi® and Bluetooth® LE coprocessor. The coprocessor is controlled via AT command over SPI interface.

The figure below illustrates the architecture of the solution:

Figure 1. X-CUBE-ST67W61 architecture view



- MW API documentation is available:
 - X-CUBE-ST67W61_V1.0.0\Middlewares\ST\ST67W6X_Network_Driver\Doc\ST67W6X_Network_Driver.chm

ST Customer Support

- [ST Support Home](#)
 - Tickets can be submitted under the part number X-CUBE-ST67W61 or ST67W611M1



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